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Integrated Process & Product Design: Process-Based Entrainer Screening For Extractive Distillation

Master Thesis

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Abstract

Extractive distillation is considered an important intensified separation processes. It is capable of conducting separation that are not feasible through conventional distillation by introducing an additional liquid compound in the column called entrainer. In fact it is applied to achieve separations beyond the azeotropic compositions granted that a suitable entrainer is present. The economic viability of such process is very much reliant on the existence of such an entrainer. Traditionally, these entrainer were found on a trial an error basis through experimentation. This is exhaustive, time consuming and might not yield any success. In recent years, computer aided tools have been constructed to assist in the search of suitable entrainers. These computer aided tools use a series of constraints and target properties that the entrainer must keep to. However most of these properties are based on heuristic measures that are not necessarily always true. An interesting prospect is to develop a process model that can be incorporated in the product and thus achieve a an entrainer based on on the evaluation of a process model. For extractive distillation these models are rare and if available they are cumbersome to implement and rely on extensive mathematical procedure. In this work, Computer-Aided Molecular Design program is developed. The program is able to generate a multitude of feasible structures and predict their pure properties as well as their properties within a mixture. As a novelty, a shortcut model is Incorporated to investigate the performance of an entrainer candidate. The model consist of the "Infinitely Sharp Split (ISS)" by Petlyuk (2004). The model uses a series of thermodynamic criteria to determine the minimum entrainer to distillate ratio. Furthermore, the method is also able to calculate the corresponding value of the reflux ratio. These two parameters can be used t evaluate the process economics. A low entrainer to distillate ratio suggest less investment in procuring entrainers and less amount to be supplied to the process. A low reflux ratio would indicate less operational cost by reducing the cost of utilities. As far the the knowledge of the authors, this is the first attempt to incorporate this short-cut model into a Computer-Aided- Molecular design framework. The Design Program is performed in the software GAMS. The constructed framework is applied in an industrial relevant case consisting of the separation of the azeotropic system acetone-methanol.

Preface

This research originally stemmed from my passion for mathematical modelling, thermodynamics and coding. This project is conducted as part of the final assessment of the M.Sc. in Chemical & Biochemical Engineering. The project was conducted at the PROSYS (PROcess and SYStems Engineering) research centre of the department of Chemical & Biochemical Engineering (KT) at the Technical University of Denmark (DTU).

Th completion of this work would not have been possible without the help of my supervisors. Their help and guidance, suggestions and meaningful discussion as well as their vast experience have been of valuable and was dearly appreciated. A special thanks go to my main supervisor throughout my M.Sc. studies, Jens Abildskov. Countless meetings and lengthy emails to plan and set the frame for my studies have truly been appreciated. I hope I was not too much of a hassle. Another special thanks goes to Stefano Cignitti. His help with issues regarding the GAMS software was life-saving, without which, the framework would not be functioning. I would also like to thank Carina Gargalo and Yuqiu Chen for their inputs about the GAMS software. Finally, I would like to express my gratitude to Ivonne Rodriguez-Donis for a fantastic talk about the possible methods to implement in this work and especially explaining the ISS method and providing working examples.

It has been a challenging period conducting this work and would have been unbearable if not for the presence the people around me. They helped me pull through. Their support has truly been invaluable. A special thanks goes to *mi hermanos*: Andres, Jorge and Nail. The constant flow of memes they supplied has brought me immense joy and lengthy laughter. Another thanks goes to Thomas, Sussie, Simone and Chinmayi for helping keep my spirit high and hearing my constant complaints. Last but not least, a big thanks goes to my parents and especially my mother. I wouldn't have been here without their support.

I hereby declare that the work presented in this document is based on research carried out by the authors:

Signature: **Date:**

Nomenclature

Mathematical symbols:

Symbol	Units	Description
x	–	Liquid mole fraction
y	–	Vapour mole fraction
x'	–	Liquid mole fraction on entrainer free-basis
y'	–	Vapour mole fraction on entrainer free-basis
T	K	Temperature
T_m	K	Normal melting point
T_b	K	Normal boiling point
T_c	K	Critical temperature
P	kPa	Pressure
P_c	kPa	critical Pressure
P^{sat}	kPa	Saturated vapour pressure
P^0	kPa	Reference presser (atmospheric pressure)
w	–	Accentric factor
E/D	–	Entrainer to distillate ratio
x^{iso}	–	composition at isovolatility
x^{min}	–	camposition corresponding to the mnimum E/D
S_{ij}	–	Selectivity
C	–	Capacity
$S^{\inf}$	–	Selectivity at infinite dilution A
N	–	Number of stages
R	–	Reflux ratio
k	–	group k
v_k	–	Valency of group k
a	–	A

Abbreviations:

PSWD	Pressure-Swing Distillation
THF	Tetrahydrofuran
EA	Ethyl Acetate
DMSO	Dimethyl sulfoxide
CAMD	Computer-Aided (or Computer-Assisted) Molecular Design
QSPR	Quantitative Structure-Property Relation
GC	Group Contribution
GAMS	The General Algebraic Modeling System
GCM	Group Contribution Method
IP	Isopropyl alcohol
DMF	Dimethylformamide
USD	U.S. Dollar
TAC	Total Annual Cost
TI	Topology Indices
SD	Signature Descriptors
MIP	Mixed Integer Problem
MILP	Mixed Integer Linear Problem
MINLP	Mixed Integer NonLinear Problem
NLP	NonLinear Problem
UNIFAC	UNIversal Functional-group Activity Coefficients
VLE	Vapour-Liquid Equilibrium
ISS	Infinitely sharp Split
min	minimum
max	maximum
ICAS	Integrated Computer Aided System

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Chapter 1

Introduction

Many of the recent scientific and technological developments in the last century have been catalysed by the development of faster and stronger computer tools. These tools are capable of solving complex problems in nearly every subject. They have been used in sociology to predict human behaviour such as the public reaction towards a specific product or policy [81]. They have also been employed in environmental sciences, where they have been able to forecast natural disasters such as the trajectory of tornadoes [61]. This has made it possible for the authorities the initiate contingency plans in time to reduce material and human loss. In physics, supercomputers have been used to analyse data from the Large Hadron Collider (LHC) where particles collide more than 500 times per second in the hope of shedding some light on a series of fundamental questions such as the deep structure of space and time. In the fields of biology and medicine, computer aided tools for the detection of diseases have been developed such as the detection of cancer types in their early stages [51]. These are but a few of the fields that have been influenced by the development of computer science. Other fields include economics, construction, mathematics and supply chain management.

The processing engineering industry and specifically the chemical engineering field is no exception from the influence of the advancement in computing system technology. The "Amundson" report issued by the National Research Council of the U.S in 1987, suggested a list of seven directions where process engineering would develop in the 21st century [62]. Computer assisted process engineering (CAPE) was among these directions. This has indeed come true since then. Each year, series of entire conferences, conference sessions and symposia are dedicated to the discussion and presentation of the newest developments within the application of computers in the chemical engineering field. Such conferences include but are not limited to "PSE: *Process Systems Engineering*","ESCAPE: *The European Symposium on Computer Aided Process Engineering*" and "FOCAPD: *Foundations of Computer Aided Process Operations / Chemical Process Control*". The extent of this influence can also be seen through the dedication of an international journal on computer applications in chemical engineering titled "Computers & chemical engineering". In fact, the use of computers in chemical engineering have become a discipline on it own that has given birth to "Computer aided" fields within chemical engineering such as computer aided process design, computer aided process control and computer aided product design. These fields mainly focus on the modelling of unit operations, the simulation of entire processes and the investigation of the process dynamics to study their operability. Other focuses also include process synthesis, process integration and process intensification. These fields have become even more important nowadays due to the increasing pressure on the available world resources. This has pushed industries to manage and reduce their energy consumption while achieving similar or higher production capacity, resulting in the development of highly specialised and complex processes. These processes are referred to as "intensified processes".

Moulijn et. al. defined process intensification as as "*Any chemical engineering development that leads to a substantially smaller, cleaner, safer, and more energy efficient technology*" [84]. These processes have such a degree of complexity that computer-aided tools are necessary to design and develop them. These tools are needed because the design methods are usually based on large algorithms that need to evaluate and solve a series of constitutive equations describing the dynamics of the system at hand. Within the chemical engineering field, separation processes have especially been targeted by process intensification, resulting in processes such as reactive distillation, divided wall columns, azeotropic distillation and extractive distillation. However, these tools are not exclusively used for process design, and are of equal importance in product design. A product can be specifically design/engineered to satisfy desired properties such as the ability to carry heat (refrigerants), to dissolve another compound (solvents) or offer protection in specific environments (coatings). The use of computer assisted tools within separation processes and chemical product design is motivated and illustrated in the following section by underlining the huge importance of these processes and the opportunities computer-aided tools offer. These will set the stage for the problem formulation in focus in this work.

1.1 Background

1.1.1 Importance Of Separation Processes

Separation processes are processes intended to separate a mixture feed into its constituents. Based on the nature of the feed and the principle by which the separation is realised, several types of separation processes can be distinguished. Mechanical separation processes such as centrifugation or sedimentation are applied if the feed mixture contains immiscible phases. However if the the feed mixture exists as a homogeneous phase other separation techniques with different nature must be used. In these processes, the constituents are forced to separate into different spacial locations with a given process equipment. All separation process can be related to either of the following techniques: separation by phase creation (e.g. distillation), separation by phase addition (e.g. extraction), separation by barrier (e.g. membrane filtration), separation by solid agent (e.g. adsorption) or separation by force field or gradient (e.g. centrifugation). However more advanced separation processes combine some of these techniques together e.g. the combination of phase creation (distillation) and phase addition (in the form of a mass separating agent) to form extractive distillation or the combination of phase creation (distillation), phase addition (mass separating agent) and a barrier (filtration) in azeotropic distillation.

In general, separation processes have been of constant interest for process intensification. The reason for this is the vital role they play in modern human life, highlighted by the amount of energy dedicated to their operating them. Separation processes in the chemical industry cover close to 15% of the world energy consumption [80]. Distillation covers a big part of this consumption. In fact, it covers about 50% of the U.S. energy consumption of separation processes [80]. In principle, distillation exploits the difference in the relative volatility between the components in a given mixture [43] and is one the most energy intensive processes as reported by the U.S. department of energy [89]. The relative energy usage of the different industrial processes can be seen in fig. 1.1.

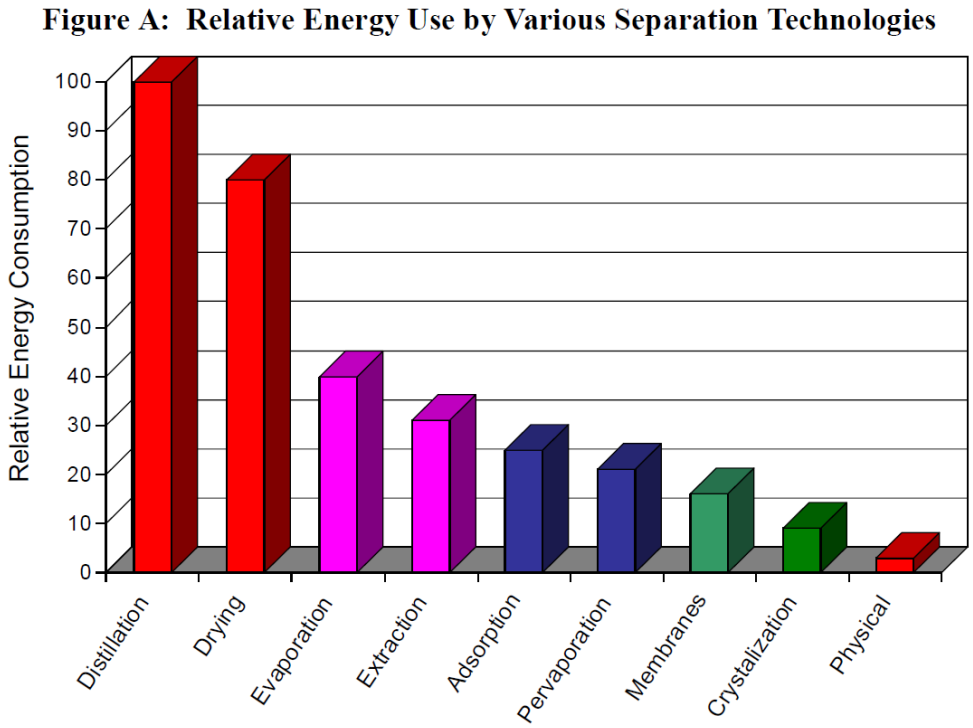


Figure 1.1: Relative Energy use by various separation processes [89].

Process intensification has opened the door for possibilities, previously unattainable by conventional processes. As an example, conventional distillation is not able to fully separate some mixtures. This is due to azeotrope formation between the components. A mixture forming an azeotrope exhibits a constant boiling point for which the vapor and liquid phases have the same composition and no further separation is attainable using conventional distillation. Azeotropes are the result of the mixture deviating from ideality. This deviation can either be due to strong attractive forces between the components which will result in maximum boiling azeotrope [55], or be the result of strong repulsive forces between the molecules in the liquid mixture [55]. This results in a minimum boiling azeotrope. The product loss resulting from this, constitute an untapped resource which is undesirable from an economic and a sustainability perspective. To tap into these potential resources, advanced distillation schemes have been developed. Such schemes include pressure-swing distillation, azeotropic distillation and extractive distillation. In the following, the working principle and the industrial application of each of these techniques are illustrated.

1.1.2 Advanced Distillation Techniques

Pressure-Swing Distillation (PSWD)

Some azeotropic systems exhibit a particular sensitivity towards change in pressure [12]. For these systems, PSWD can be an attractive method of separation. Particularly because it avoids contamination effects associated with the addition of a separating agent. The process configuration for this type of distillation consists of two columns operating at different pressures. If the system forms a minimum boiling azeotrope, the operating pressure of the first column is lower than than of the second column [22]. The inverse is

true for the maximum boiling azeotropic systems. The jump in the azeotropic composition between the two columns results in obtaining 2 pure component streams along with a third stream containing a mixture of azeotropic composition. This stream can then be recycled as feed to the first column. Such column configuration can be seen in fig. 1.2.

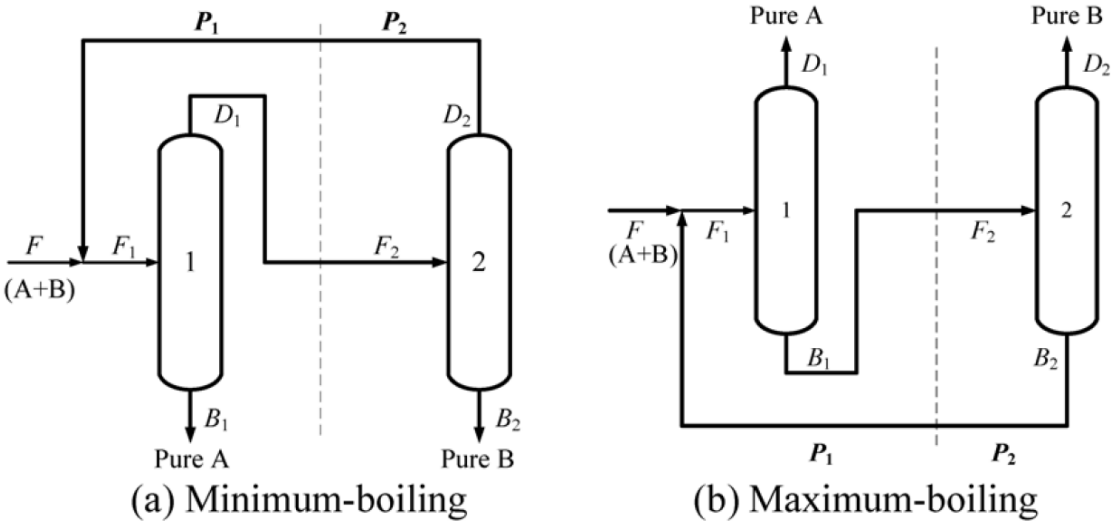


Figure 1.2: Column arrangement in pressure-swing distillation for minimum and maximum boiling azeotropes [22].

Some examples of the application of PSWD in industry are shown in table 1.1.

Table 1.1: Examples of industrial application of PSWD

Mixture	Process discritpion	Ref
Acetone - Methanol	The mixture forms a minimum boiling azeotrope at 1 atm at a temperature of 328K. Increasing the pressure to 10 atm, the azeotrope is shifted to a 37.5 mol% of acetone at a 408 K. Purity of 99.5 mol% for methanol and 99.4 mol% for acetone were achieved.	[22]
Water - tetrahydrofuran (THF)	This system forms an azeotrope at 1.01 bar with 82.3 mol% THF. At 7.9 bar, the composition is shifted to 63.9 mol% THF. PSWD has produced a product purity of 99.9 mol% and 99.0 mol% for THF and water respectively.	[48].
Ethyl acetate (EA) - ethanol	A minimum boiling azeotope is formed at a composition of at 55.5 mol% EA and at a temperature of 344.9 K. Deploying a first column operating at 6 atm and a second column operating at 0.5 atm, product purity beyond 99.5 %mol is achieved for both compounds.	[100]

One major concern in using PSWD is the fact that not all azeotropes are susceptible to significant changes with pressure. PSWD is also associated with high compressor duties needed to displace the azeotrope. To overcome these disadvantages, azeotropic or extractive distillation can be used. These two processes are characterised by the addition of an extra mass separating agent known as an "entrainer" to the process. The main purpose of this component is to alter the relative volatility existing between the components forming the azeotrope [43].

Azeotropic Distillation

Depending on whether the added agent causes the formation of an immiscible liquid phase, the azeotropic distillation can be distinguished as homogeneous or heterogeneous [12]. In heterogeneous azeotropic distillation, the entrainer forms a low boiling azeotrope and a phase split (two liquid phases). The entrainer is mixed with the feed and fed into a distillation column [43]. Due to the formation of minimum boiling azeotrope, the heavy key can be easily separated as the bottom product [12], while the light key along with the entrainer leave the column as top product. Since the top products are not miscible, the components can be separated using a decanter. The entrainer can thus be recycled back to the feed of the distillation column along with a make-up stream, since small losses of the entrainer is inevitable. An illustration of such configuration can be seen in fig. 1.3. For homogeneous azeotropic distillation, two mass separating agents are employed. The entrainer is fed with the azeotropic mixture. The entrainer forms a low boiling azeotrope with one of the components and is separated as overhead product in the first column. The bottom product is a pure stream of the remaining component. The entrainer is separated in the overhead stream using liquid-liquid extraction and thereby the introduction of the second mass separating agent. The new resulting mixture should then be separated using a second column. It is a requirement that the volatility is sufficiently different so the can be separated into pure product streams. The solvent can then be recycled for the liquid-liquid extraction process [47] [43]. A schematic of a typical azeotropic distillation can be seen in fig. 1.3.

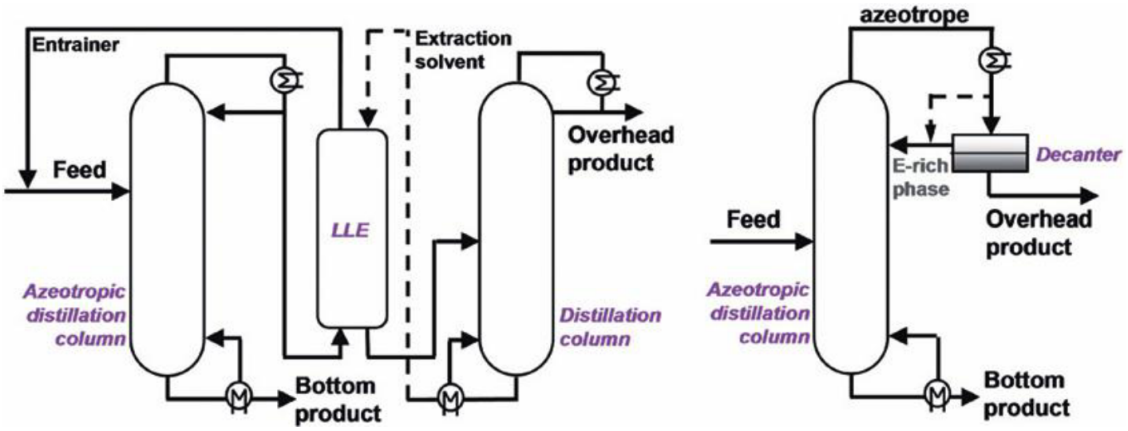


Figure 1.3: Typical Homogeneous (left) and heterogeneous (right) azeotropic distillation schematics [43].

There are many examples of the use of azeotropic distillation to overcome the challenges presented by azeotrope formation. An important application is the production of anhydrous ethanol which is of special importance in the production of biofuels. The ethanol-water mixture forms a minimum boiling azeotrope at atmospheric pressure. The azeotropic composition is 89 mol% ethanol. Cyclohexane can be used as a heterogeneous entrainer. The process schematic is identical to the one presented in fig. 1.3. The bottom product of the first column mainly comprises of ethanol with a composition of 99.6 mol%. The bottom product of the second column is pure water with very small traces of ethanol and the solvent. The top product which contains ethanol and water is recycled to the first column [8].

Extractive distillation

Similarly to azeotropic distillation, extractive distillation necessitates the addition of an entrainer. The entrainer alters the relative volatility of the azeotropic components without creating any new azeotropes or allowing phase split when it is added [12]. In extractive distillation there is a distinction between whether the entrainer used has a higher, an intermediate, or a lower boiling temperature compared to the the original mixture. This distinction results in different process configuration based on the type of entrainer. The typical extractive distillation configurations for a binary azeotropic mixture comprises two successive distillation columns. The first one is called the "extractive" column, while the second is called the "recovery" column [32]. The entrainer is fed into the extractive column on a feed stage located close to the top of the column. This is due to the fact that the altering of the relative volatility predominantly occurs in the liquid phase and also to insure that it is present in appreciable concentration in every tray [12]. The component having the highest volatility in presence of the entrainer leaves the extractive column at the top as a pure stream. The entrainer and the remaining component are fed into the recovery column where the entrainer is recovered and recycled to the extractive column. Depending on whether the entrainer is the heavy boiling component or not, the entrainer can either leave the top of the recovery column or the bottom of the recovery column. Heavy boiling entrainers are usually more desirable due to the fact that no additional energy is needed to evaporate them in the recovery column [49]. The typical process configuration for extractive distillation can be seen in fig. 1.4.

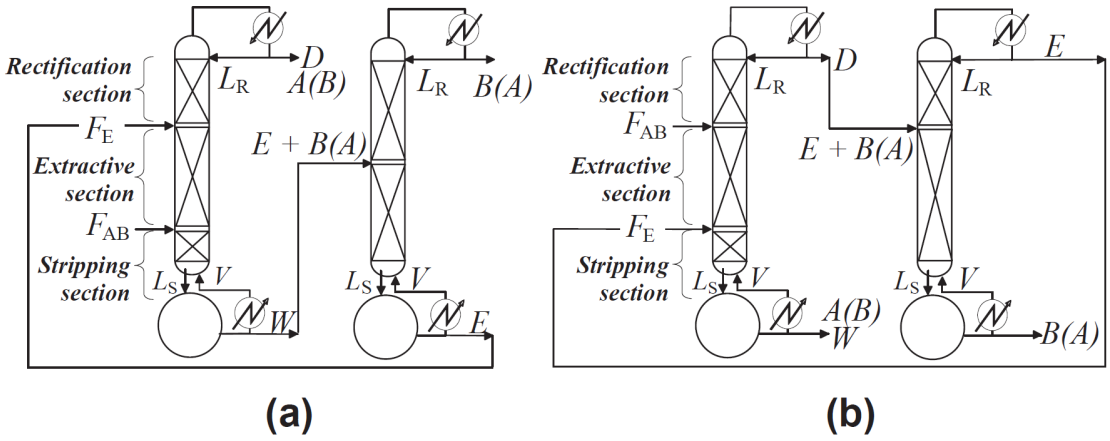


Figure 1.4: extractive distillation configuration (left:heavy entrainer, right:light entrainer) [32]

An example is the acetone-methanol system system, which forms an azeotrope at 328 K with a 77.6 mol% acetone. A series of solvents have been used in extractive distillation to achieve high purity (minimum 99.5 mol % for both components) [17]. The entrainers used include water, DMSO and chlorobenzene. An extensive overview of the different systems to which extractive distillation is applied and the respective entrainers used can be found in the literature review section or in the paper by Gerbaud et al. (2018) [92].

All three separation techniques offer the possibility to achieve product purity beyond the azeotropic composition. Nevertheless, each process is associated with some advantages and disadvantages. Pressure-swing distillation is only applicable if the azeotrope is pressure-sensitive and in some cases a very high pressure is needed to displace the

azeotrope significantly. Thus resulting in high compressor operating cost. On the other hand, azeotropic distillation and extractive distillation can be operated at atmospheric pressure. The addition of the entrainer, however, results in additional cost: cost of the entrainer and an additional distillation column to recover the entrainer. The use of an entrainer could also increase the risk of contamination. Several studies, however, have shown that if a suitable entrainer is available, then extractive distillation could prove to be the most advantageous of these technologies from an operational and economic perspective [55]. The choice of a suitable entrainer, constitute a major challenge for extractive distillation as it is considered the deciding factor on the economic feasibility of the process. Therefore, many efforts have been dedicated to address this issue. The process of designing or selecting an entrainer is considered a chemical product design problem.

1.1.3 Chemical product design

By definition, product design is the process of translating a set of needs into a product [11]. Chemical product design is therefore the procedure "*of finding a chemical product that exhibits certain set of functional properties (the needs)* [25]. In terms of chemical products, the product can either be a single chemical, a mixture of chemicals or a formulation of active ingredients and additives [27]. The identity of the product is unknown but its properties are well defined [24]. This type of design is referred to as "reverse property prediction". In essence, it is the ability to determine the structure of the molecule based on its properties [24]. The product function or performance can be related to some of its properties. An example of this is refrigerants. The heat capacity of refrigerants have major impact on their ability to carry heat. However in some cases, a clear-cut relation between properties and performance is not always available. Therefore the product candidates are usually applied in the process they are intended to an evaluated based on that. Adjustments can also be implemented on the intended process to accommodate the obtained product. This underline the importance of being able to measure the suitability of a product not only by its properties but also by a performance metric. The interconnection between the product design and process design is illustrated in fig. 1.5.

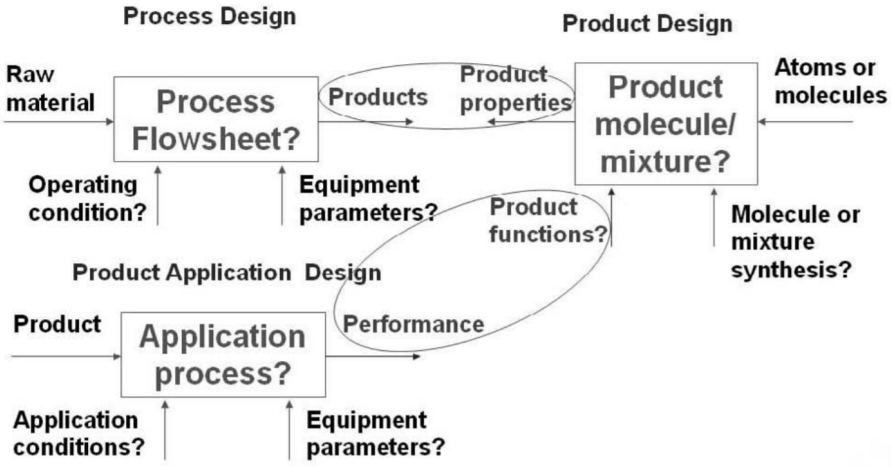


Figure 1.5: Interconnection between product and process design/application [27]

Different approaches have been developed to solve the product design problem. These approaches include experiments, databases, heuristics or model based approaches [99]. A schematic of the different product design approaches can be seen in fig. 1.6.

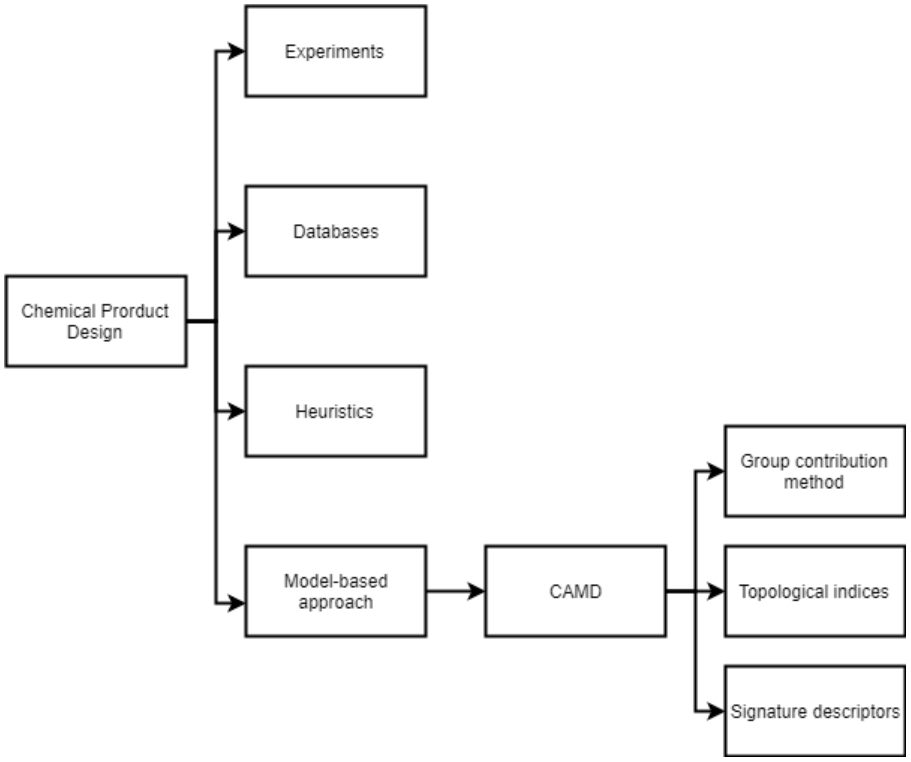


Figure 1.6: A schematic of the different approaches to the chemical product design problem

Each approach has its advantages and disadvantages. Conducting experiments has usually been based on trial and error, which can be tedious, time consuming and success is not always guaranteed. In addition, this approach can be limited by the financial resources available to acquire such products. Even if this approach is successful in generating a suitable candidate, there is no guarantee that it is the optimal solution. The experimental approach is however a real reflection of the reality and the real performance of a given product and is thus vital for validating of the obtained product [99]. Using databases offers access to a large amount of available chemical products. Most registered properties are reliable and can generate candidates fast based on desired properties. A disadvantage is that it is mainly pure component properties that are available in databases, while mixture properties, such as as selectivity are rarely found in databases [99]. Heuristics offer a fast, easy and inexpensive solution to a given design problem. The downside is the need for "know-how" concerning the process and product. Additionally, heuristic measures lack theoretical basis and are thus of limited applicability [99]. The last approach is a model-based search technique that applied mathematical models to describe the design problem in terms of properties. This requires the existence of validated mathematical models to estimate all the required properties [27]. This approach is capable of generating a list of feasible candidates that can be tested and evaluated with regard to process performance in a fast and efficient manner. These models represent computer aided molecular design (CAMD) tools. The models used are based on the concept of relating the structure of of a compound to its properties, also known as QSPR (Quantitative Structure-Property

Relationships (QSPRs). QSPRs are semi-empirical mathematical models that link physical, chemical and thermodynamic properties to the molecular structure [99]. TA few of these methods are signature descriptors, topology indices and group contribution methods [2]. The ability to describe the product design problem in terms of mathematical models offers another advantage which is the ability to evaluate the performance of the product in a given process. This requires the availability of a process model. Should this be the case, then the design problem can be formulated as an integrated product-process design problem, where the product is not solely evaluated through its properties but also by its performance.

In summary, the importance of separation processes and especially extractive distillation has been demonstrated. Some of the challenges associated with extractive distillation have been presented, of which the entrainer selection is the most important. The entrainer selection is a product design problem for which computer-aided tools might offer significant help. It is in this context that the problem formulation in the following section is considered.

1.2 Problem Formulation

1.2.1 Problem Statement

The feasibility of extractive distillation is heavily dependent on the availability of an efficient entrainer capable of breaking the azeotrope existing between the components of a given mixture. The entrainer must posses a series of properties that qualifies it be used in the extractive distillation processes. It is therefore of paramount importance to be able to quickly screen the available candidates for those that clearly do not have these properties and thus reducing the search space considerably. These candidates should then be ranked based on some process metric that reflects either its performance or the economics of the process. Therefore, the objective of this work is the develop a CAMD tool to screen a defined search space for promising entrainers to used in extractive distillation. This screening should be based not only on the product properties but also on its performance in the process (integrated product-process design). The tool will be developed on a conceptual level and then implemented in a suitable software. The software chosen for this work is GAMS (The General Algebraic Modeling System), which is a software used to formulated mathematical models and solve complex numerical problems through advanced algorithms.

1.2.2 Project plan

The project problem will be divided into sub-problems or sub-tasks to manage the complexity of the problem but also to provide an overview of the different parts and stages of the project. The project work-flow will be reported in a Gantt chart and the deadline for the partial reports will be set within. The project tasks are defined in the following.

Task 0: Problem formulation

The first part consist of formulating the project problem and establish the aims and objectives of the projects.

Task 1: Literature Review

The second part of the project will consist of a literature review on the uses of extractive distillation and the existing methods for entrainer selection. This will serve to establish an overview on existing methods to perform the molecular design and the different target properties.

Task 2: Design framework

A framework to handle the entrainer design problem is to be established. This framework will provide a systematic approach to manage the complexity of the integrated product-process design problem. The formulation of the entrainer design problem must be developed and the different methods to formulate and solve the design problem must be identified. The following sub-tasks are planned.

a Establishing a Design framework suitable for the entrainer design problem.

b Apply the framework on the entrainer design problem.

c Identify the methods to describe the entrainer from a product design perspective.

d Identify the methods to describe the entrainer from a process design perspective.

Task 3: Implementation & validation of the CAMD tool

The formulated design problem and the supporting methods and models are implemented in GAMS and validated through sanity checks, data from the litterature and cases studies. The second part of the project is dedicated to the extractive distillation process design. The process design covers the extractive column as well as the recovery column. An investigation is performed of short-cut methods that can provide process insights for economic evaluation. The following sub-tasks are planned.

a Model Implementation.

b Model validation.

c Solving an entrainer design problem through a case study.

1.2.3 Gantt Chart

The decomposed problems have been reported in a Gantt chart to monitor and plan the progress of the work at hand. The Gantt chart can be seen in fig. 1.7.

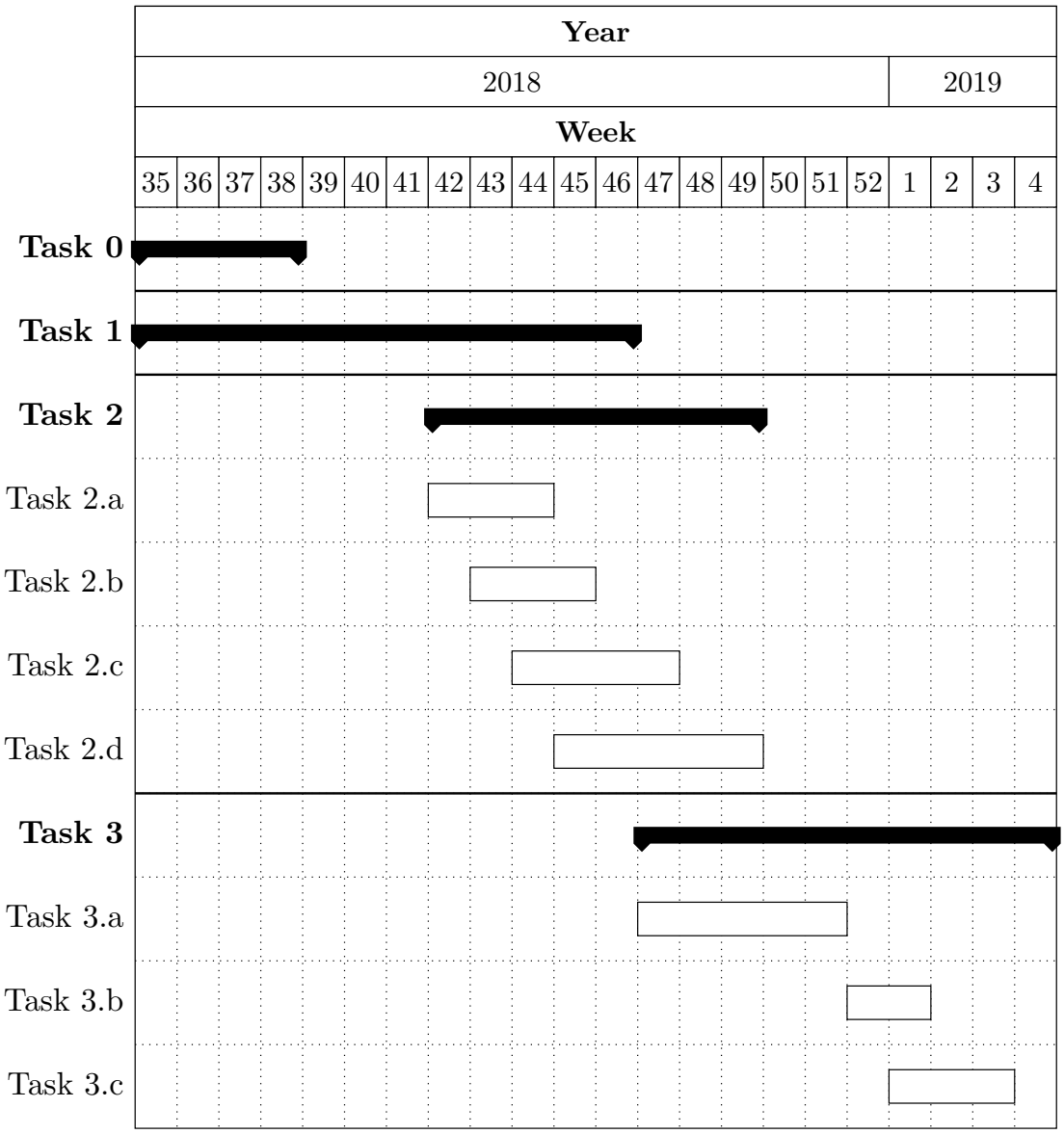


Figure 1.7: Gantt Chart illustrating the project plan.

Chapter 2
Literature Review

This chapter serves as a short literature review of the general aspects of extractive distillation and especially the solvent selection procedure as well as the different CAMD techniques applied in the chemical product design field. The advantages and disadvantages of the different selection criteria and the design techniques will be discussed. This will set the stage for the selection of suitable methods for determining entrainers.

2.1 Extractive Distillation

In this section, the technology of extractive distillation is reviewed. The different process configurations and types of extractive distillation are presented. A more thorough overview of the application of this process in an industrial context is also given.

2.1.1 Types Of Extractive Distillation

A distinction between the operating mode of the extractive distillation can be made; the process can either operate in a semi-batch or as a continuous process. The configuration in both operating modes changes depending on the nature of the the entrainer used. This is discussed in the following.

Semi-Batch Operation

The entrainer used in an extractive distillation process can either have a significantly higher boiling point than the azeotropic pair and thus be denoted"heavy entrainer" or have a significantly lower boiling point and thus be called "light entrainer". Based on the type of entrainer used, two configurations are suggested. For the use of a light entrainer, a batch stripper is used. The entrainer is fed in an intermediate stage of the column. The section above the feed stage makes up the extractive section, whereas the section below makes up the stripping section. One of the components is stripped from the original mixture, while the remaining compound can be removed from the bottom [32]. This configuration is usually used when the original mixture contains high-boiling-point components or the components are heat sensitive. When using a heavy entrainer, a batch rectifier column is used. Similarly to the other configuration, the entrainer is fed at an intermediate location. The section above is called the rectification section, while the the section below is the extractive section. The mixture is fed into the reboiler. This type is used if the mixture has a low boiling point [32]. An illustration of the process configuration can be seen in fig. 2.1. Both configurations have been extensively studied and used. Table 2.1 contains a short overview of the applications where batch extractive distillation has been employed.

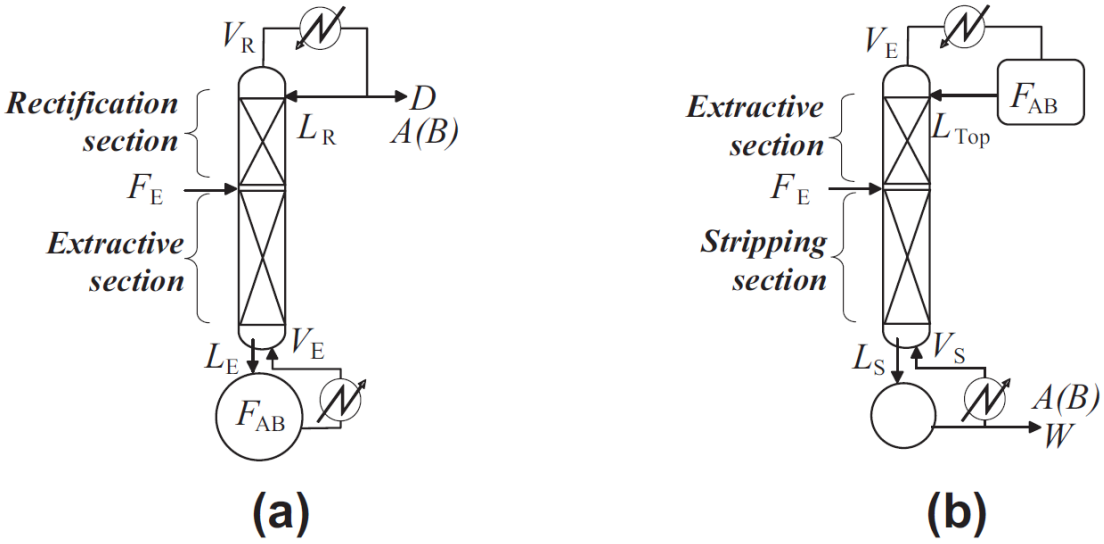


Figure 2.1: Semi-Batch configuration for extractive distillation (left: heavy entrainer, right: light entrainer)[32]

Table 2.1: Industrial application of extractive distillation operated as semi-batch

Mixture(A-B)	Entrainer (E)	Reference
ethanol-water	methanol	[91]
IPA-water	DMSO	[96]
methyl acetate-cyclohexane	carbon tetrachloride	[50]
chloroform-ethyl acetate	2-chlorobutane	[50]
cyclohexane-1-propanol	DMF	[85]
ethanol-water	ethylene glycol	[39]
acetone-methanol	water	[75]

Continuous Extractive Distillation

Extractive distillation operated as a continuous process is the core focus of this thesis. Thus, a more extensive review is presented. Firstly, a description of the different process configurations is given followed by an overview of the industrial application. Finally, the comparison between continuous extractive distillation with the remaining technologies used to separate azeotropic mixtures is presented. This comparison will motivate the interest in the technology at hand.

The distinction between using a heavy or a light entrainer is visible from the column configurations of the processes. The important feature of extractive distillation is that the entrainer feed location differs from the azeotropic mixture feed stage. For both entrainer types, the extractive column is divided into three parts: an extractive section that is located between the two feed stages, a rectification section that is located above the extractive section and a stripping section that is located below the extractive section. The figure presented in the previous chapter is revisited below.

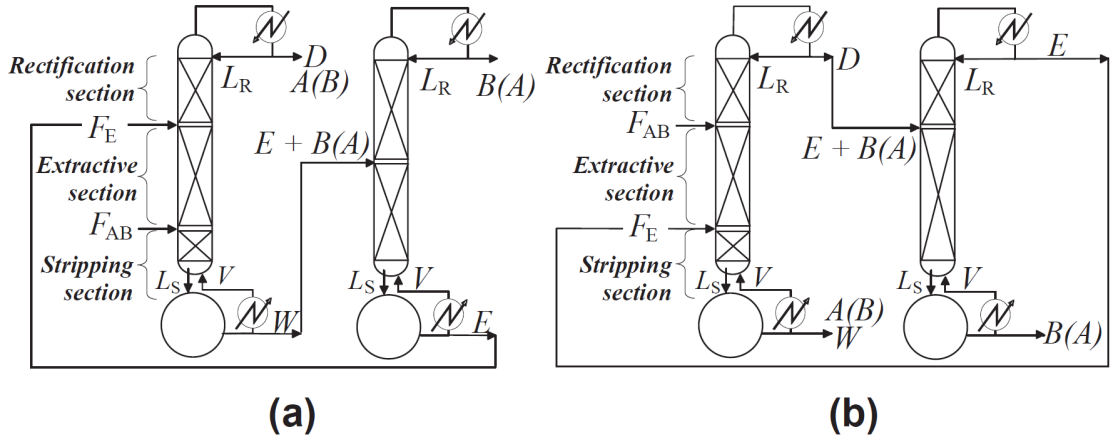


Figure 2.2: Typical configuration for extractive distillation (left: heavy entrainer, right: light entrainer) [32]

In configuration (a), the heavy entrainer interacts with one of the compounds and reduces its relative volatility which makes it possible to obtain the other component as a pure product at the top of the extractive column. The entrainer and the remaining compound are then separated in the recovery column where the entrainer is the bottom product. The advantage of using a heavy entrainer is the fact that there is no need for extensive energy supply to evaporate it. The entrainer is then recycled into the extractive column. It is essential that the top product in this column is as pure as possible. Impurities in the bottom product of the recovery column are not of concern as this stream proceeds to the extractive column.
In configuration (b), the light entrainer interacts with one of the azeotropic components and increases its relative volatility. This will result in a top product consisting of the entrainer and one of the components of the original mixture. The remaining component from the original azeotropic mixture can then be obtained as the bottom product of the extractive column. In the recovery column, the entrainer is obtained as the top product and recycled to the extractive column while the remaining component from the original mixture is removed from the bottom.
The popularity and importance of this process is highlighted by the numerous systems and mixtures it has been applied to in an industrial context. In fact it has been used to separate both maximum- and minimum-boiling azeotropes using two types of entrainers. Table 2.2 presents a list of mixtures forming maximum boiling azeotropes and where a light entrainer has been used.

Table 2.2: Example of maximum boiling azeotropes and applied light entrainers

Mixture	Azeotrope	Entrainer	Purity	References
Acetone (A) - chloroform (B)	x_A=0.36	dichloromethane	0.99 0.99	[73]
Methyl isobutyl ketone (A) - Propanoic acid (B)	x_A=0.57	dimethylformamide	0.99 0.99	[73]

Table 2.3 shows an overview of systems forming maximum boiling azeotropes where a heavy entrainer have been used. The table also showcases the azeotropic composition of the mixtures and the product purity obtained by using the entrainer.

Table 2.3: Example of maximum boiling azeotropes and applied heavy entrainers

Mixture	**Azeotrope**	**Entrainer**	**Purity** x_A x_B	**References**
Acetone(A)-chloroform(B)	x_A=0.96	Benzene	0.99 0.99	[72]
		Toluene	0.99 0.99	[72]
		DMSO	0.995 0.995	[55]
		Dimethylformamide	0.995 0.995	[1]
water(A)-ethylene diamine(B)	x_A=0.45	1,3-propandiol	0.997 0.997	[98]
		1,2-propandiol	0.997 0.997	[98]
		1,4-butandiol	0.997 0.997	[98]

Table 2.5 and Table 2.4 show minimum boiling azeotropic systems where heavy and light entrainers have been applied respectively.

Table 2.4: Example of minimum boiling azeotropes and applied light entrainers

Mixture	**Azeotrope**	**Entrainer**	**Purity** x_A x_B	**References**
ethanol(A)-water(B)	x_A0.8970	methanol	0.99 0.99	[91]
		acetone	0.998 0.998	[46]
ethanol(A)-toluene(B)	x_A=0.8152	acetone	0.99 0.99	[73]
2-propanol(A)-water(B)	x_A=0.6740	methanol	0.998 0.998	[46]

Based on the industrial applications presented above, it is evident that the same system can be separated by either using a light or a heavy entrainer. In a similar way, a light entrainer for a specific system can act as a light entrainer in another system. In all applications, a product purity beyond the azeotropic composition is obtained. This not only underline the wide spread usage of the process but also the versatility of the process (different entrainers same process). To further underline the benefits of using extractive distillation, a case based comparison between the different technologies to separate azeotropic mixtures is provided in the following.

Table 2.5: Example of minimum boiling azeotropes and applied heavy entrainers

Mixture	**Azeotrope**	**Entrainer**	**Purity** x_A x_B	**References**
ethanol(A)-water(B)	x_A=0.8970	ethylene glycol	0.995 0.995	[54]
		Propylene glycol	0.995 0.995	[63]
		glycerol	0.995 0.995	[29]
tetrahydrofuran (A)-water (B)	x_A=0.817	propylene glycol	0.995 0.995	[95]
		ethylene glycol	0.999 0.005	[18]
		DMSO	0.9999 0.9999	[33]
		1,2-propandiol	0.9975 0.9975	[33]
		glycerol	0.985 0.985	[33]
Acetonitrile (A)-water (B)	x_A=0.6900	DMSO	0.999 0.999	[78]
		dichloromethane	0.9995 0.9995	[87]
		ethylene glycol	0.999 0.999	[78]
Acetone (A)-methanol (B)	x_A=0.7600	water	0.994 0.995	[88]
		2-propanol	0.995 0.995	[45]
		ethanol	0.995 0.995	[45]
		DMSO	0.994 0.995	[54]
		Butandiol	0.995 0.995	[45]
		propandiol	0.995 0.995	[45]
		ethylene glycol	0.995 0.995	[45]
		p-xylene	0.995 0.995	[82]

Extractive distillation: the go to process for azeotropic separation

In many cases, extractive distillation offers an attractive option regarding cost and energy savings in comparison with pressure swing and azeotropic distillation. Luyben made a comparative study of the cost of separating acetone and chloroform using extractive distillation (entrainer: DMSO) and pressure-swing [55]. The estimated annual cost of

operating the pressure-swing process was found to be over 4 billion USD , while extractive distillation could be operated with less than a quarter of that amount [55]. This is not surprising since the cost of compression and maintaining the column under pressure will result in major equipment and energy costs. Luyben made another study using the example of methanol recovery using water in the TAME reactive distillation process [56]. This study showed that the extractive distillation would offer a reduction of the capital investment of 40% (from 4.25 billion USD to 2.44 Billion USD) while the energy saving would reach 60% (from 6.92 billion USD to 2.22 billion USD). The total annual cost is thus reduced by 65% from 8.34 billion USD to only 30.04 billion USD [56]. Modla et. al. [59] made the comparison based on the acetone-methanol azeotrope. Using water as the entrainer, the extractive distillation still offered lower total annual cost than pressure swing. Even though extractive distillation does in general offer cost savings compared to pressure swing, it is not always the case. This can be due to two reasons: the azeotrope is very pressure-sensitive or the entrainer used is too expensive. Such case were studied by Lladosa et al.[53] for the separation of di-n-propyl ether and n-propyl alcohol [53]. In this case extractive distillation uses 50% more in capital investment and resulting in a total annual cost reduction of 30% (0.64 million Euros to 4.45 Million euros) for 12,000 ton/year plant[53]. The same trends are seen when comparing azeotropic distillation with extractive distillation. Zhao et al.[101] studied the economics for the separation of a ternary azeotrope made of ethanol, toluene and water using both processes. The entrainer for the extractive distillation was glycerol, while the heterogeneous azeotropic distillation used the toluene present as entrainer itself. For the proposed capacity of 1 ton/h of the azeotropic mixtures, the total annual cost of the the extractive distillation process was 30% less than the one for azeotropic distillation [101]. The ethanol dehydration process has also been used as a benchmark to evaluate the cost difference between azeotropic and extractive distillation (using ethylene glycol) [8]. With a plant capacity of 300 cubic meters per day, the study showed that savings of around 30% in the capital cost (from around 3 million to 2 million USD) [8]. An important feature of the extractive distillation that can result in cost savings is the use of heavy entrainers that do not need to be evaporated in the recovery column.

Even though extractive distillation does offer some attractive savings on the capital and operational costs, there are some challenges associated with this technology. These challenges reside mainly in the choice of a suitable entrainer. In fact, this can be the deciding factor in the feasibility of a process [49]. In the industrial case of the separation of water and tetrahydrofutan (THF), three entrainers were evaluated [40] for a 100 kmol per hour production capacity. The evaluated entrainers were ethylene glycol (EG), glycerol (GL) and dimethyl sulfoxide (DMSO). The total annual cost associated with EG was the lowest. Compared with DMSO, it offered 16% reduction of the TAC, while comparaed with GL it offered 60% savings [40]. For the separation of acetonitrile from water, a total of six entrainers were evaluated (DMSO, 1,2-ethandiol, DMF, Glycerol, Butyl acetate and pyridine) [77]. Three of them (DMF, butyl acetate and pyridine) resulted in thermodynamically infeasible separations due to the formation of azeotropes and heterogeneous regions. The energy savings were evaluated and it was found that using glycerol offered 6.5% and 13% savings in reboiler duties compared to DMSO and 1,2.ethandiol [77]. It is therefore important to highlight and review the criteria used in the literature to overcome the challenge of selecting a suitable entrainer.

Design methods

A brief overview of the methods used to design the extractive distillation process is presented in the following. The main advantages and disadvantages are also highlighted.

The conceptual design of extractive distillation is a challenging task [69]. This is due to multiple reasons: the extractive column has multiple feeds (feed stage location, feed conditions, entrainer to feed ratio, reflux ratio, number of stages), the fact that the first column exhibits different phenomena (rectification, stripping and extraction) and the presence of recycle streams from the recovery column to the extractive one[68]. All these particularities of the process result in greater degrees of freedom from a design perspective.

Commercial simulators have been applied extensively in the design and optimisation of extractive distillation processes and have usually been based on "trial and error" [69]. This procedure can be cumbersome, however it permits rigorous design and the usage of different thermodynamic packages. De Figueiredo et al (2011) proposed a systematic procedure to design and optimise extractive distillation columns using a commercial simulator Aspen Plus[20]. The procedure was applied on the dehydration of ethanol. The aim was to minimise the ratio of reboiler duty to the ethanol production rate. An initial feasible design must be constructed at first and the objective function along with the free variables must be indicated (solvent flow ratio, distillate flowrate, feed stages number of stages). Aspen plus optimiser toolbox was then used to fix the solvent flowrate, the distillate flowrate and the reflux ratio. The number of stages and the feed stages are determined using a sensitivity analysis[20]. The objective function used in by De Figueiredo et al (2011) might not reflect the whole process economics and therefore might not provide an optimal solution. Another concern is the fact that the interplay between the flow rates and the feed stages/number of stages is not investigated since they are determined in a sequential manner.

In contrast to rigorous models, short-cut methods usually provide much easier and simpler performance metrics that makes it possible to compare different designs. Since the entrainer chosen for a given separation have huge impact on the process design, these methods can be used to compare the performance of the entrainers. Extensive efforts have been dedicated to develop methods to study the feasibility of extractive distillation processes from a thermodynamic point of view [69]. These methods usually are based on on a series of analysis of the ternary diagrams and distillation residue curve maps. Such efforts include the works by Doherty et al.(1985) in which they used ternary residue curves to determine if a separation of a given mixture was possible using extractive distillation [13]. Wahnschaft et al. (1993) developed a geometric approach based on residue curve maps to determine the feasibility of a split given a specific feed composition and target product purity [93]. More recent work include the work by Rodriguez-Donis et al. (2009) in with the a method based on the analysis of the uni-volatility lines was developed to study the feasibility of batch extractive distillation [73][72].
Studies on the thermodynamic feasibility of splits in extractive processes are numerous, however not many methods have been developed for the process design part. Kossack et al (2008) proposed a shortcut method to determine the minimum entrainer flow rate and the minimum reflux ratio[45] based on the rectification body method by Bausa et al. (1998) [3]. The principle of this method is to approximate the column profiles through "rectification bodies" that are determined from pinch points. The minimum entrainer flow rate is then determined through a pitchfork analysis of the pinch points. Results ob-

tained with this method enables the ranking for different entrainers based on the amount of entrainer required but also on the possible energy demand through the reflux ratio [45], which proved to be in accordance to rigorous designs obtained through process simulators [45]. Despite its reliability, the method have had limited usage. This is due to the complex mathematical approach and the numerical instability it exhibits. Another short-cut method is based on the principle of infinitely sharp split developed by Petlyuk et al. (2015) [69]. Similarly to the method by Kossack et al. (2008), this method is capable of determining the limiting value of the entrainer to distillate flowrate as well as the minimum reflux ratio [69]. this method is considered much simpler as it it only requires knowledge about the feasibility of a given split and no tedious mathematical algorithms are needed to determine the metrics of interest. The results obtained from this method were in good agreement with the results obtained from a commercial process simulator. A last example of short-cut methods used for extractive distillation is the application of the Fenske equation [63]. This methods is predominantly used in multicomponent conventional distillation to determine the number of stages at infinite reflux. However, it has been used in extractive distillation to illustrate the ease of separation, resulting from the addition of a fixed amount of entrainer.

2.2 Entrainer Selection

In this section, the desired properties of an entrainer for extractive distillation is highlighted. The section will also present a review on the different criteria applied in CAMD for entrainer screening.

The objective of adding an entrainer is to separate an azeotropic pair beyond the azeotropic composition and should therefore not form an azeotrope with the mixture compounds[45]. In the case of azeotropic distillation, this criteria might be disregarded as the recovery step is not necessarily done using a distillation column. The added entrainer should also no create a phase split[4]. Otherwise, less energy intensive processes can be used[72]. The entrainer should not only make the original separation feasible, it should also ensure that the recovery process is easy by having suitable enthalpy of vaporisation and boiling point[4]. The entrainer should also be safe to handle (not cause corrosion, should not harm operators) [49]. All these criteria should be determined and evaluated for a given entrainer candidate. Entrainers have predominantly been screened based on a series of heuristic rules that are rooted in either thermodynamic perspective (ability to alter the relative volatility) or a molecular perspective (ability to form hydrogen bounds[49]).
From a heuristic point of view, the ability to alter the relative volatility is related to the ability of the entrainer to associate preferentially with one of the azeotropic components in the original mixture (e.g. through hydrogen bounds). Blass (1989) [4] suggests that molecules can be classified into five categories. This classification is based on comparing the Trouton's constant with the enthalpy of vaporisation of an entrainer. The reasoning supporting this approach is that higher values for the enthalpy of vaporisation would suggest stronger interactions in the liquid phase. In contrast, a lower value suggests the formation of associates in the vapor phase. This classification can be seen in table 2.6). Based on observations and experience working in the field, authors usually agree that compounds for class I and class II are often successful when selected as entrainers [49].

The selection of an entrainer can also take roots in the thermodynamics occurring when adding an entrainer. The main objective of the entrainer is to alter the relative volatility

Table 2.6: Molecule classification based on the ability of forming hydrogen bounds

Class	Description	Example
I	Capable of of forming three dimensional networks of strong Hydrogen bounds. Highest deviation fro Trouton's constant	water, glycerol hydroxy-acids, amides
II	Capable of forming one dimensional chains by hydrogen bounding. Effect of hydrogen bounds is not a dominant as Van der waal forces	ethanol acetic acid
III	Capable of acting as hydrogen bound acceptors. The enthalpy of vaporisation is close to Trouton's constant	ethers, ketones nucleophelic aromatics
IV	Capable of acting as hydrogen bound donors	chloroform, dichlorethane trichloroethane
V	Not capable of acting neither as hydrogen bond donor nor acceptor.	Tetrachloroethane aromatics

of one of the compounds forming an azeotrope. Assuming ideal vapour phase, the relative volatility has been defined as in equation eq. (2.1), where x and y are the mole fraction in the the liquid and vapour phase respectively, P^{sat} is the vapour pressure, γ is the activity coefficient calculated from a thermodynamic model and "i","j" are the binary pair in the mixture where "j" denotes the solute (the component on which the solvent/entrainer act).

$$\alpha_{ij} = \frac{y_i/x_i}{y_j/x_j} = \frac{\gamma_i \, P_i^0}{\gamma_j \, P_j^0} \tag{2.1}$$

When an azeotrope is formed, the relative volatility equals unity. Therefore, the aim of the addin the entrainer is to displace the relative volatility far from unity. For a fixed temperature and pressure, the influence of the entrainer on the relative volatility can be examined predominantly based on the ratio of the activity coefficients. This ratio in the presence of the solvent is called "selectivity" eq. (2.2). This metric has been used throughout the literature in the selection of entrainers [12] [32] [45] [49].

$$S_{ij} = \left(\frac{\gamma_i}{\gamma_j}\right)_S \tag{2.2}$$

Since the activity coefficients are dependent on the liquid phase composition and the fact that the entrainer effect increases proportionally with the concentration, it is usually evaluated infinite dilution instead [45]. The selectivity at infinite dilution represents the maximum selectivity that can be possibly achieved and is evaluated for a binary mixture where the entrainer composition is close to unity [45]. The expression of the selectivity at infinite dilution can be seen in eq. (2.3).

$$S_{ij}^{\infty} = \left(\frac{\gamma_i^{\infty}}{\gamma_j^{\infty}}\right)_S \tag{2.3}$$

Another parameter that has been considered in the literature is the "capacity" [4]. The capacity of the entrainer towards the solute is defined in eq. (2.4).

$$C_{j,S}^{\infty} = \frac{1}{\gamma_j^{\infty}} \tag{2.4}$$

The capacity is an expression of the interaction between the solute and the entrainer. The lower the activity coefficient, the higher the interactions between the two compounds and thus the higher the capacity. A high interaction between the entrainer and one of the azeotropic compounds is a desired property. However, it has been documented

that entrainer candidates exhibiting high capacities also exhibits low selectivity values [4] [45]. The common practice is to drop the the capacity from consideration and only consider the selectivity at infinite dilution [49]. Candidates are usually ranked based on this parameter and those possessing the highest value are deemed the most promising. The general expectation is that those with the highest selectivity offer the most cost-effective process [4] [60]. However, this is not always the case. In fact multiple studies have been conducted to evaluate the accuracy of using selectivity as basis for solvent screening [45] [60]. Momoh(1991) evaluated the selectivity as a ranking parameter for a number of entrainers [1] used on three different azeotropic systems. The following systems were considered:

1. Mixture A: 2-Methyl-1-Butene/Isoprene.
2. Mixture B: n-Butane/Trans-2-Butene.
3. Mixture C: n-Hexane/Benzene.

For each of the aforementioned mixtures, the author calculated the total annual cost (TAC) of the extractive distillation process including both the operating cost and the capital cost for the extractive column and the recovery column. The calculations were based on an entrainer to feed ratio of two and and equimolar feed of the mixtures. The TAC was then evaluated with regard to the calculated selectivity. The results can be seen in fig. 2.3, fig. 2.4 and fig. 2.5 for mixtures A, B and C respectively.

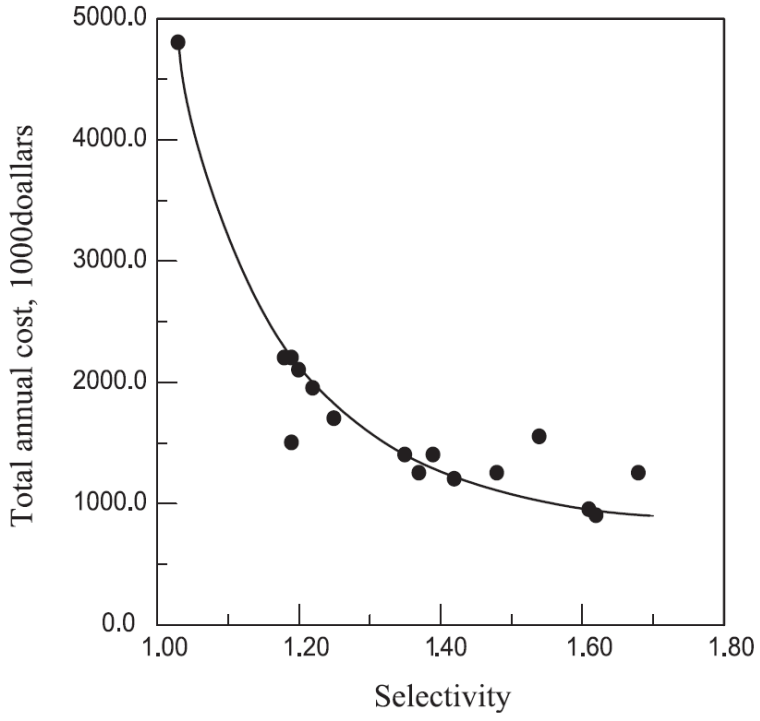


Figure 2.3: Influence of entrainer selectivity on the TAC of the extractive distillation process for (2-methyl-1-butene/Isoprene) [60]

[1] The nature of these entrainers are not important in this discussion. They can be found in the references [60] and [49]

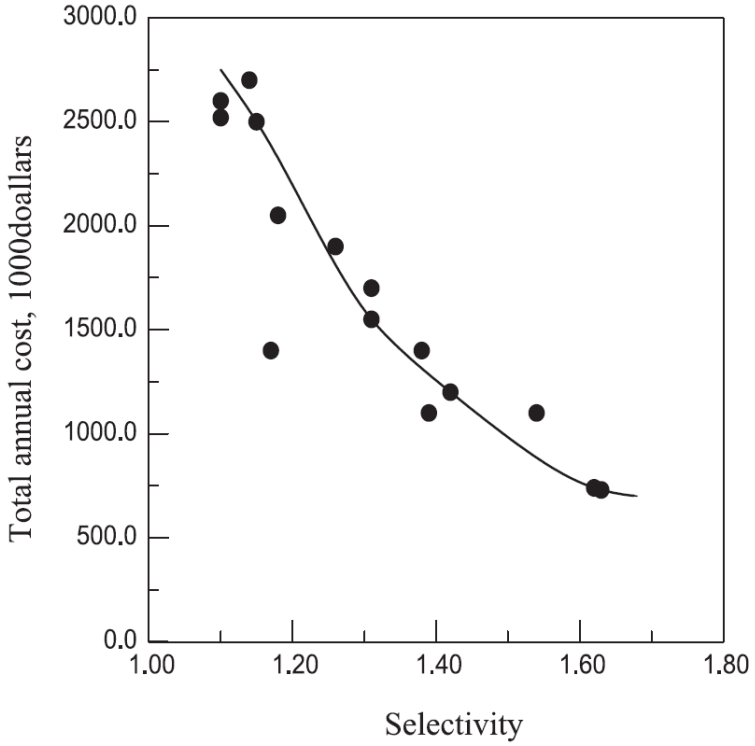


Figure 2.4: Influence of entrainer selectivity on the TAC of the extractive distillation process for (n-butane/Trans-2-butene) [60]

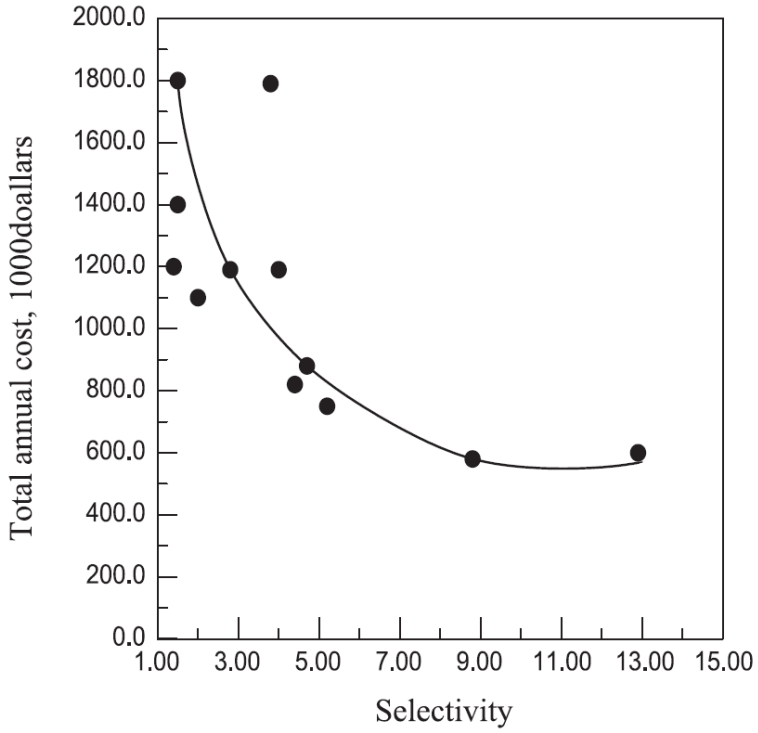


Figure 2.5: Influence of entrainer selectivity on the TAC of the extractive distillation process for (n-Hexane/Benzene) [60]

Based on the results obtained by Momoh (1991) and shown in fig. 2.3, fig. 2.4 and fig. 2.5, it seems that the TAC is inversely proportional to the selectivity; that is, the TAC decreases with increasing selectivity [60]. This trend can be explained by the fact that with increasing relative volatility, the reflux ratio decreases which will result in a decrease in the number of stages needed to achieve the separation in the extractive column [60]. Additionally, the decrease in the reflux ratio reflects a decrease in the vapour flow rate. This could therefore suggest a reduction in the utilities used for the reboiler and the condenser. For low values of the selectivity, the TAC is dominated by the cost of the extractive column. As the selectivity increases, the separation in the first column becomes easier and the TAC decreases. Eventually, the TAC becomes dominated by the cost of the recovery column thus the graphs becomes more flat. Although the trend of inverse proportionality between the selectivity and the TAC, the figures do not suggest that the highest selectivity corresponds to the lowest TAC. In mixture A, the highest selectivity does not correspond to the lowest TAC, in fact it is approximately 30% higher than the lowest TAC value. In mixture C, the highest TAC does not correspond to the lowest selectivity either (2nd lowest).

Using selectivity as a basis for screening is by no means a bad criteria, but concerns about the accuracy are evident. It could be suggested that the selectivity could be used as a "cut-off" criteria where a minimum value for the selectivity is set. In fact Momoh(1991) made a comparative study between the TAC of the extractive distillation process using the different entrainer candidates and the TAC of the ordinary distillation set-up [60]. It was shown that to achieve a TAC lower than that of the conventional distillation, a minimum selectivity can be suggested. These can be seen in table 2.7. It is important to highlight that the suggested values are system-dependant and will vary for various systems.

Table 2.7: Minimum selectivity to achieve lower TAC compared to conventional distillation [60]

Systems	**Minimum Selectivity**
Mixture A	1.1760
Mixture B	1.1060
Mixture C	4.3250

Another study conducted by Kossack et al.(2003) investigated the TAC including the energy demand for the separation of a water/acetone mixture using different entrainers [45]. The energy demand for the extractive column and the total process as function of the selectivity can be seen in fig. 2.6. Based on these results, it is clear that for low selectivities, the energy demand is not predicted accurately and it seems that this demand does not change significantly for higher values of the selectivity.

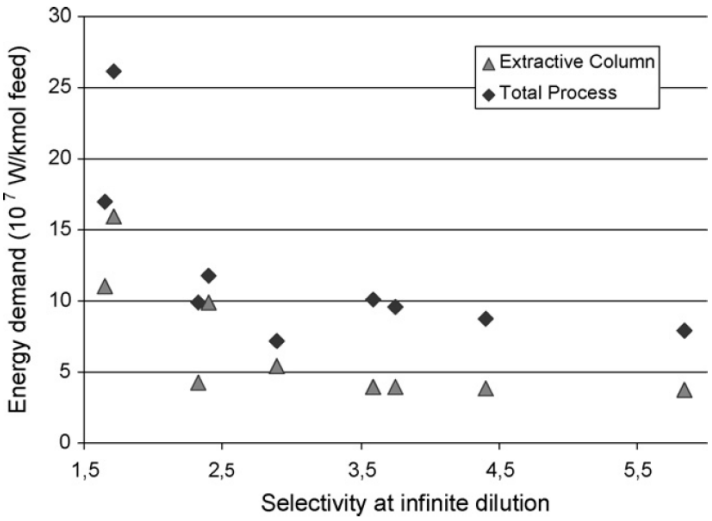


Figure 2.6: Energy demand as function of the selectivity [45]

The TAC as a function of the selectivity was also investigated [45]. The results can be seen in fig. 2.7. These results show that there is a poor correlation between the two parameters.

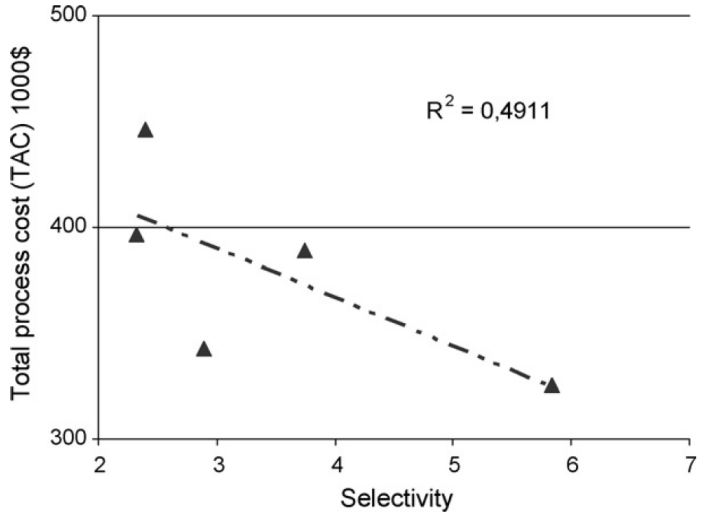


Figure 2.7: Total annual cost as a function of selectivity[45]

The authors however suggest that a better agreement is observed between the TAC and the product of the selectivity and the capacity of the entrainer. The parameter is defined as follows and the obtained results can be seen in fig. 2.8.

$$C_{j,S}^{\infty}\dot{S}_{ij}^{\infty} = \frac{\gamma_i^{\infty}}{(\gamma_j^{\infty})^2} \tag{2.5}$$

It is important to note that these criteria are based on observation and are thus heuristics and do not always hold as previously demonstrated. To be able to more precisely assess the performance of an entrainer, some process parameters must be included in the evaluation of the entrainer.

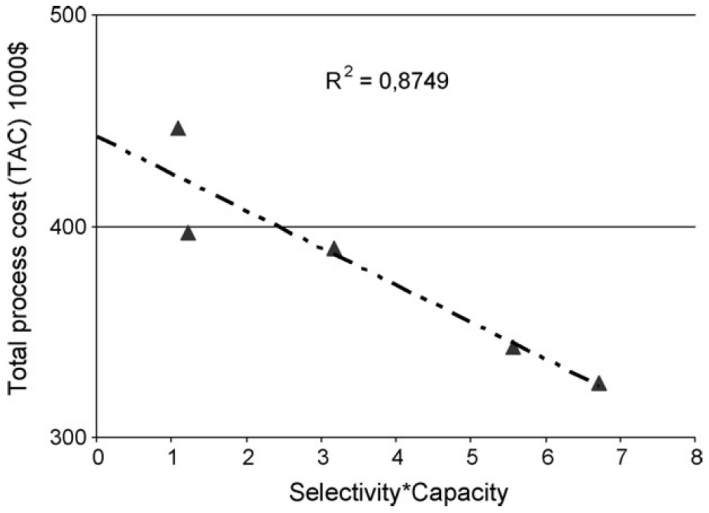


Figure 2.8: Total annual cost as a function of capacity-times-selectivity [45]

2.3 Computer-Aided Molecular Design: CAMD

As touched upon in the background section, computer-aided molecular design have been applied in chemical product design and also in the entrainer design problem. In this section, the most important CAMD techniques are presented and reviewed along with the theory and principles behind the this approach.

2.3.1 CAMD Techniques And Formulations

Computer aided molecular design applies mathematical models to reverse engineer the molecular structure of a chemical entity based on its properties. These models are usually semi-empirical, of which the most applied are:

1. Group contribution methods.
2. Signature descriptors.
3. Topological indices.

In the following, the principle supporting each QSPR techniques is presented.

Group Contribution Method (GCM) The Group contribution method is the most popular QSPR model used in CAMD [2]. Its popularity is the result of the simplicity and universality of the concept. This technique is especially attractive in a conceptual design stage and as part of pre-design stages [44] as it is not computationally heavy. The working principle of GC methods is that the property of a given molecule can be related or predicted based on the number of occurrence of a series of predetermined building blocks forming the molecule [2]. These building blocks could either refer to atomic fragments (occurrence of atoms), bond fragments (occurrence of a specific bound between two atoms) or as group fragments (specific arrangement of atoms) in the molecule. An illustrative example can be seen in fig. 2.9 for a molecule of ethanol.

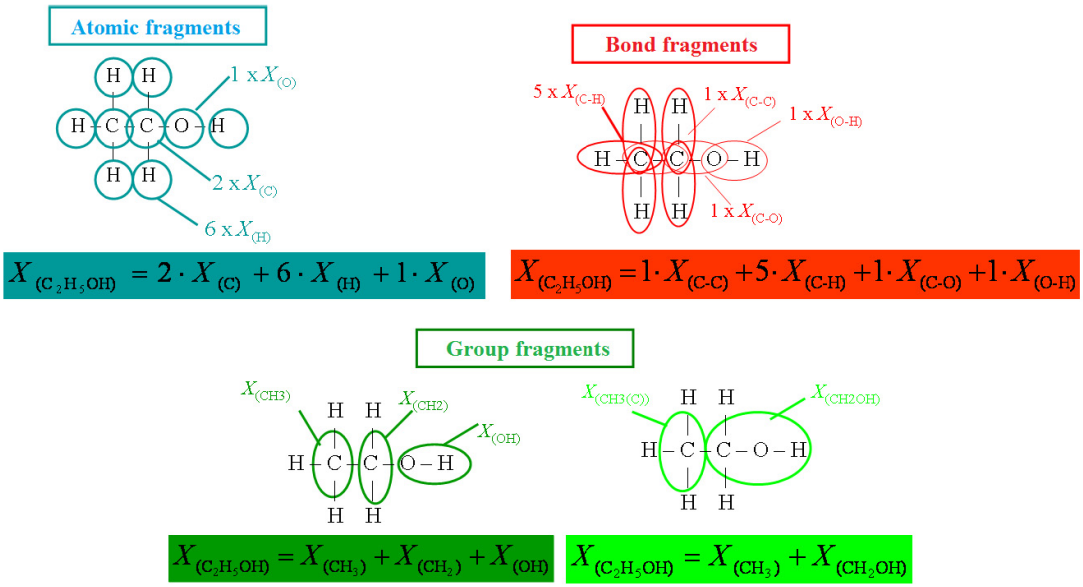


Figure 2.9: Decomposition of ethanol into fragments

In GCM, each group has a "contribution" towards the overall property of the molecule. The methods takes into account the type of groups present and their number of occurrence [2]. A generic formulation can be seen in equation eq. (2.6) where P is the property, C_i is the contribution of group g and N_i is a vector representing the number of occurrence of the groups.

$$P = \sum_i N_i \, C_i \tag{2.6}$$

Many different GC methods have been developed throughout history. One for the early and popular version is the GC method developed by Joback and Reid [2][41]. They suggested that some properties should be described as non-linear (e.g. logarithm, power function) since not all properties could be captured with a linear function. A generic functional transformation of eq. (2.6) can be seen in equation eq. (2.7)), where f is the functional transformation.

$$P = f\left(\sum_i N_i \, C_i\right) \tag{2.7}$$

An important limitation to this method is the fact that it only accounts for the occurrence of groups and can therefore not distinguish between isomers. Constantinou and Gani (1994) proposed a new GC method that uses two levels of contribution [9]. The first level is made of a set of first order groups while the second level is made of a much smaller subset of second order groups that describe a larger arrangement of groups. The second order groups serve to provide more information of the overall structure of the molecule. The introduction of a new structural level has also contributed to capturing the group proximity effects. Proximity effect is a result of the interaction between two groups in a molecule which alters the contribution of either of the groups. This method has resulted in more accurate prediction for a wide range of properties such as the boiling point temperature, the critical point, critical pressure and melting point [9]. Marrero and Gani (2001) extended the model of Constantinou and Gani (1994) by introducing a new structural level and third order groups to increase the accuracy of predicting properties for polycyclic compounds [57]. The generic form of the new GC method is shown in eq. (2.8)

where C_i, D_j, E_k are contribution of the first, second and third groups i,j,k respectively. N_i, M_j, O_k are integer vectors accounting for the occurrence of the groups. w and z are binary parameters that take the value one if the level is considered, otherwise they are zero.

$$P = \sum_i N_i C_i + w \sum_j M_j D_j + z \sum_k O_k E_k \tag{2.8}$$

A wide range of properties has been predicted using GC methods [2]. Table 2.8 presents a small overview of some of these properties for ehich the mentioned methods have been applied. For a much larger overview, may consult be found in the work of Austin et al. (2016) and kolska et al (2012) [2] and [44].

Table 2.8: Properties predicted by GCM

GC methods	**Properties**	**Ref**
Joback and Reid (1987)	Boiling point, critical pressure, enthalpy of formation enthalpy of fusion, enthalpy of vaporizarion, heat capacity Gibbs energy of formation, viscocity	[41]
Constantinou and Gani (1994)	Boiling point, accentric factor, critical pressure, critical temperature, enthalpy of formation, critical volume enthalpy of vaporisation, melting point	[9]
Marrero and Gani (2001)	Boiling point, critical pressure, critical temperature, enthalpy of formation, critical volume,enthalpy of vaporisation, melting point	[57]

Topological Indices (TI) Topological indices is based on "chemical graph theory". The basic concept of this idea is that "*atoms and bounds forming a molecule can be considered nodes and edges in a graph*" [2] and thus the molecule can be represented in terms of vertices and edges. Based on these elements (vertices and edges) of the graph (the molecule) different graphical features can be determined such as the Wiener index, edge index and connectivity index [2]. All these features are then incorporated in QSPR model to predict a wide range of properties. Some of these properties and the corresponding TI method by which they are predicted can be seen in table table 2.9. A larger overview can be found in the work by Austin et al. (2016) [2].

Table 2.9: Properties predicted by TI

Property	**TI method**	**Reference**
Aqueous solubility	Hall et al. (1975)	[35]
Boiling point	Hall et al. (1975)	[35]
Heat capacity	Yao et al. (2003)	[97]
Flash point	Patel et al. (2009)	[67]
Enthalpy of vaporization	Galvez et al. (1994)	[23]
Density	Estrada(1995)	[16]
Biodegradability	Beothling(1986)	[5]

Signature Descriptors (SD) This method has been developed much more recently than the two other methods and can be considered hybrid method combining some aspects of GC and TI [2] [19]. It combines the atomic connectivity from the GCM with the chemical graph theory from TI. In fact, methods have been developed to translate TI

method formulation into SD formulations and therefore the properties predicted by this method are similar to those predicted by TI [19].

Each of the presented QSPR techniques exhibits a series of advantages as well as shortcomings. Depending on the GC method used, a distinction of isomers might not be possible. This problem does not occur when using TI or SD. However GC methods are much simpler, more intuitive and more flexible compared to the other methods [2]. It is also able to represent a much larger design space. Additionally, GC methods provide a simple mathematical formulation that is easy to formulate but also easy to apply [2].

2.3.2 CAMD applications

CAMD have been used to solve a wide range of design problems. They have been used to identify replacements to already existing products in an effort to make a given process more sustainable or more safe. Some of the CAMD application fields and examples can be seen in table 2.10.

Table 2.10: Fields of CAMD application

Applications	**Method**	**Reference**
Replacement solvent for liquid liquid extraction of phenol from wastewater	GC	[37]
Replacement solvent for benzene	GC	[37]
Design of extractive distillation agent	GC	[37]
Replacement for refrigerant	GC	[26]
Emulsifier design	GC	[26]
Optimal solvent for liquid-liquid extraction	GC	[7]

As can be seen in the table, GCM is the method of choice in solving CAMD problems.

2.3.3 Solution Strategies

After reviewing some of the techniques to formulate the CAMD problem, it is important to identify the conceptual strategies used to solve such problems. Each of these strategies are explained in the following.

Generate And Test

The generate and test method consists of systematically reducing the number of alternatives. At first, all possible molecular structures from a predefined search space are generated. The properties of interest are then evaluated for all the candidates and compared to the desired specs [37]. Those candidates that do not exhibit the desired property are eliminated from further considerations. It is important to note that this procedure is only applicable in a limited design space where some structural limitations have been defined. The advantage of this solution strategy is the fact that it is not computationally heavy since most properties are not described by complex correlation and are thus easy to evaluate [2]. Another advantage is the fact that this method can generate an optimal solution in a given design space given that all possible candidates within this space were generated [71]. This method has been applied in many case studies such as the design of solvent for organic synthesis [28], the design of polymers for heat resistant films [76] and for the design of cosmetics [94].

Mathematical Optimisation

If the desired CAMD problem can be formulated as an optimisation problem, then a numerical solver can be applied to solve it and produce possibly a candidate for the design problem [71]. Depending on the property models and the process models used, the complexity of the formulation and the performance of numerical solvers might differ and in some cases be too computationally heavy [71]. The resulting mathematical formulation is usually a mixed integer problem that can be either linear or non-linear depending on the models used. It is important to note that for these types of problems, a global optimum is not always guaranteed and the procedure can be computationally heavy. To overcome this, some decomposition methods have been proposed to reduce the MIP to either linear problems or non-linear problems. Mathematical optimisation has been used to solve design problems related to polymer [90] design and refrigerant design [14]. In some cases, heuristic algorithms can be used to solve such problems. These algorithms are suitable for problems that are multi-dimensional, have a large design space and exhibit strong non-linearity in the process or property models. The working principle of such algorithms includes high-level selection strategies to generate different starting point which are then evaluated with regard to the objective function [2]. The memory of each solution is retained so as not to repeat the same solution again [2]. Examples of this type of algorithm that have been reported in the literature are: Genetic algorithms (algorithms that tries to imitate the idea of natural selection), ant colony algorithms (imitates the behaviour of ants) and simulated annealing [2]. Table 2.11 contains examples in which these algorithm have been used.

Table 2.11: Heurtic algorithms and their application on CAMD problems

Algorithm	**Problem**	**Reference**
Genetic algorithm	Two-phase reactions and gas absorption problems	[102]
Ant colony	Liquid-liquid extraction	[30]
Simulated annealing	Refrigerant replacement	[66]

Chapter 3

Methods

In this chapter, the solution procedure to the entrainer screening problem is presented in the form of a CAMD framework. This will cover the principle and the methodology employed to construct the computer aided product design, the formulation of the design problem and the solution approach adopted. In the first section, the overall framework used and the solution approach are presented. In the second section, the property model and the thermodynamic model used within the framework are motivated.

3.1 CAMD Framework

In this chapter, a step-by-step procedure to formulate and solve the CAMD problem is presented. Since the CAMD is based on a series of mathematical models consisting of the QSPR models, it can be formulated as a mathematical problem. The following framework is presented in a general aspect that is independent of the QSPR model used.

CAMD Framework

Cignitti et al. (2015) proposed a systematic framework framework for formulating and ultimately solving various types of CAMD problems [6]. The framework serves to reduce the complexity of the design problem into a series of sub-problems. A schematic of the various steps in the framework and the information network between the steps can be seen in fig. 3.1. Each step of the framework is explained in the following.

Step 1: Problem Definition

The problem is defined by determining the needs, the type along with the process or application the chemical product is intended. The needs can be expressed in terms of a series of properties, environmental impact potentials or the cost of the product. These needs are then translated into target properties that can be identified. These properties can either be primary properties, secondary properties, functional properties or mixture properties [24]. Primary properties are constant and are directly related to the molecular structure. This type of properties include the critical temperature and the normal boiling point. Secondary properties are dependent on other properties and include the enthalpy of vaporisation at the normal boiling point and the eccentric factor. Functional properties vary with temperature or pressure e.g. liquid viscosity, heat of vaporisation and thermal conductivity. The mixture properties that characterise the mixture in which the product is present[24]. These properties can either characterise the overall bulk of the mixture such as the composition or the temperature. These are called bulk properties. Other mixture properties are related to the phase equilibrium. This could include volatilities, saturation composition and solubility. After fixing the considered properties, a target must be set for them in terms of lower and upper bounds. These bounds can be derived from the intended application or some safety consideration. In some cases these can also be suggested by heuristic methods or past experience.

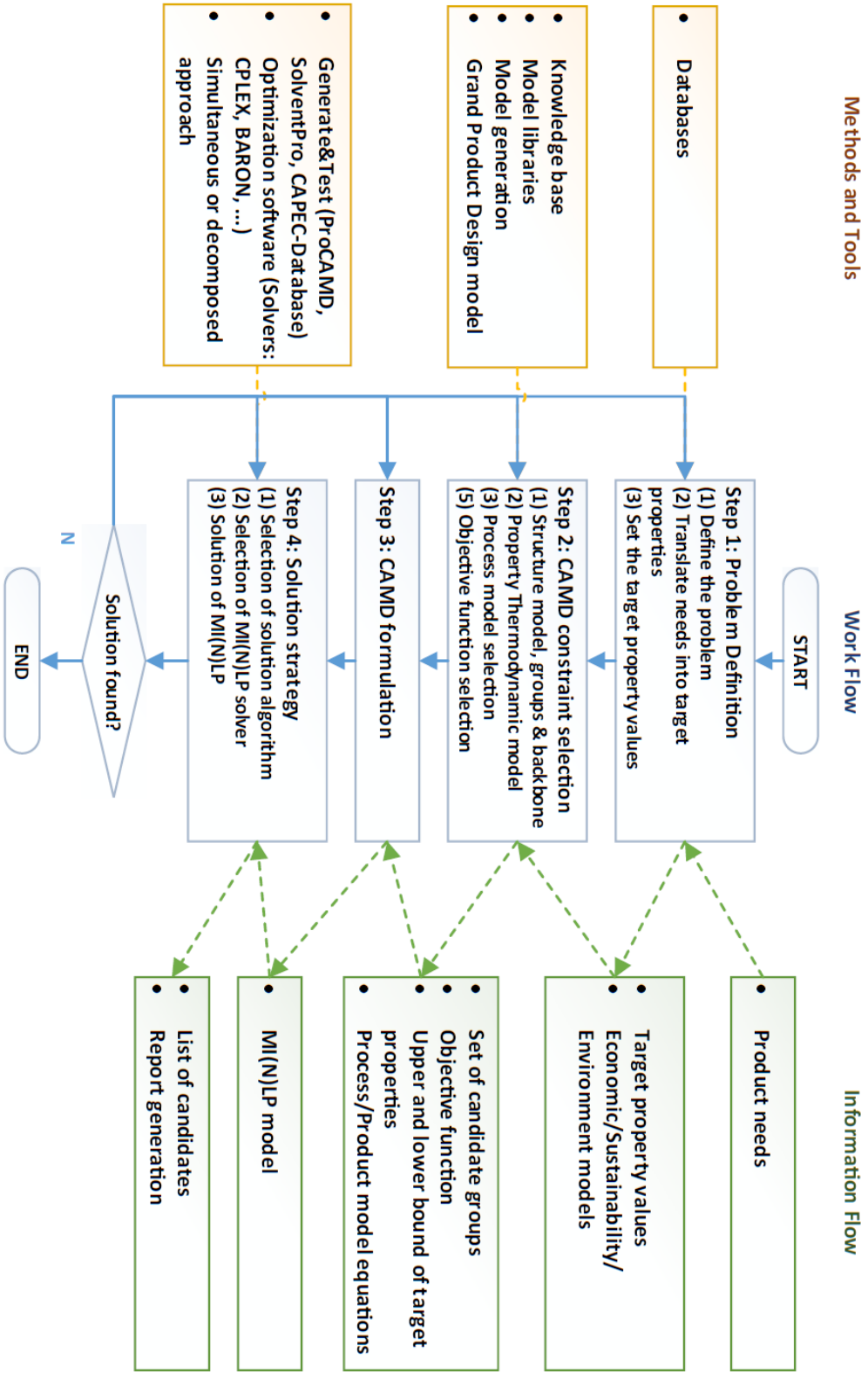


Figure 3.1: CAMD framework proposed by Cignitti et l. [6]

Step 2: CAMD Constraint Selection

In this step the constraints for the CAMD procedures are set. This includes the selection the building blocks of the product as well as the structural model used to assemble these building blocks into meaningful structures. The property models and thermodynamic models used to determine the pure properties and mixture properties of the product are selected. The property models are usually QSPR models such as the GCM and TI [6]. Thermodynamic models are used to calculate the phase equilibrium between the compounds of a mixture and are also used to determine mixture properties such as solubility[7]. IF applicable, a process model to evaluate the performance or the application of the product should also be selected. This allows for a simultaneous product and process design since the product is selected based on process performance and not only based on its properties. A target objective function should also be selected in this step. This function could include some indication of the performance of the product derived from the process model. The objective function is then the subject of an optimisation based design where the final design should provide an optimum value for the objective function.The objective function could also be disregarded either due to the fact that they are too complex or hard to determine or describe. If the objective function is not considered, the design problem is then considered a feasibility problem that produces all available candidates that abide by the previously set constraints.

Step 3: CAMD Problem Formulation

The CAMD problem is the formulated as a Mixed Integer Problem (MIP). Depending on the structural and property models used, the problem can either be linear (MILP) or nonlinear (MINLP). A generic CAMD formulation taking into account all the previously developed constraints and models is presented in the following:

$$\begin{aligned}
\min / \max \quad & F(\mathbf{X}, \mathbf{N}) && (3.1)\\
\text{Structural Constraints} \quad & g_1(\mathbf{N}, \mathbf{Y}) \leq 0 && (3.2)\\
\text{Pure component property Constraints} \quad & g_2(\mathbf{N}) \leq 0 && (3.3)\\
\text{Mixture property Constraints} \quad & g_3(\mathbf{N}) \leq 0 && (3.4)\\
\text{Process model or other constraints} \quad & g_4(\mathbf{X}, \mathbf{N}) = 0 && (3.5)\\
& \mathbf{X} \in \Re^n \, , \, \mathbf{N} \in \mathbf{Z}^m \, , \, \mathbf{Y} \in \{0,1\}^q &&
\end{aligned}$$

The CAMD problem is formulated as an optimisation problem, eq. (3.1) is the objective function, which is a target for maximisation or a minimisation depending on the nature of the function. If the objective function reflects the energy usage in a given processes, it is usually desired to minimise this. If the objective function reflects the revenue a given product might ensure, then it is usually maximised. "g_1" is a set of linear equations reflecting a set of structural constraints. These constraints reflect the feasibility of a suggested structure and if it is a meaningful structure. Additional structural constraint might include restriction on the occurrence of specific building blocks or the total allowed number of blocks in a structure. "g_2" is a set of constraints concerning some of the primary properties of the product. "g_3" are a set of constraints applied on the mixture properties. The constraints related to the properties are usually expressed by upper and lower bounds for a given property. "g_4" is a set of constraints or model equation reflecting the process in which the product is intended or some kind of performance indicator. The process model is usually made of a series of constitutive equations in the form of mass and energy balances that must be satisfied. $\mathbf{N}$ is a vector of integer variables which represent the number of

occurrences of the structural building blocks forming the product. **Y** is the adjacency matrix that reflects the molecular structure and the interconnection between the different building blocks. **X** is a vector of continuous variables representing the properties or the process variables.

Step 4: Solution Strategy

Depending on the process and thermodynamic models used, the CAMD formulation can lead to highly non-convex and non-smooth problems [6]. The solution for these kinds of problems is not always straight forward and a solution is not guaranteed. Karunanithi et al. proposed a decomposition approach to convert the MIP into a series of smaller problems that are handled in a sequential manner [42]. Each of these sub-problems represent a subset of the CAMD constraints. This approach will result in a solution space that is continuously reduced after each sub-problem. This is illustrated in fig. 3.2.

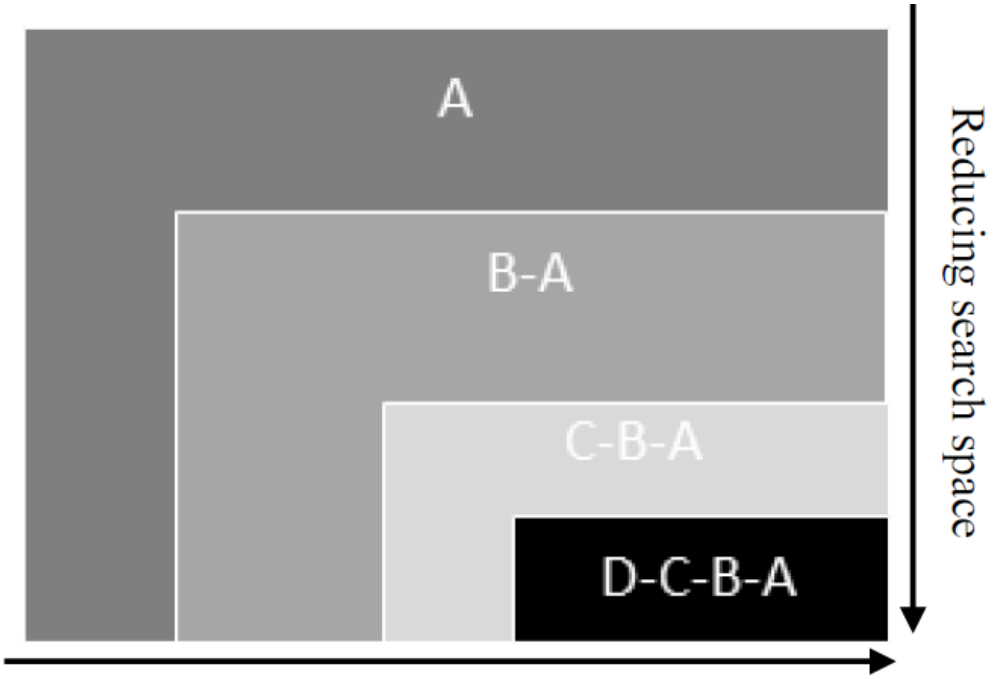


Figure 3.2: Continuous reduction of the solution space

By handling each set of constraints sequentially, the nature of the CAMD problem is changed. Only the first sub-problem consisting of generating feasible structures is a MIP [6]. After the structures are determined, they are all evaluated against the remaining constraints which would either be Linear problems or nonlinear problems depending on the models used. Cignitti et al. incorporated the decomposition principle by Karunanithi et al. into the CAMD framework described earlier[6] which can be seen in fig. 3.3.

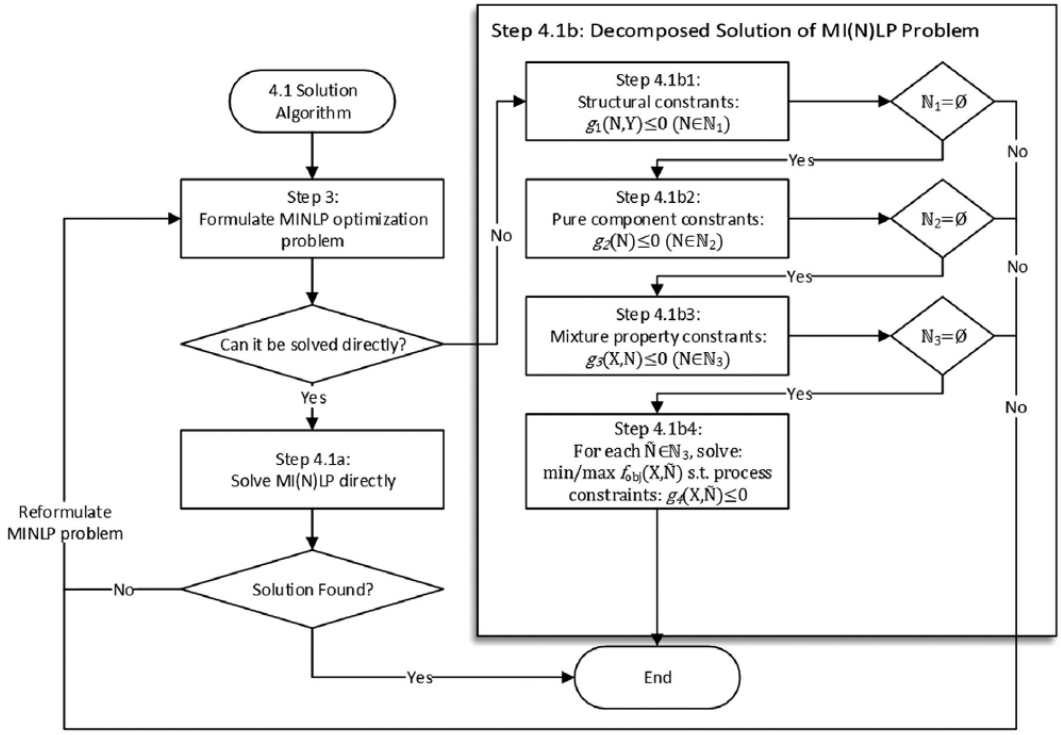


Figure 3.3: CAMD Framework by Cignitti et al.[6] integrated with decomposition approach by Karunanithi et al. [42]

3.2 Model Selection

The models generating feasible molecular structure, determining their primary properties as well as the thermodynamic model enabling phase equilibrium calculations largely determine how the CAMD is formulated. In this section the models considered in this work are introduced. The working principle as well as the motivation for choosing these model are also presented in this section.

3.2.1 Structure Model: Odele and Macchietto (1993)

The structure model used in this work is was developed by Odele and Macchietto in 1993 [65]. This model provides a simple relation to describe the structural feasibility by considering the number of valances a structural group has. The valences can be considered as the number of connection a given building block posses. A feasible structure should have no unoccupied connections. This is also known as "the octet rule". Therefore for each molecular building block, the number of valency is given. An additional constraint is formulated for this model which states that "*two adjacent groups may not bi linked by more that one bond* [65]. An implication of this constraint is that if a group of valency four is to be included in the solution, at least four additional groups must be present. This will therefore eliminate the possibility of creating double or triple bounds between the molecular building blocks. This rule is called the "connectivity constraint" [7]. The building blocks chosen are the first order groups presented by Constantinou and Gani in 1994 [9].

3.2.2 Property Model: GCM by Constantinou and Gani (1994)

The model used to determine the pure compound properties is a GCM. They relatively simple to implement, intuitive to understand from a chemical engineering point of view and are capable of determining a wide range of properties (e.g. boiling point, critical points). The method chosen was developed by Constantinou and Gani in 1994 [9]. The model is able to provide both first and second order predictions. The second order groups serve to increase the accuracy of the prediction as well as enable the differentiation between isomers. However in this work, only first order groups are considered. The reason for excluding second order groups is that this work is concerned with the proof of concept of developing a decisions making tool for fast screening of entrainer candidates. The increased accuracy and complexity might not be merited at this stage. One drawback is that the method will only provide the identity of possible entrainer candidates by the pool of groups present and will provide no information on the sequencing and arrangement of such groups. The groups forming the search space will be the first order groups containing Carbon (C), Oxygen (O) and hydrogen (H). An additional motivation behind the selection of this method is the fact that the model uses the same first order groups considered in the UNIFAC thermodynamic model [6]. This makes it possible to generate structures and evaluate the mixture properties based on the same set of groups.

3.2.3 Thermodynamic Model: UNIFAC by Fredenslund (1975)

Mixture properties is of great importance in the entrainer design problem. It is therefore necessary to conduct phase equilibrium calculations. The extractive distillation processes involves vapour-liquid phase equilibrium calculations as is the case for all distillation processes. Since the processes does not involve high pressures and the mixture properties re not to be evaluated beyond the critical points, a γ-ϕ approach is used for VLE calculations, where activity coefficient models are used for both the gas and the liquid phase. The criteria for vapour-liquid equilibrium is that the fugacity "f" of each phase is equal as shown in eq. (3.6) where superscript "V" denotes the vapour phase and "L" denotes the liquid phase [15].

$$\hat{f}_i^V = \hat{f}_i^L \tag{3.6}$$

The fugacity can be expressed as shown in eq. (3.7) and eq. (3.8) for the liquid phase and vapour phase respectively. For these equations, "y" is the vapour mole fraction, "x" is the liquid mole fraction, P is the total pressure, P^{sat} is the saturated vapour pressure, ϕ is the fugacity coefficient and γ is the liquid activity coefficient [15].

$$\hat{f}_i^V = y_i \, \hat{\phi}_i \, P \tag{3.7}$$

$$\hat{f}_i^L = x_i \, \hat{\gamma}_i P_i^{sat} \tag{3.8}$$

Since the phase equilibrium will be evaluated at low pressures the gas phase will be assumed ideal (ϕ_i=1). To model the interaction in the liquid phase, the original UNIFAC model developed by Fredenslund et al (1975) [21] with the revision and extensions by Hansen (1991) [36]. The thermodynamic model is also based on the principle of group contribution and the groups considered are identical to the first order groups used in the GCM by Constantinou and Gani (1994). This is makes the development of the CAMD procedure more convenient since the same groups are used in predicted both the pure compound property as well as the mixture properties. For the vapour pressure calculation needed for the VLE calculation, the Antoine equation is used.

3.2.4 Mixture Properties

The mixture properties of interest of an entrainer is the ability to fully mix with the components of the azeotropic mixture, also called stability condition, and the ability to not form any azeotropes with any of the compounds in the mixture. These properties necessitates the vapour-liquid and the liquid liquid equilibrium calculations. The method for ensuring these properties are upheld are presented in the following.

Stability Conditions by Conte et al. (2011)

The miscibility of a compound in a mixture is related the trend of the Gibbs free energy of mixing "ΔG_{mix}" [10]. The most characteristic behaviours of this function can be seen in fig. 3.4.

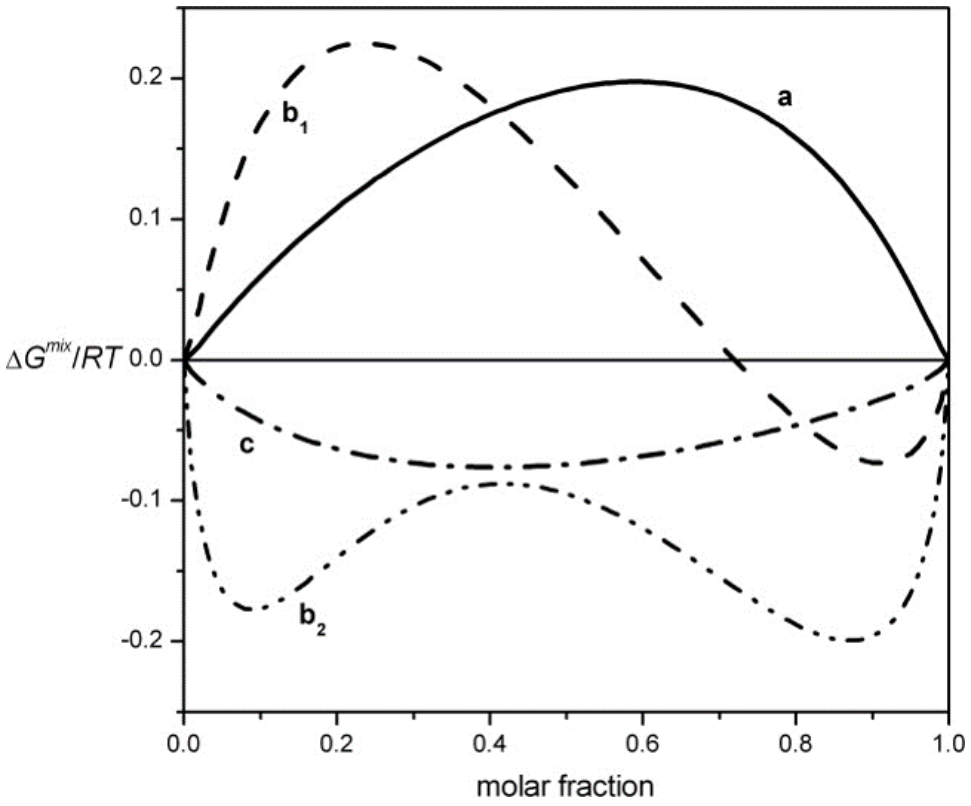


Figure 3.4: The most common shapes for the function ΔGmix/RT. From [10].

Based on the graphical trends in fig. 3.4, a series of observations can be made. The curve (a) represents a case where there is complete immiscibility in the whole composition range and is expressed by the Gibbs energy of mixing being positive. Curve (b_1) represents the case where there is partial miscibility. This means that for some certain compositions, no phase split occurs. The mixing process is spontaneous if for the whole composition range, the Gibbs energy of mixing is negative. This correspond to curve (c) in fig. 3.4. Curves with similar trends as curve (b_2), indicates that there is a miscibility gap, even though for the whole composition range the Gibbs energy of mixing is negative. One way to identify this case is to ensure that the Gibbs energy of mixing is convex and exhibits only one extremum. This can be identified through calculating the second derivative of the function Gibbs energy of mixing. For the whole composition range, the second derivative

must be positive [10]. These conditions have previously been used to perform stability checks in the design of formulated products by Conte et al. (2011) [10].

"No azeotrope" Condition

Systems forming an azeotrope are characterised by the great variation of the relative volatility as a function of the composition [64]. The relative volatility between two components is defined as the ratio of the equilibrium constants of these components in the mixture. The equilibrium constant is expressed as the ratio between te vapour and liquid molefraction. For systems forming an azeotrope, the relative volatility of the binary mixture crosses the diagonal line in an x-y diagram [64]. Typical representations for a minimum boiling and maximum boiling azeotropes can be seen in fig. 3.5.

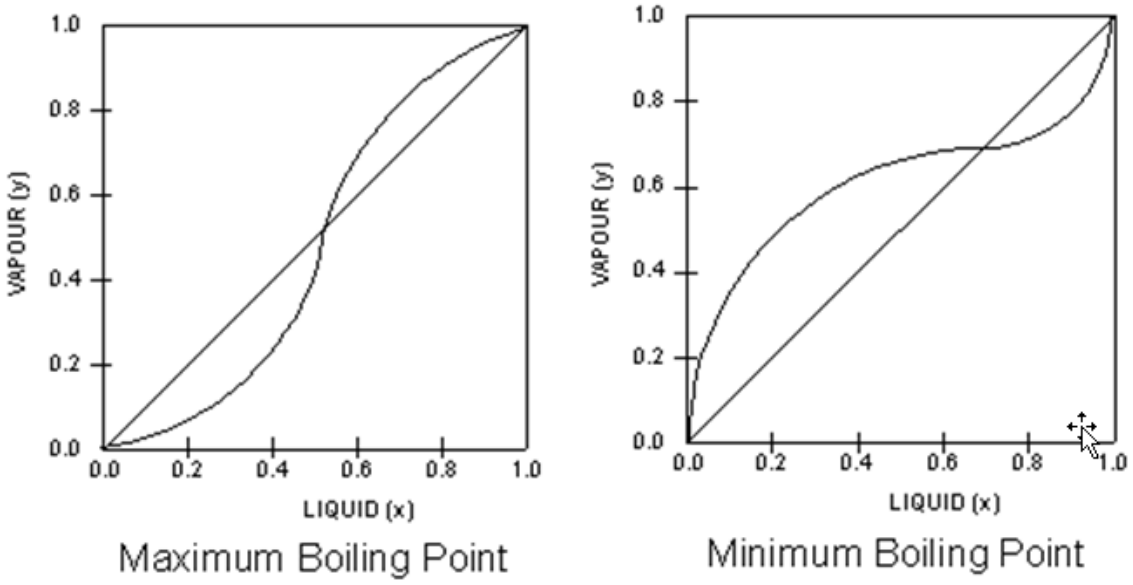


Figure 3.5: Schematic of x-y diagram for maximum boiling and minim boiling azeotropes

For minimum boiling azeotropes, the relative volatility is greater than unity at low concentration while it decreases below unity at high concentrations. For maximum boiling azeotropes, the case it is the opposite [64]. The point where the the interception between the relative volatility occur with the diagonal corresponds to the azeotropic composition. Based on this description if a binary system exhibits composition values for which the relative volatility are located on both sides of unity, they form an azeotrope.

3.2.5 Process Model

In this section, the mathematical model describing the performance of the entrainer is developed. The model chosen produce a series of process metrics that can be used as a basis for comparing the efficiency of the entrainer candidates. It is important to note that these models are short-cut models i.e. they are simplified version of the real process model. the process model considered in this work are: the Fenske equation for determining the minimum number of stages in a distillation column and the infinitely sharp split method by Petlyuk for determining the entrainer to distillate flow rate and the reflux ratio [69].

Fenske Equation (1932)

The minimum equilibrium stages needed in a column corresponds to the number of stages for a column operating at total reflux. The number of stages can be determined using the Fenske equation eq. (3.9), where N_{min} the minimum number of stages, "S" is the separation factor in the column and α_m is the mean relative volatility [79].

$$N_{min} = \frac{\log S}{\log \alpha_m} \tag{3.9}$$

The separation factor S is calculated as shown in eq. (3.10) , where x is the product mole fraction, the superscripts "B" and "D" denotes the bottom and distillate products and "i" denotes the light key while "j" denotes the heavy key.

$$S = \frac{x_i^D}{x_i^D} \frac{x_j^B}{x_i^B} \tag{3.10}$$

Since the relative volatility is not constant throughout the column, the geometric mean of the top and bottom stage is calculated as shown in eq. (3.11). "N" denotes the top stage while "1" denotes the bottom stage.

$$\alpha_m = \sqrt{(\alpha_{i,j})_N \ (\alpha_{i,j})_1} \tag{3.11}$$

The Fenske equation is applied under the assumption of total reflux. Total reflux is achieved by charging the column with the feed and operating it until steady state is achieved. During operation no feed addition or product removal occurs. In the case of extractive distillation, the method can be applied by considering the three component system as a pseudo-binary system composed of the heavy key and light key [63] [52]. It is assumed that the effect of the entrainer is present throughout the column from top to bottom and that the flow of entrainer is constant on all stages [63]. Te pseudo-binary system is obtained by converting the molefraction in presence of the entrainer into entrainer free molefractions. the molefractions can then be used to calculate separation factor and the minimum number of stages.

Infinitely Sharp Split method (ISS)

Petlyuk et al (2015) proposed a method to determine limiting parameters related to the operation of the extractive column. These parameter consist of the minimal entrainer flow rate and the minimum reflux ratio[69]. The column is assumed to operate under the assumption that a sharp split occurs in the top section of the column. Considering a column divided into three section; a top section, an extractive section and a bottom section as shown in fig. 3.6. the limiting mode of operation is also know as the "Infinitely Sharp Split (ISS) mode". It is assumed that each section is of infinite height and tat a sharp split occurs in the top section of the column. A sharp split is achieved if one of the components is only present at one end of a section. The sharp split in this case occurs in the top section where the top of the top section only contains component A.

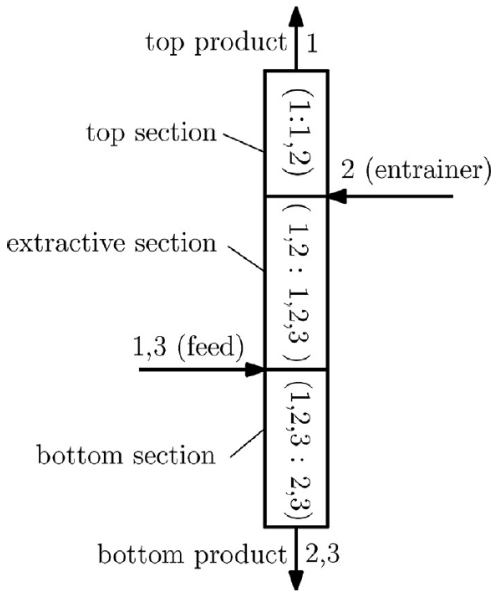


Figure 3.6: Component splits in the extractive column [69].

In contrast to the Fenske equation for which the theory and the derivation are known, the ISS mode and how the limiting parameters are calculated are more obscure. Therefore, the derivation and calculation of these parameters is explained in the following.

Two control volumes are identified to perform a mass balance: the top section and the combination of the top section and the extractive section. These control volumes along with in-going and out-going streams can be seen in fig. 3.7.

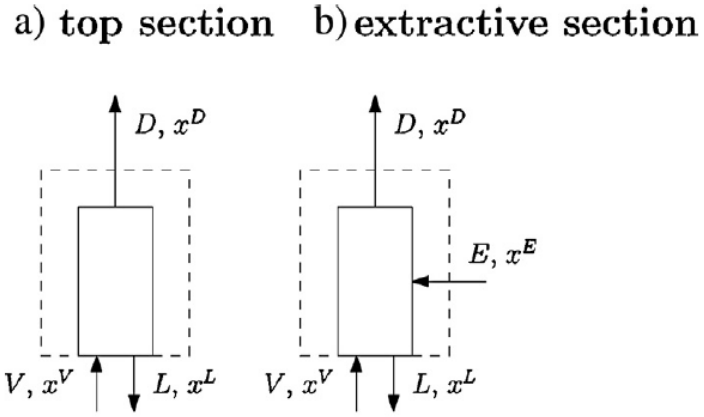


Figure 3.7: Control volumes for the mass balance [69].

The total mass balance for the top and the combined top and extractive section can be seen in eq. (3.12) and in eq. (3.13) respectively.

$$\text{Top section:} \qquad V = D + L \tag{3.12}$$

$$\text{Top + extractive section:} \qquad V = (D - E) + L \tag{3.13}$$

The component mass balance can be seen in eq. (3.14) and eq. (3.15) for the top section

and the combined section respectively.

$$\text{Top section:} \qquad V\,y^V = D\,x^D - L\,x^L \tag{3.14}$$

$$\text{Top + extractive section:} \qquad V\,y^V = (D\,x^D - E\,x^E) + L\,x^L \tag{3.15}$$

At this point the concept of delta composition "x^{Δ}" also known as difference point is introduced. The delta composition is a virtual composition that "*represent a composition of difference between the distillate and the entrainer feed to the column*" [69]. This concept have been discussed in details by Tapp et al. (2004) and Haun et al. (200)[86][38]. This composition is used to rewrite the component mass balance in the combined top and extractive section as shown in eq. (3.16).

$$\text{Top + extractive section:} \qquad V\,y^V = (D - E)\,x^{\Delta} + L\,x^L \tag{3.16}$$

Considering component "1" (the top product): since a sharp split occurs, only component "1" is present in the top product (Distillate) therefore:

$$x_1^D = 1 \qquad\qquad x_1^E = 0$$

eq. (3.15) can then be rewritten as

$$V\,y_1^V = D + L\,x_1^L \qquad \Rightarrow \qquad D = V\,y_1^V - L\,x_1^L \tag{3.17}$$

Turning attention to eq. (3.16), an expression for the delta composition x^{Δ} can be derived. This expression can be seen in eq. (3.18)

$$x^{\Delta} = \frac{V\,y^V - L\,x^L}{D - E} = \frac{D}{D - E} = \frac{1}{1 - E/D} \tag{3.18}$$

Another concept is now introduced, which is the pinch points. For columns with infinite height, the vapour and liquid at same height are in equilibrium. In addition, at these point in the distillation trajectory, the compositions-called pinch composition- (denoted x^r) of the phases do not change. Each section has at least one pinch point since all sections are of infinite height. Considering eq. (3.16) at the pinch composition and by substituting the total mass balance into the equation, as shown int eh following an expression of the delta point can be obtained.

$$\text{Introduce the pinch composition:} \qquad V\,K_i\,x_i^r - L\,x_i^r = (D - E)\,x_i^{\Delta}$$

$$\text{Substitute with the total mass balance:} \qquad V\,K_i\,x_i^r - L\,x_i^r = (V - L)\,x_i^{\Delta}$$

$$\text{Substitute with the total mass balance:} \qquad V\,K_i\,x_i^r - L\,x_i^r = (V - L)\,x_i^{\Delta}$$

The expression for the delta point is shown in eq. (3.19).

$$x_i^{\Delta} = x_i^r\ \frac{K_i - L/V}{1 - L/V} \tag{3.19}$$

Considering component "3" which is neither present in the distillate or the entrainer feed, the balance in the combined top and extractive section eq. (3.15) can be reduced to eq. (3.20).

$$V\,y_3^V = L\,x^L \qquad \Rightarrow \qquad y_3^V / x_3^L = K_3 = L/V \tag{3.20}$$

An since the amount of component "3" in this column section is very small, the K value derived corresponds to that of infinite dilution of "3" in component "1" and "2" ($K_3 = K_3^{\infty 1,2}$).
By combining eq. (3.18) and eq. (3.19) and rearranging the obtained expression, the following expression for E/D is obtained.

$$\frac{E}{D} = 1 - \frac{1 - L/V}{x_1^r \left(K_1 - L/V\right)} \tag{3.21}$$

Introducing the result obtained in eq. (3.20), a reduced expression for E/D is obtained and is shown in

$$\frac{E}{D} = 1 - \frac{1 - K_3}{x_1^r (K_1 - K_3)} \tag{3.22}$$

By analysing the column behaviour and the trajectories of the column, Petlyuk deduced that the E/D possess a minimum value for an x_1^r value below a certain value of the pinch branch composition. This composition corresponds to the one obtained at the univolatility line at the 1-2 side of the ternary diagram[69][68]. This value is obtained when the condition in is fulfilled.

$$K_1^{1,2} = K_3^{\infty 1,2} \tag{3.23}$$

The composition of component "1" for which eq. (4.38) represent a singularity point for which the assumption of ISS is not valid and thus the composition for E/D is minimal must be located below that value. The above mentioned theory is supported by a series of graphical constructions that Petlyuk et al. (2004) conducted on the behaviour of pf the K values and the E/D [68]. The trend of the equilibrium constant and the E/D ratio can be seen in fig. 3.8 and fig. 3.9 respectively. Based on the figures, it is evident that a minimum value for E/D is present at a composition below that of the isovolatility between component 1 and 2.

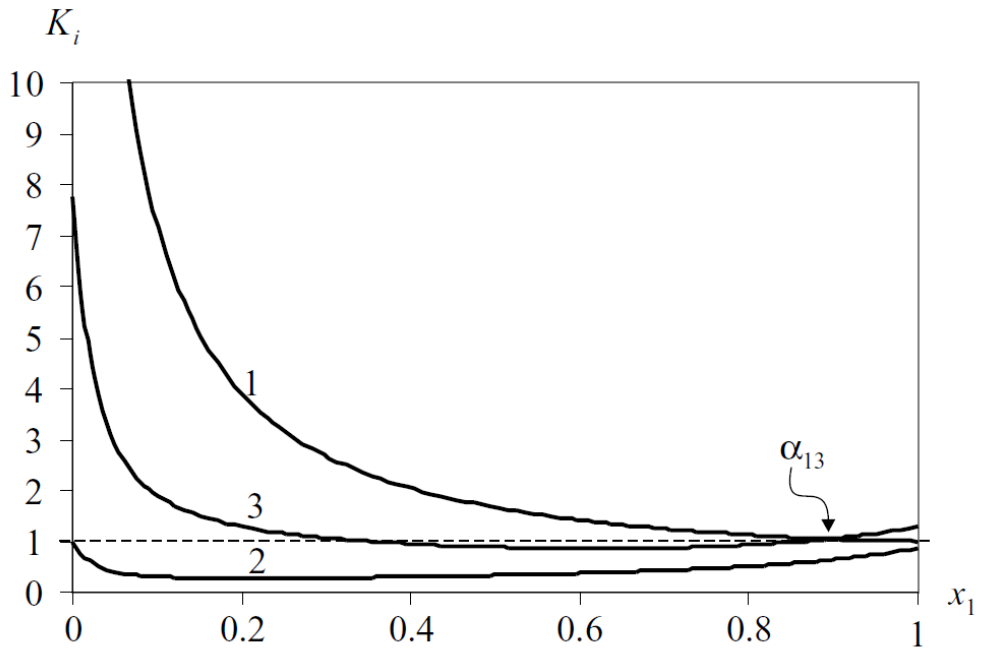


Figure 3.8: K-values as function x1 on the binary side 1-2

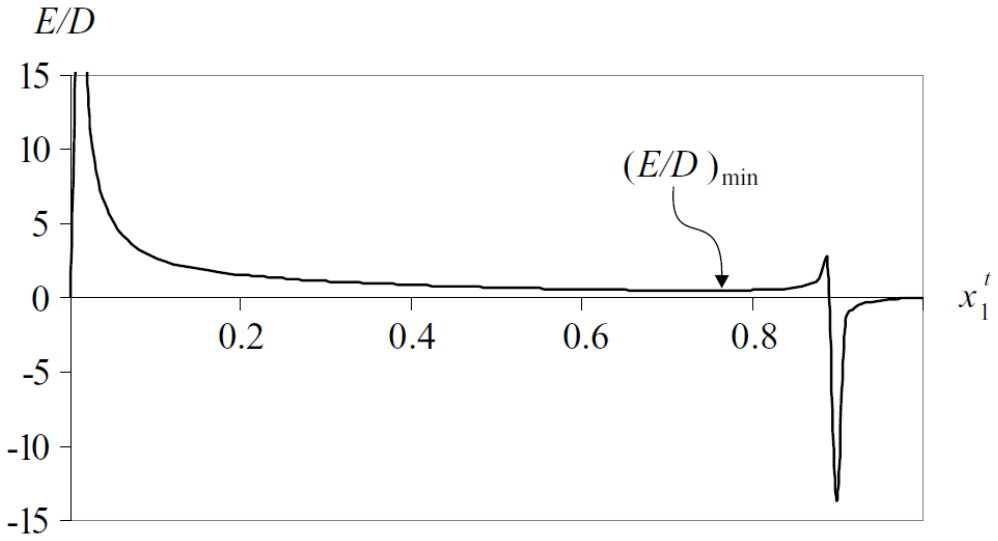


Figure 3.9: E/D as a function of x_1^r

Finally, by using the mass total mass balance derived in eq. (3.13) and the obtained expression for E/D, an expression for the corresponding reflux ratio can b derived and is shown in eq. (3.24).

$$R = \frac{K_3^{\infty 1,2} - E/D}{1 - K_3^{\infty 1,2}} \tag{3.24}$$

Chapter 4
Implementation & Validation

In this chapter, the various implementations of the methods and tests used in the CAMD problem are presented. Each implementation is described in details and how it was performed. Each implementation is validated and tested.

4.1 Structural model

4.1.1 Implementation

A subset of first order groups are chosen to limit the design space. this is called the basis set. The first order groups considered are those considered by Fredenslund (1975) and Hansen (1991) [21] [36] and contain Carbon, hydrogen and oxygen.
A molecular structure is described with a vector "n" comprised of the integers "n_k" accounting for the occurrence of group "k" in the proposed structure. The valency number of each group "k" is denoted "v_k" and describes the number of connections a group can form with other groups. The structure model by Odele and Macchietto (1993) is then formulated as following.

The Octet Rule: The octet rule is expressed as shown in equation (4.1), where "m" is a parameter to express the type of structure generated. For acyclic structures "m=1", for monocyclic structures "m=0", for bicyclic structures "m=-1". In this work, only acyclic compounds were considered and thus "m" is set to unity [65]. The reason for considering only acyclic compounds is to limit the design space considered.

$$\sum_k (2 - v_k)\, n_k = 2\, m \tag{4.1}$$

The Connectivity rule: The connectivity rules is expressed as follows in eq. (4.2) [65] [2].

$$\sum_i n_i \geq n_j(v_j - 1) + 2 \tag{4.2}$$

In the implementation of the structural model, additional constraints might be considered to limit the design space. These constraints could either limit the length of the generated structures as in eq. (4.3) or limit the amount of occurrence of certain groups as in eq. (4.4).

$$n^l \geq \sum_k n_k \geq n^u \tag{4.3}$$

$$n_f^l \geq \sum_k n_k - n_{CH_3} - n_{CH_2} - n_{CH} - n_C \geq n_f^u \tag{4.4}$$

In eq. (4.3), a maximum length n^u set to six and a minimum length of n^l three. eq. (4.4) restricts the number of functional groups in the structure. The lower limit n_f^l and the

upper limit n_f^u are set to one and two respectively. This was done in an effort to limit possible proximity effect.

Generating the feasible structures is a Mixed Integer Problem. The problem involves integer values through "n_k" and all the structural models or constraints are linear. Such problems can be solved The problem is solved using the branching method. This method is best illustrated in fig. 4.1. From an initial point (k=CH_3) all possible paths are investigated. Linking "CH_3" to another "CH_3" will satisfy the octet rule and the connectivity rules. However due to the fact that the structure breaches the minimum requirement for structure size, the structure fails and the search continues from the initial point again. Branching to a"CH_2" group and then an "OH" group result in a feasible structure satisfying all constraints the structure deemed feasible. When all possible paths have been exhausted, a new initial point is chosen and the search continues.

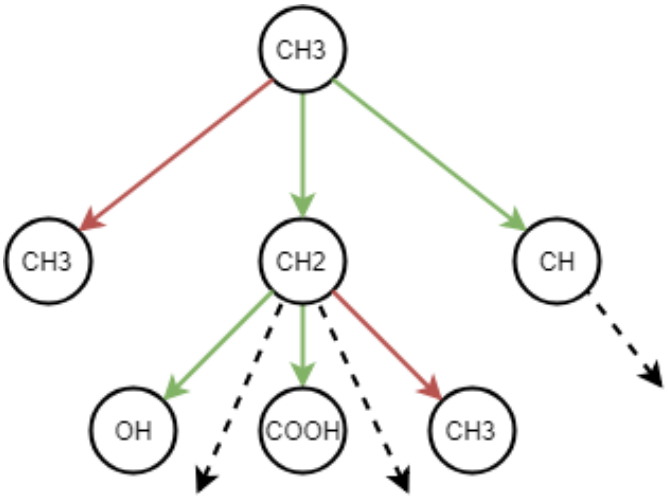


Figure 4.1: Illustration of the branching method for MIP

In terms of the software chosen, such algorithm is already in place. The "BARON" (Branch And Reduce Optimization Navigator) is a global solver able to handle MINLP problems. Since the algorithm is formulated as an optimization tool in GAMS, an objective function must be provided. However, since we are interested in all feasible structures within the design space chosen, a constant objective function is provided to the algorithm. Since a constant function is always valid, all candidates that abide by the constraints will be determined.

4.1.2 Validation

To validate the structural model, a series of structures will be tested for survival. The feasibility of the structures is known beforehand thus validating whether the model have been implemented correctly. A small basis set of the groups by Fredenslund (1975) and Hansen (1991) are chosen and presented in along with their valency intable 4.1 [21] [36].

Table 4.1: Subset of first order groups and their number of connections

Groups, k	CH_3	CH_2	OH	H_2O	CHO
$\mathbf{v}_k$	1	1	1	0	3

	Groups					Octet rule	Connectivity rule	Comments
	CH3	CH2	OH	H2O	CHO	$\sum_k (2 - v_k)\, n_k = 2$	$\sum_i n_i \geq n_j(v_j - 1) + 2$	
n(A)	1	1	1	0	0	$(2-1)*1+(2-2)*1+(2-1)*$ $1+(2-0)*0+(2-3)*0=2$	$3 \geq 1*(1-1)+2=2$ $3 \geq 1*(2-1)+2=3$ $3 \geq 1*(1-1)+2=3$ $3 \geq 0*(0-1)+2=2$ $3 \geq 0*(3-1)+2=2$	Both the octet and the connectivity rules are satisfied. The structure corresponds to ethanol with the chemical formula CH3-CH2-OH
n(B)	1	1	0	0	0	$(2-1)*1+(2-2)*1+(2-1)*$ $0+(2-0)*0+(2-3)*0=1$	$2 \geq 1*(1-1)+2=2$ $2 \geq 1*(2-1)+2=3$ $2 \geq 0*(1-1)+2=2$ $2 \geq 0*(0-1)+2=2$ $2 \geq 0*(3-1)+2=2$	The structure is not feasible as it has one unused valency. This is detected by the octet rule. Additionally, since the "CH2" group has two valences, at least two additional groups must be present. Since this is not the case, the connectivity rule captures that.
n(C)	1	3	0	1	1	$(2-1)*1+(2-2)*3+(2-1)*$ $0+(2-0)*1+(2-3)*1=2$	$6 \geq 1*(1-1)+2=2$ $6 \geq 3*(2-1)+2=5$ $6 \geq 0*(1-1)+2=2$ $6 \geq 1*(0-1)+2=1$ $6 \geq 1*(3-1)+2=4$	Although both rules are satisfied, it is clear that the proposed molecule does not exist since it contains a whole water molecule.

The proposed molecular structures "A", "B" and "C" are suggested for the verification of the structure generation model. The model have been able to detect that "A" is a meaningful structure, which is not surprising since it represent an ethanol molecule. "B" failed the structural constraints since it is an incomplete structure. "C" however passed through all constraints although the structure is not possible. The reason for this is the presence of "H2O" group. Since the model was not intended to handle these type of groups from Fredenslund (1975) and Hansen (1991) that have no valency, they must be handled separately. These groups will be called "category 1" groups while the remaining groups will be referred to as "category 2" groups. "Category 1" groups will henceforth bypass the structural constraints since they are already meaningful structures.

4.2 GCM

4.2.1 Implementation

The properties needed in this work are: the normal boiling point, the melting point, the critical temperature, the critical pressure, acentric factor and enthalpy of vaporisation. The correlations by Constantinou and Gani (1994) are formulated explicitly in terms of the desired property and considering only the first order groups in this work. The correlation can be seen in table 4.3.

Table 4.3: Pure property GCM correlations by Constantinou and Gani (1994) [9].

Property	Correlation		
Normal melting point, T_m	$T_{m_0}\ \log \sum_k n_k\, T_{m_1,k}$		T_{m_0} Adjustable parameter $T_{m_1,k}$ Contribution of grp "k" to T_m
Normal boiling point, T_b	$T_{b_0}\ \log \sum_k n_k\, T_{b_1,k}$		T_{b_0} Adjustable parameter $T_{b_1,k}$ Contribution of grp "k" to T_b
Critical temperature Tc,	$T_{c_0}\ \log \sum_k n_k\, T_{c_1,k}$		T_{c_0} Adjustable parameter $T_{c_1,k}$ Contribution of grp "k" to T_c
Critical pressure Pc	$P_{c1}\left[\sum_k n_k P_{c1,k}\right]^{-2}$		P_{c_1} Adjustable parameter $P_{c_1,k}$ Contribution of grp "k" to T_c
Acentric factor, ω	$\omega_{01}\left[w_{03}\ \sum_k n_k w_{1,k}\right]^{w_{02}}$	+	ω_{01} 1^{st}Adjustable parameter ω_{02} 2^{nd}Adjustable parameter ω_{03} 3^{rd}Adjustable parameter $\omega_{1,k}$ Contribution of grp "k" to T_c
Standard enthalpy of vaporisation ΔHv_{298K}	$h_{v_0} + \sum_k n_k\, h_{v_1,k}$		h_{v_0} Adjustable parameter $h_{v_1,k}$ Contribution of grp "k" to ΔHv

The first order contributions to each of the aforementioned properties for "Category 2" groups can be found in the original publication by Constantinou and Gani (1994) [9]. The value of the adjustable parameters are reported in table 4.4 [9].

Table 4.4: Value of adjustable parameter of the GCM by Constantinou and Gani (1994) [9].

Parameter	Value	
T_{m0}	102.425	K
T_{b0}	204.359	K
T_{c0}	181.128	K
P_{c1}	1.3705	bar
ω_{01}	0.4085	
ω_{02}	0.505	
ω_{03}	1.1507	
h_{v0}	6.829	kJ/mol

To be able to accommodate "Category 1" groups in the developed framework, their properties must be determined in a similar way using the correlations in table 4.3. Using the same adjustable parameters, their contributions are determined using the corresponding experimental value. An example is seen below for the contribution to the boiling point for water (H2O):

$$T_b = 373.15K = T_{b_0} \log T_{b_1}$$
$$\Downarrow$$
$$T_{b_1,H_2O} = \exp \frac{T_b}{T_{b_0}} = \exp \frac{373.15}{204.359} = 6.208$$

This is done for all "Category 1" groups and the properties considered above. The experimental values used to determine the contribution were extracted from the commercial simulator Aspen Plus.

4.2.2 Validation

The implementation of the GCM by Constantinou and Gani (1994)[9] is verified. This is done by calculating the pure compound properties of interest for some random molecular structures and comparing the with properties calculated using ProPred as well as experimental data if available. ProPred is a property estimation toolbox integrated with the ICAS software developed by CAPEC DTU. The software is able to calculate properties based on the GCM used in this work. The validation is performed on the following properties: Normal boiling point, melting point, critical temperature and pressure, enthalpy of vaporisation and acentric factor. The results from the implementation, ProPred and the experimental values from NIST (National Institute of Standards and Technology) for the properties of interest can be seen in table (4.5) .

Table 4.5: Comparison between the Properties of interest for certain molecular structures

Structure			Name	Properties																	
				Tm [K]			Tb [K]			Tc [K]			Pc [bar]			Hv [at 298K] [kJ/mol]			w		
Nr.	Groups	Occurrence		Imp	ProPred	Exp	Imp	ProPred	Exp	Imp	ProPred	Exp	Imp	ProPred	Exp	Imp	ProPred	Exp	Imp	ProPred	Exp
1	H2O	1	water	373.15	NA	273.15	373.15	NA	273.15	647.00	NA	647.00	220.64	NA	220.64	43.98	NA	43.98	0.345	NA	NA
2	DOH	1	Ethylene glycol	260.15	NA	260.15	470.45	NA	470.45	719.70	NA	719.70	85.38	NA	85.38	66.25	NA	66.25	0.487	0.846	NA
3	CH3 CH3CO	1 1	Acetone	171.62	171.62	178.35	305.37	305.37	329.15	490.11	490.11	508.2	48.8	48.8	47	29.94	29.94	30.84	0.332	0.332	NA
4	CH3 CH2 OH	1 1 1	Ethanol	164.57	164.57	159.05	330.01	330.01	351.35	489.28	489.28	513.92	55.57	55.57	61.47	40.12	40.12	42.32	0.527	0.527	NA
5	CH3 CH2COO	2 2	NA	213.52	213.53	NA	439.00	439.00	NA	433.99	433.99	NA	31.00	31.00	NA	50.58	50.58	NA	0.567	0.567	NA
6	CH2 HCO CH3COO	1 1 1	NA	228.40	228.40	NA	409.33	409.33	NA	592.12	592.12	NA	43.60	43.60	NA	47.10	47.10	NA	0.607	0.607	NA

For water and ethylene glycol, the experimental results are identical with the calculated values. This is due to the fact they belong to the groups for "Category 1" and their contribution were directly determined from the experimental values as explained in the implementation. ProPred is not able to generate structures for water. Since water cannot be described by the GCM by Constantinou and Gani (1994). Since ethylene glycol has its own group in the UNIFAC, it was considered as a "Category 1" group and a new contribution was calculated to match the experimental values. For acetone and ethanol, some discrepancies can be noted in terms of all properties, this reflects the accuracy of the predictive model used. The highest deviation can be seen for the boiling point and the critical temperature of acetone. For structure Nr. 5, the name cannot be determined since it can have different isomers. It is however important to note that the calculated boiling point is higher than the critical point for this structure. This cannot be the case in reality. This reflects yet again some of the weaknesses with using predictive models. It is then important to sort out such structures from further analysis in the framework. Therefore an additional constraint should be added stating that the boiling point of a generated structure must not be higher than its critical point. The enthalpy of vaporisation for water and ethylene glycol matches the experimental ones. This is yet again due to the fact that they have been determined manually based on a single group contribution. The acentric factor was only compared t the values obtained through ProPred.

Based on the previous analysis and the data in table (4.5), we can conclude that the GCM by Constantinou and Gani (1994) have been successfully implemented in this work as it matches the results obtained by ProPred.

4.3 VLE-Calculation

4.3.1 Implementation

UNIFAC

The liquid phase activity coefficients are calculated using UNIFAC using the extended and revised groups of Hansen et al. (1991) [36]. For component "i" the activity coefficient γ_i can be calculated from eq. (4.5), where γ_i^C and γ_i^R are the combinatorial and residual parts of the activity coefficient.

$$\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R \tag{4.5}$$

The combinatorial part can be calculated through eq. (4.6) where x_i is the liquid mole fraction of component "i", z is a constant equal to 10. Φ_p is the segment fraction, θ_i is the area fraction and l_i is a parameter.

$$\ln \gamma_i^C = \ln \frac{\Phi_i}{x_i} + \frac{z}{2} q_i \ln \frac{\theta_i}{\Phi_i} + l_i - \frac{\Phi_i}{x_i} \sum_j x_j l_j \tag{4.6}$$

The l_i parameter is calculated as in eq. (4.7), where r_i and q_i are measures to the molecular van der Waals volumes and molecular surface areas.

$$l_i = \frac{z}{2}(r_i\, q_i)(r_i - 1) \tag{4.7}$$

The area fraction θ_i can be calculated by eq. (4.8).

$$\theta_i = \frac{q_i x_i}{\sum_j q_j x_j} \tag{4.8}$$

The segment fraction determined be eq. (4.9).

$$\Phi_i = \frac{r_i x_i}{\sum_j r_j x_j} \tag{4.9}$$

q_i is related to the pure property of compound "i" by eq. (4.10) where Q_k is the area parameter of group k.

$$q_i = \sum_k v_k^{(i)} Q_k \tag{4.10}$$

r_i is also related to the pure property of compound "i" by eq. (4.11) where R_k is the surface parameter of group k.

$$r_i = \sum_k v_k^{(i)} R_k \tag{4.11}$$

The residual part γ_i^R i calculated through eq. (4.12), where Γ_k is the group residual activity coefficient and $\Gamma_k^{(i)}$ is the residual activity coefficient of group k in a reference solution only containing group k. These are calculated as following in eq. (4.13) and eq. (4.14) respectively.

$$\ln \gamma_i^R = \sum_k v_k^{(i)} (\ln \Gamma_k - \ln \Gamma_k^{(i)}) \tag{4.12}$$

$$\ln \Gamma_k = Q_k \left(1 - \ln \sum_m \Theta_m \Psi_{m,k} - \sum_m \frac{\Theta_m \Psi_{m,k}}{\sum_n \Theta_n \Psi_{n,m}} \right) \tag{4.13}$$

$$\ln \Gamma_k^{(i)} = Q_k \left(1 - \ln \sum_m \Theta_m^{(i)} \Psi_{m,k} - \sum_m \frac{\Theta_m^{(i)} \Psi_{m,k}}{\sum_n \Theta_n^{(i)} \Psi_{n,m}} \right) \tag{4.14}$$

The area fraction fraction of group m Θ_m is calculated as in eq. (4.15), where χ is the mole fraction group m in the mixture. This mole fraction is calculated by eq. (4.16).

$$\Theta_m = \frac{Q_m \chi_m}{\sum_n Q_n \chi_n} \tag{4.15}$$

$$\chi_m = \frac{\sum_i v_m^{(i)} \, x^{(i)}}{\sum_i \sum_n v_m^{(i)} \, x^{(i)}} \tag{4.16}$$

The corresponding parameters for the reference solutions can be determined by eq. (4.17) and eq. (4.18).

$$\Theta_m^{(i)} = \frac{Q_m \chi_m(i)}{\sum_n Q_n \chi_n(i)} \tag{4.17}$$

$$\chi_m(i) = v_m(i) \tag{4.18}$$

Ψ is a measure of the group interaction parameter between group "m" and "n", while $a_{m,n}$ is the interaction parameter between group m and n calculated by eq. (4.19). This parameter is evaluated experimentally and through data regression. T is the temperature.

$$\Psi_{m,n} = \exp\left(\frac{-a_{m,n}}{T} \right) \tag{4.19}$$

If the identity of the compounds present in the mixture are known, then only two degrees of freedom remain in the UNIFAC model. As a simplification, the model can then be represented as the activity coefficient as a function of the mole fraction and the temperature as seen in eq. (4.20)

$$\gamma_i = \gamma_i(x_i, T) \tag{4.20}$$

4.3.2 Validation

The UNIFAC implementation is validated by producing the liquid activity coefficients for a binary mixture of Diethylamine (DEA) and n-heptane (HEP) at 308.15 K. The system was chosen as since it has been previously used to demonstrate the application of the model [83]. The two compounds are described in terms of groups as shown in table 4.6.

Table 4.6: DEA and HEP in terms of groups

Compound	Groups		
	CH3	CH2	CH2NH
DEA	2	1	1
HEP	2	5	0

The liquid activity coefficients obtained through the implementation are compared to those obtained for the same system, under the same condition and using the same model in Aspen Plus. The results for the whole composition range are shown in fig. 4.2.

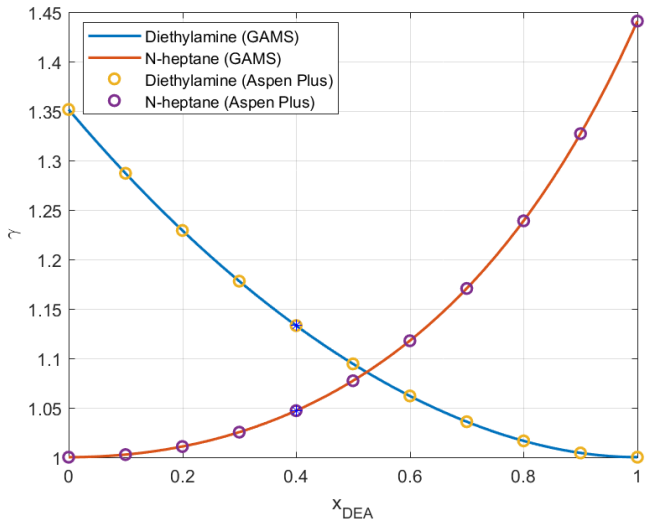


Figure 4.2: Logarithm of the activity coefficients of the binary system diethylamine(1)-n-heptane(2) at T=308.15 K as a function the mole fraction of diethylamine

The results from the implementation are identical to the ones obtained through the commercial software. Furthermore, the case study performed in the literature obtained the results shown in table 4.7, which are identical to the result in fig. 4.2. Furthermore, the trend observed in fig. 4.2 suggests that as the composition approaches unity, the slop approaches zero as required by the Gibbs/Duhem equation.
Based on the results obtained through Aspen Plus and the literature case study, it can be concluded that UNIFAC was successfully implemented. Data produced by GAMS and Aspen for the validation can be seen in appendix.

4.3.3 Implementation of the saturated pressure calculation

The saturated vapour pressure is calculated through the Antoine equation [79] which can be seen in eq. (4.21), where P^{sat} is the saturated pressure in bar, and A, B and C are

Table 4.7: Liquid activity coefficients for DEA and HEP at 308.15 K at x_{DEA}=0.4 [83]

Compound	x	γ
DEA	0.4	1.133
HEP	0.6	1.047

constant that are experimentally determined. T is the temperature in Celsius.

$$P^{sat} = A - \frac{B}{T + C} \tag{4.21}$$

Since the identity of the generated structure is not known and in the absence of a database containing the Antoine parameter, a procedure to determine these is implemented. For all compounds the following three relations must hold true: the saturated vapour pressure at the normal boiling point must correspond to the atmospheric pressure P_0 eq. (4.22), the saturated vapour pressure at the critical point must correspond to the critical pressure P_c eq. (4.23) and the the definition of the Pitzer acentric factor eq. (4.24) [70].

$$P_0 = A - \frac{B}{C + T_b} \tag{4.22}$$

$$P_c = A - \frac{B}{C + T_c} \tag{4.23}$$

$$\omega = -1 - \log_{10} P^{sat}_{T_r=0.7} \tag{4.24}$$

The A, B and C parameters are determined through an optimisation procedure minimising the square errors of the P_c, P_0 and ω determined by the Antoine equation and the the GCM. The optimisation problem is formulated in the following eq. (4.25).

$$\min_{A,B,C \in Z} F_{obj}(\text{A,B,C}) = R(P_c)^2 + R(P_0)^2 + R(\omega)^2 \tag{4.25}$$

$$\begin{aligned} s.t \quad & R(P_c) = P_c(\text{GCM}) - P_c(\text{Antoine}) \\ & R(P_0) = P_0(\text{GCM}) - P_0(\text{Antoine}) \\ & R(\omega) = \omega(\text{GCM}) - \omega(\text{Antoine}) \end{aligned}$$

In terms of the software used and since the problem is nonlinear (presence of power functions), a nonlinear must be chosen. The solver chosen is "CONOPT". The solver has been described as the "go-to" solver for nonlinear problems and was thus chosen.

4.3.4 Validation of he Saturated vapour calculation

To validate the procedure for calculating the saturated vapour pressure, four compounds were chosen: diethylamine, n-heptane, water, ethylene glycol. Based on the calculated critical points, normal boiling temperature and the acentric factor calculate by GCM, the Antoine parameter were calculated. The data used in and obtained from the optimisation can be seen table 4.8. The obtained Antoine parameter successfully recreated the data from which they were obtained. The residual error (F_{obj}) is very small in each case.

Table 4.8: Data used and obtained from the procedure for determining the Antoine constants

	by GCM				by Antoine equation			F_{obj}	Antoine Parameters		
	T_b	T_c	P_c	ω	P_0	P_c	ω		A	B	C
DEA	341	516	39,96	0,326	1,013	39,96	0,325	1,4 e-7	4,423	1367,537	241,40
HEP	379	549	28,26	0,337	1,013	28,26	0,337	8,4 e-18	2,56	334,773	25,117
H2O	373	647	220,6	0,345	1,013	220,6	0,345	1,1 e-9	5,27	1809,921	243,43
DOH	470	719	84,38	0,487	1,013	84,38	0,487	6,6 e-12	4,04	1104,657	76,215

A further verification is done by calculating the vapour pressure curve of water using the obtained Antoine constants and comparing the results with those obtained from the implementation. The results obtained can be seen in fig. 4.3.

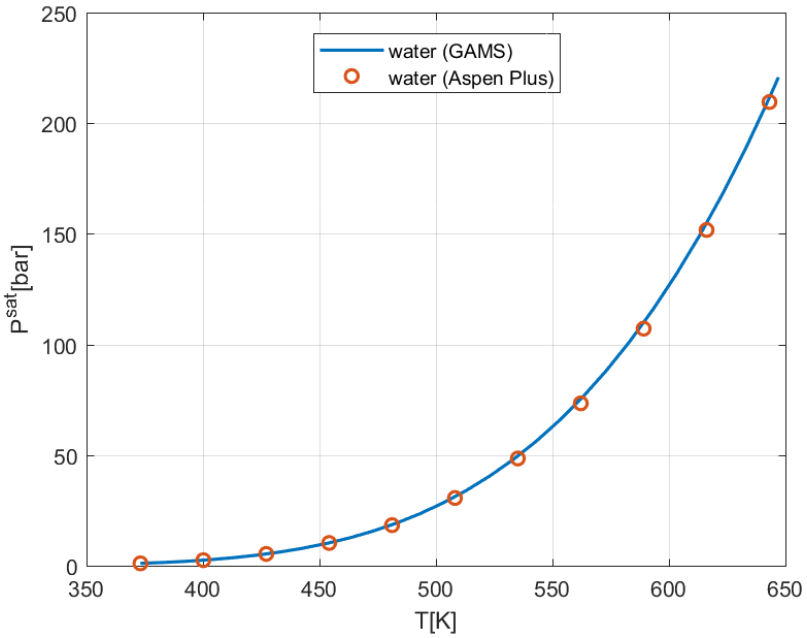


Figure 4.3: Saturated vapour pressure of water obtained through the current implementation and through Aspen Plus

The good accordance between the the results obtained from this work and those from Aspen Plus is a further proof of the correctness of the implementation. It must be noted however that Aspen uses an extended Antoine equation as shown eq. (4.26).

$$\log P^{sat} = C_1 + \frac{C_2}{T + C_3} + C_4 T + C_5 \log T + C_6 T^{C_7} \tag{4.26}$$

4.3.5 Implementation of VLE calculation

The mixture properties of interest are closely related to the vapour-liquid equilibrium occurring during the extractive distillation process. The vapour liquid equilibrium is

described by eq. (3.6) while the fugacity coefficients are given by eq. (3.7) and eq. (3.8).

$$\hat{f}_i^V = \hat{f}_i^L$$
$$\hat{f}_i^V = y_i \, \hat{\phi}_i \, P$$
$$\hat{f}_i^L = x_i \, \hat{\gamma}_i P_i^{sat}$$

Since the vapour phase will be considered ideal, the equilibrium condition will be rewritten in terms of the vapour mole fraction y_i as in eq. (4.27)

$$y_i = x_i \, \hat{\gamma}_i \, \frac{P_i^{sat}}{P} \tag{4.27}$$

The saturated vapour pressure P^{sat} is only a function of temperature, while the total pressure P is fixed to the operating pressure of the column. This is set to atmospheric pressure. By applying a degree of freedom analysis, it is clear that eq. (4.27) has three variables: the mole fractions x_i, y_i and the temperature T. By supplying the liquid mole fractions x_i, the degree of freedom is reduced by one. The current implementation is formulated as a bubble point calculation where the liquid composition for the whole concentration range is provided.
An additional relation can be formulated, where the sum of the the mole fractions must be equal to unit as shown in eq. (4.28).

$$\sum_i y_i = 1 \tag{4.28}$$

$$\sum_i x_i = 1 \tag{4.29}$$

However since the liquid mole fraction is provided by the user, eq. (4.29) is not considered. This will increase the number of equations in the system by one which will result in a well defined problem with two equations (eq. (4.27) and eq. (4.28)) and two variables (T and $y_1 = 1 - y_2$).

4.3.6 Validation of the VLE calculation

To verify the implementation of the VLE calculation, a series of the system Acetone, methanol and water is considered in pairs and the corresponding x-y diagrams are presented. The diagrams are produced with the current implementation and by Aspen Plus to provide a basis for comparison. The acetone-methanol system is a known azeotropic mixture that is used as a benchmark case in many articles concerning extractive distillation and water has been used as an entrainer for the mixture [45].

System1: acetone(1) + methanol(2): The x-y data obtained for the VLE calculation implemented can be seen in fig. 4.4. As can be seen in fig. 4.4, the performed calculation does not show the formation of an azeotrope. By careful examination, it was concluded that the deviation was due to the vapour pressure calculation. These are dependant of the pure component properties and as pointed out earlier, there was a notable deviation between the experimental and the GCM calculated properties of acetone.

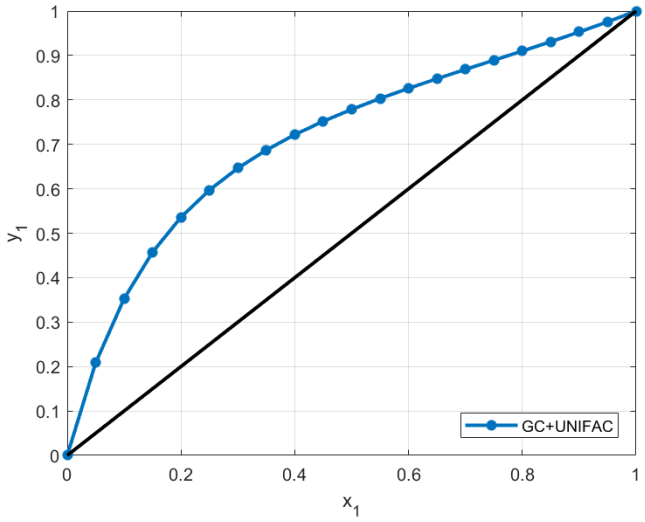


Figure 4.4: x-y diagram for acetone(1)-methanol(2) using GC+UNIFAC

The problem described can be remedied by manually inserting the experimental values of T_b,T_c,P_c for acetone. The implementation was able to reproduce the x-y data from Aspen Plus as can be seen in fig. 4.5 [1]. The data for acetone used in both calculations can be seen in table 4.9

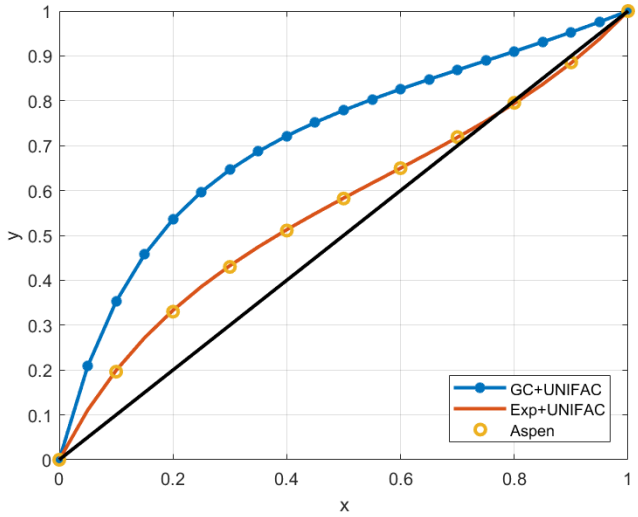


Figure 4.5: x-y diagram for acetone(1)-methanol(2) with different techniques: Aspen, GC+UNIFAC and Exp+UNIFAC

As can be seen in 4.5, the accuracy of the VLE calculation is much closer to the one provided by Aspen Plus. This results underline the importance of having accurate pure compound properties. Therefore, since the azeotropic pair to separate is known, the pure compound properties for these will be inserted manually.

[1] EXP: indicates that the experimental values for Tc, Pc, Tb were inserted for the binary A-B

Table 4.9: Physical properties for acetone

Property	GCM	Experimental
Tb [K]	305.37	329.15
Tc [K]	490.11	508.20
Pc [bar]	48.83	47.20

system 2: methanol(1)-water(2) The VLE calculation is shown as x-y diagram which can be seen in figure(4.6). In the case of the binary system methanol water the VLE data calculated by the implementation match well with the data from Aspen. Additionally, there is no difference between data entered through the GCM or manually. This is due to the fact that the two compound are described by a single group for which the contribution was calculated through the experimental values.

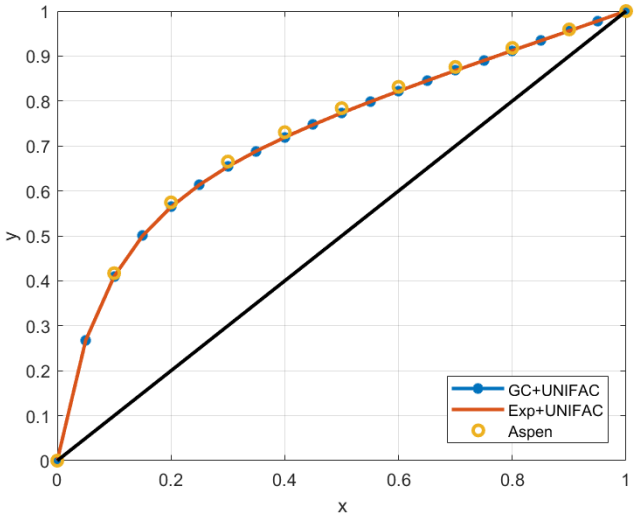


Figure 4.6: x-y diagram for methanol(1)-water(2)

System 3: Acetone-water The x-y diagram resulting from the VLE calculation between water and acetone can be seen in fig. 4.7. The results from the implementation are in good agreement with those obtained from Aspen Plus.

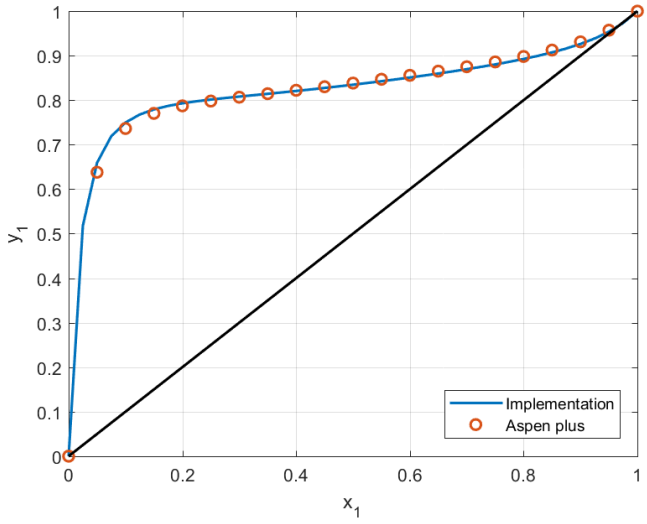


Figure 4.7: x-y diagram for acetone(1)-water(2)

The results obtained by the implementation match the ones obtained by the implementation. fig. 4.7 suggest that the separation might become very difficult approaching pure acetone. To investigate this further, a closer look at high liquid mole fraction for acetone is done. The obtained results can be seen in fig. 4.8.

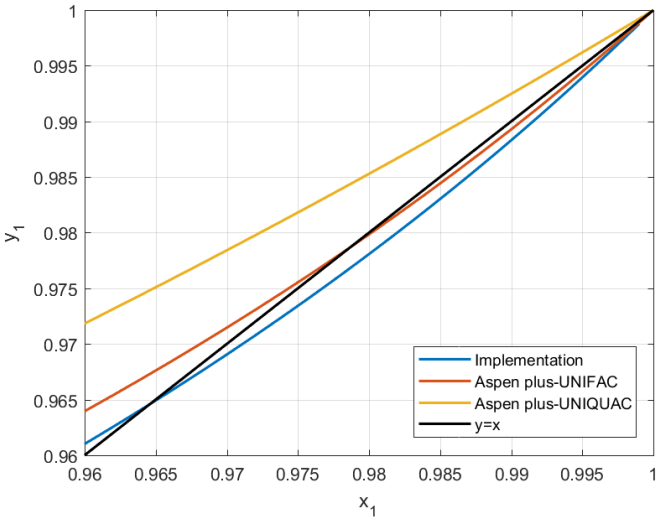


Figure 4.8: Portion of x-y diagram for acetone(1)-water(2) (x=0.96 to x=1) by various methods

The implementation and the UNIFAC in Aspen Plus both suggest the formation of an azeotrope between water and acetone. This is however not the case as cited in the literature [45] [32] [31]. This is also supported by the results obtained from Aspen Plus using UNIQUAC. This incident is simply due to the model and is not related to the implementation.

Based on the various cases inspected, it can be concluded that the VLE calculation has been implemented correctly.

4.3.7 Implementation of mixture Property calculations

The mixture properties of interest is selectivity and capacity at infinite dilution. Additionally, a test for the formation of azeotropes as well as phase stability must be established.

Selectivity At Infinite Dilution The selectivity at infinite dilution is determined as indicated in eq. (2.3).

$$S_{ij}^{\infty} = \left(\frac{\gamma_i^{\infty}}{\gamma_j^{\infty}} \right)_E$$

In the current implementation, the activity coefficients at infinite dilutions are calculated at the boiling point of the entrainer E and with a molefraction predominantly made of the entrainer as shown in table table 4.10.

Table 4.10: Data for calculating the activity coefficients at infinite dilution

	T	x_E	x_i	x_j
γ_i^{∞}	T_b(E)	0.9999999	0.0000001	0
γ_j^{∞}	T_b(E)	0.9999999	0	0.0000001

In a similar way, the capacity is calculated as in eq. (2.4).

$$C_{j,S}^{\infty} = \frac{1}{\gamma_j^{\infty}}$$

The validation of the of the calculation of the selectivity and the capacity are assumed proven by the previously proven accuracy of the VLE calculation.

Check For The Presence Of Azeotropes The entrainer E must not create an azeotrope with neither A nor B. Azeotropic mixtures are characterised by the fact that the relative volatilities α_{A-E} and α_{B-E} exhibits values on both signs of unity. The proposed implementation is to consider two points at the extremities of the composition range. The first point approaches $x_E \approx 1$ while the second approaches $x_E \approx 0$. The relative volatilities are calculated at these points. Based on the analysis shown in the method section, the relative volatility at the two points will be on both sides of unity is an azeotrope is formed. The suggested detection method is to calculate the product of the the logarithm of the activity coefficients at the suggested two points. A positive product indicates that the calculated relative volatilities are on the same side of unity (either both logarithms are positive or they are both negative). This is evaluated for both binary pairs A-E and B-E. Considering the binary A-E, the calculation is performed as following in eq. (4.31).

$$\log \alpha_{AE}\Big|_{x_E \approx 0} \times \log \alpha_{AE}\Big|_{x_E \approx 1} \tag{4.30}$$

$$\Downarrow$$

$$\log\left(\frac{K_A}{K_E}\right)\Big|_{x_E \approx 0} \times \log\left(\frac{K_A}{K_E}\right)\Big|_{x_E \approx 1}$$

$$\Downarrow$$

$$\log\left(\frac{1}{K_E}\right)\Big|_{x_E \approx 0} \times \log\left(\frac{K_A}{1}\right)\Big|_{x_E \approx 1}$$

$$\Downarrow$$

$$\log\left(\frac{1}{\gamma_E\, P_E^{sat}}\right)\Big|_{x_E \approx 0\,,\, T \approx Tb(A)} \times \log\left(\gamma_A\, P_A^{sat}\right)\Big|_{x_E \approx 1\,,\, T \approx Tb(E)} \tag{4.31}$$

In a similar manner, the expression for the binary pair B-E as shown i eq. (4.32)

$$\log\left(\frac{1}{\gamma_E\, P_E^{sat}}\right)\Big|_{x_E \approx 0\,,\, T \approx Tb(B)} \times \log\left(\gamma_A\, P_B^{sat}\right)\Big|_{x_E \approx 1\,,\, T \approx Tb(E)} \tag{4.32}$$

The verification of the azeotrope detection is performed on the Acetone(A)-water(E)-methanol(B) system considered previously. The obtained results can be seen in table 4.11

Table 4.11: Results for azeotrope detection implementation in the acetone-water-methno system

Pair	$\log \alpha_{ij}\Big\|_{x_j \approx 0}$	$\log \alpha_{ij}\Big\|_{x_j \approx 1}$	Product	Comment
A-B	0.0131	-0.0142	≤ 0	The implementation successfully detect the azeotrope formed between Acetone and methanol as confirmed by the VLE calculations.
A-E	3.7895	-0.2394	≤ 0	The implementation detects an azeotrope between Acetone and water, which was also confirmed by the VLE calculation although this is not the case in reality.
B-E	2.0916	0.8658	≥ 0	No azeotropes are detected between water and methanol.

Based on the results obtained above, it can be concluded that the azeotrope detection method works. Although it does detect an azeotrope between acetone and water, this is the fault of the model used and the risk associated with using the predictive models.

Check For Phase Stability The phase stability is closely related to the Gibbs energy function of mixing. This is calculated as in eq. (4.33) where R is the gas constant, T is the temperature and G^E is the Gibbs excess energy[10].

$$\frac{\Delta G^{mix}}{RT} = \frac{G^E}{RT} + \sum_i x_i \log(x_i) \tag{4.33}$$

The Gibbs excess energy is calculates as in eq. (4.34)

$$\frac{G^E}{RT} = \sum_i x_i \log(\gamma_i) \tag{4.34}$$

Combining the two relations yields the full expression of the Gibbs energy of mixing

$$\frac{\Delta G^{mix}}{RT} = \sum_i x_i \log(\gamma_i) + \sum_i x_i \log(x_i) \tag{4.35}$$

The second derivative of the Gibbs energy of mixing is determined by performing central differentiation using the points obtained from eq. (4.35). The expression thus becomes as in eq. (4.36). Here the subscript "dp" indicates the the total number of data points describing the liquid composition in the binary pair i-j.

$$\frac{\partial^2 \left(\frac{\Delta G^{mix}}{RT}\right)}{\partial x_i^2} = \frac{\Delta G^{mix}}{RT}\Big|_{x_{dp+1,i}} - 2\,\frac{\Delta G^{mix}}{RT}\Big|_{x_{dp,i}} + \frac{\Delta G^{mix}}{RT}\Big|_{x_{dp-1,i}} \tag{4.36}$$

For all entrainer and for the whole liquid composition range eq. (4.35) must always be negative, while eq. (4.36) should always be positive. The stability check implementation is tested on the acetone-methanol mixture and compared to the same results obtained through Aspen Plus as shown in fig. 4.9.

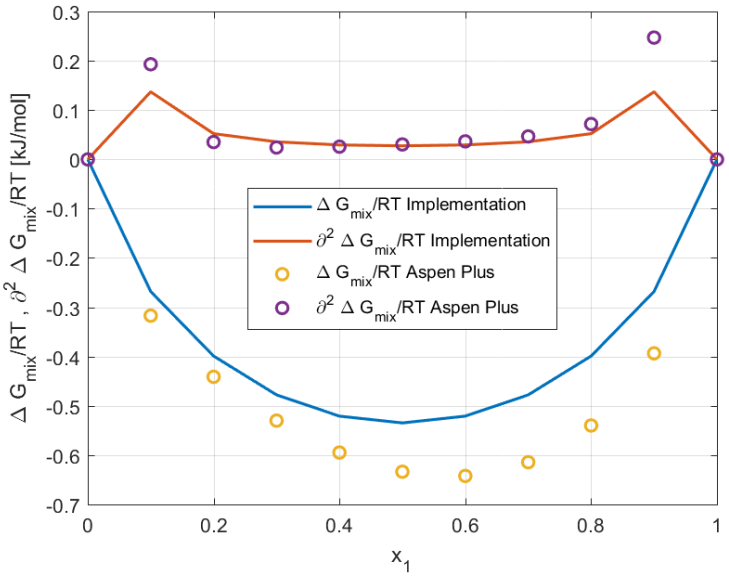


Figure 4.9: Gibbs free energy of mixing and its second derivative for Acetone (1)-water(2)

The obtained result suggest some discrepancy between the values calculated by the implementation and those obtained by Aspen Plus. The discrepancy is highest for the extremities of the composition. The model however is able to confirm that the mixture acetone-water does not create a second liquid phase and that the mixture is stable.

4.4 Process model

4.4.1 Fenske model

The Fenske equation is applied to illustrate how easy a separation is in the presence of the entrainer. The calculations must be performed on entrainer free basis. A simplified process model is considered consisting of an extractive column an a recovery column. an entrainer to feed ration of 1 is chosen to ensure that enough entrainer is present to break the azeotrope. This value has previously been reported by Kossack et al (2008) [45] and Momoh (1991) [60]. Additionally, the following specifications are made.

- All of component A leaves the extractive column at the top.
- The distillate composition is 0.995 of A and 0.005 of B on entrainer free basis.
- The top product of the recovery column is comprised of 0.995 B and 0.005 E.
- Entrainer to feed ratio : 1.
- Feed composition of 0.75 A and 0.25 B.

The entrainer free fractions denoted x' of component A and B can be calculated as shown in eq. (4.37) [79].

$$x'_j = \frac{x_j}{1 - x_E} \qquad \text{for} j = A, B \tag{4.37}$$

Using eq. (4.37) and the indicated process specs, a total mass balance can be performed on the column to determine the actual compositions as well as the entrainer free compositions. The obtained results for the molefraction are reported in table 4.12.

Table 4.12: Process stream mole fractions

	A	B	E
xF	0.375	0.125	0.500
x'F	0.75	0.25	0.000
xD1	0.427	0.003	0.570
x'D1	0.995	0.005	0.000
xB1	0.000	0.200	0.800
x'B1	0.000	1.000	0.000
xD2	0.000	0.995	0.005
xB2	0.000	0.001	0.999

For the first column, the molefraction where the entrainer is present are used in to calculate the corresponding vapour molefractions. These are then converted into entrainer free fraction using eq. (4.37). The obtained entrainer free molefractions are then used to determine the geometric average of the relative volatility using eq. (3.11). For the second column, the mixture is only binary (B+E) and therefore the molefraction used are the actual molefractions.

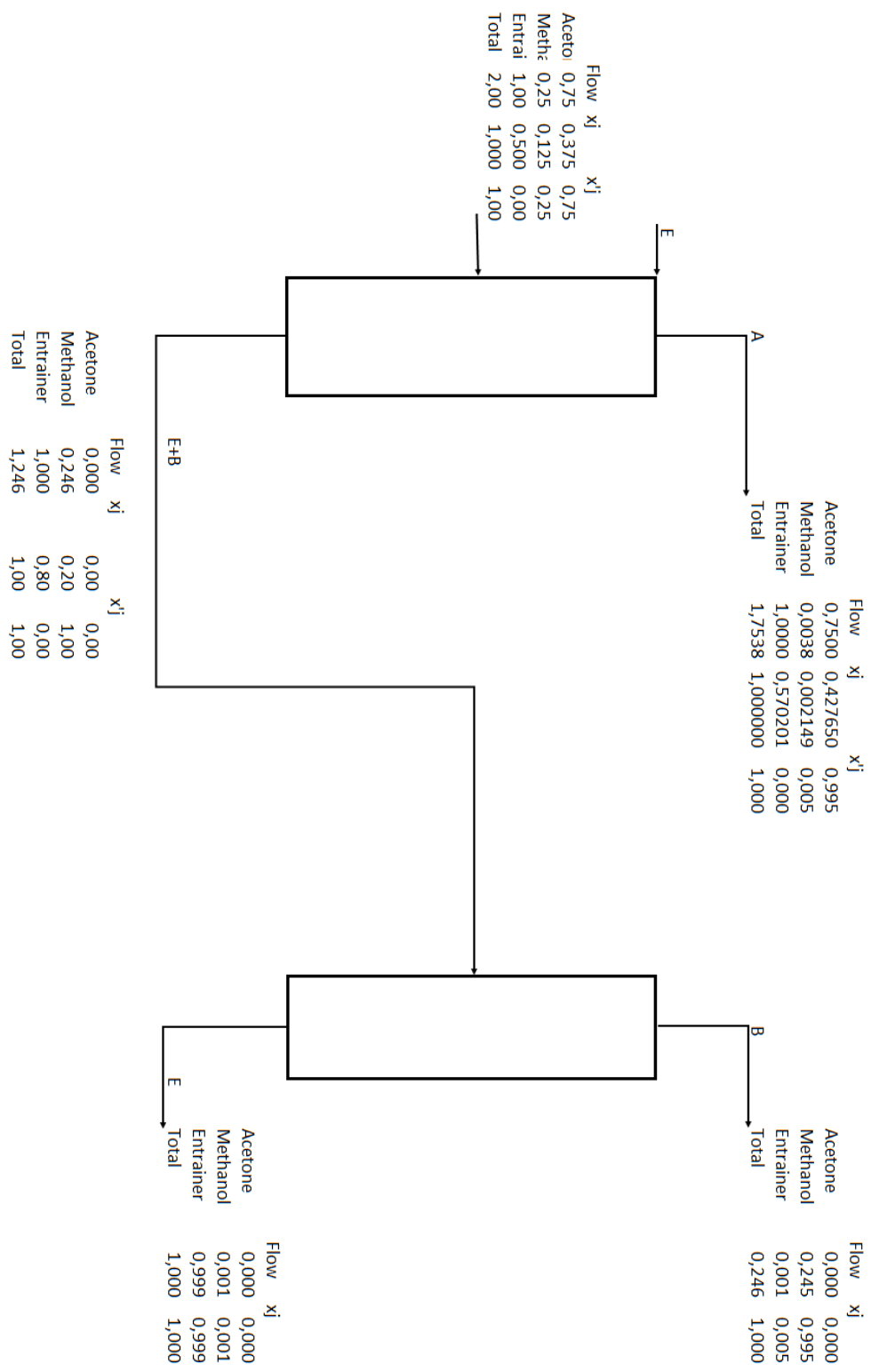


Figure 4.10: Process configurations and stream specs considered in the process model

Calculations regarding the Fenske model are related to the mass balance over the columns and the VLE caulcations. The model equations are provided in the methods section eq. (3.9) and eq. (3.11). Therefore the validation of the Fenske is not done.

4.4.2 ISS method

Implementation

The process model is implemented as an optimisation problem, where the E/D is minimised. In the implementation of the ISS method there are two compositions of interest: the composition for which the isovolatility line intersects with the binary composition A-E and the composition at which E/D is minimal. In both cases "B" is is considered to be present at infinite dilution [2]. The first step is thus to determine the composition for which the the equilibrium constant of A is equal to that of B (B is in infinite dilution).

$$\frac{K_A(x_A^{iso}, T)}{K_B^{\infty A,E}(x_A^{iso}, T)} = 1 \tag{4.38}$$

The composition x^{iso} satisfying eq. (4.38) is unique as proven by Petlyuk et al. (2004) [68]. E/D can thereafter be calculated as part of an optimisation procedure formulated as shown below in eq. (4.39):

$$\begin{aligned} &\min_{x_A^{min} \in Z} F_{obj}(x_A^{min}) = E/D \\ s.t \quad &\frac{E}{D} = 1 - \frac{1 - K_B\infty(x_A^{min}, T)}{x_A^{min}(K_A(x_A^{min}, T) - K_B^{\infty}(x_A^{min}, T))} \\ &R = \frac{K_B^{\infty}(x_A^{min}, T) - E/D}{1 - K_B^{\infty}(x_A^{min}, T)} \\ &\frac{K_A(x_A^{iso}, T)}{K_B^{\infty A,E}(x_A^{iso}, T)} = 1 \\ &x_A^{min} \leq x_A^{iso} \end{aligned} \tag{4.39}$$

The problem formulation in eq. (4.39) is a nonlinear problem. In terms of the software used, the same solver is used as for determining the Antoine constants.

Validation

The validation of the ISS method is not straight forward, in fact a thorough search of the relevant literature yielded only two related works which are the articles with proof of concept of the method [68] [69] and this is one of the reasons this method is being investigated. In the absence of working examples published in the literature, the only way to perform a validation is to investigate the trend of E/D over the entire binary composition A-E. A correct implementation should yield similar trends as the ones shown in fig. 3.9. The system Acetone-methanol-water is again considered. The implementation

[2] $x_B \approx 0$

yields the following results:

$$x_A^{iso} = 0.958$$
$$x_A^{min} = 0.927$$
$$(E/D)_{min} = 0.020$$
$$R_{min} = 3.61$$

The obtained values match the graph fig. 4.11.

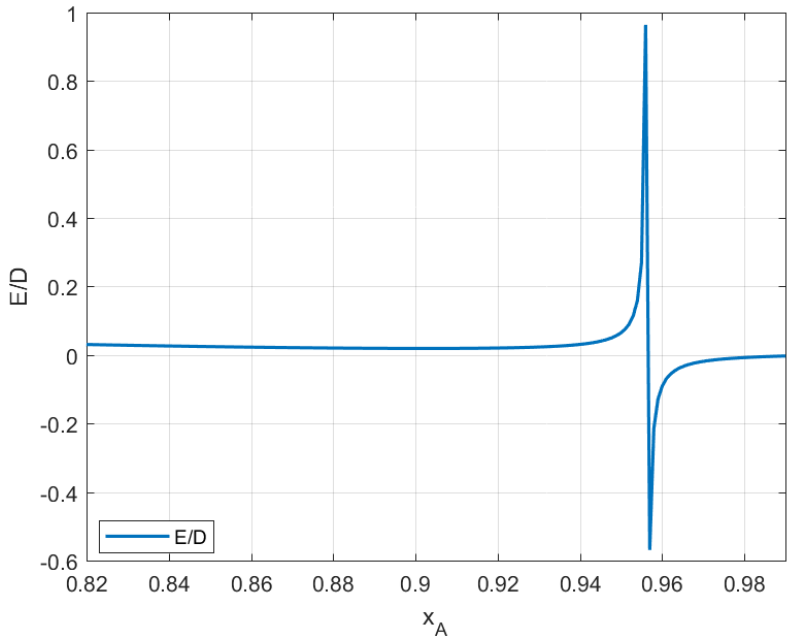


Figure 4.11: Trend of E/D obtained through the implementation

The trend of (E/D)min infig. 4.11 matches well with one constructed by Petlyuk fig. 3.9 [68].

Chapter 5

Case Studies

In the following chapter, the cases studies on which the constructed CAMD is used are presented. The mixture of interest in both cases is acetone-methanol. Acetone and methanol are major compounds of the aqueous product resulting from hydrocarbon synthesis by the Fischer-Tropsch process through the usage of carbon monoxide and hydrogen [34]. The acetone(A)-methanol(B) mixture exhibits a minimum boiling azeotrope at 328.29 K with an acetone composition of x_A=0.79 at atmospheric pressure [74]. Another reason this mixture is of interest in this work is that it has been studied extensively in the literature.

The first case studies consist of evaluating the benchmark entrainer used for the separation of acetone and methanol. These entrainers are reported by Kossack et al. (2008) and consist of: water, DMSO, and ethylene glycol [45]. It is therefore of interest to see if these survive the different constraints formulated in this work. The second Case study consist of applying the CAMD procedure in it entirety to determine suitable entrainer to the acetone-methanol case. At first, the CAMD constraints are formulated to accommodate the acetone-methanol case and thereafter, it is tested.

5.1 CAMD Framework: Acetone-methanol separation

5.1.1 Problem definition

It is desired to obtain heavy entrainer candidates capable of separating the azeotropic mixture Acetone-methanol through extractive distillation. It is furthermore desired to obtain acetone as the top product of the extractive column. The entrainer should therefore associate preferentially with methanol. The entrainer must be in liquid phase during operation. It should also be easy to separate the entrainer-methanol mixture in the recovery column. The entrainer should also not result in any hazardous gases nor cause extensive energy consumption.

The needs are then translated into target properties as shown in table 5.1. For each of these properties, target values are set to satisfy the needs. These target needs are reported in table 5.2.

Table 5.1: Entrainer needs and corresponding target properties considered

Need	Target property	Comment
Entrainer	melting point	An entrainer indicates that the product must be liquid.
Heavy entrainer	Boiling point	The boiling point of the entrainer must be sufficiently high so there is no risk it will evaporate
Reduce energy consumption	Enthalpy of vaporisation	Difficult to predict as it depends on the pressure, boiling point and enthalpy of vaporization However in this work it is assumed that a low enthalpy of vaporisation directly translate to less energy usage
Affinity towards methanol	Selectivity at infinite dilution	The litterature suggest that a high selectivity translates to a high affinity to associate with methanol and can thus be removed as the bottom product of the extractive column
Miscibility	Gibbs energy of mixing	A spontaneous mixing process correspond to a negative Gibbs energy of mixing.
No miscibility Gap	Gibbs energy of mixing	a concave profil for the Gibbs energy of mixing suggest the absence of mscibility gaps.
No azeotrope formation	relative volatility	The entrainer should not form any new azeotropes. Thus its corsponding relative volatility must not cross unity at any point.
Safety consideration	Chemical groups	Safety considerations can be applied through the choice of groups to build the entrainer.

Table 5.2: Target property values in the sepration of acetone-methanol

Target property	Target value	Comment
Melting point	greater than 298 K	The melting point of the solvent should be below room temperature to be in the liquid phase during processing.
Boiling point	greater than 368K and less than 500K	the highest boiling compound of the azeotropic mixture is methanol. Thus the a temperature difference of about 30K is chosen between the entrainer and methanol. The upper bound is dictated by the lowest critical temperature in the mixture.
Enthalpy of vaporization	less thn 67 kJ/mol	There is no solid reasoning behind the value set other than to avoid compiund with verry high entalpy of vaporisation
Selectivity at infinte dilution	above 2	Based on the observation by Kossack et al. (2008) they concluded that entrainer with selectivity above two have potential to be good candidates.
Gibbs energy of mixing	below zero	To reflect the spontaneous mixing of compound
second derivative of Gibbs energy of mixing	above 0	to reflect the trend of the mixing function
relative volatility	only on one side of unity along the whole compsition range	T ensure that the relative volatility does not cross unity.
Basis set	No sulphur, no flour, no chloride and no bromine	These chemical groups are usually associated with hazardeous gases.

5.1.2 CAMD constraint selection

The groups considered to form the backbone of the entrainer are reported in table 5.3. The considered groups all contain exclusively Carbon, hydrogen and oxygen except DMSO. This compound is included in the analysis since it is a benchmark entrainer for the separation of acetone and the methanol.

Table 5.3: Basis set for the CAMD

	Groups
Category 1	H_2O - HCOOH - DOH
Category 2	CH_3 - CH_2 - CH - C - OH
	CHO - HCO - CH_3COO
	CH_2COO - HCOO - CH_3CO
	CH_2O - COOH - COO

The selected models to incorporate into the CAMD can be seen in table 5.4 and include the d´structure model, the property model, the thermodynamic model as well as the process model.

Table 5.4: Models selected for the CAMD formulation

Type	Model	Reference
Structure model	Octet rule	Odele and Macchietto (1993) [65]
	Connectivity rule	Odele and Macchietto (1993) [65]
Property model	GCM	Constantinou and Gani (1994) [9]
Thermodynamic model	UNIFAC Original	Fredenslund (1975) [21]
		Hansen (1991) [36]
Process model	Fenske equation	Fenske (1932) [79]
	ISS method	Petlyuk et al. (2015) [69]

The objective function chosen is the entrainer to distillate ratio (E/D) determined by the ISS method by Petlyuk et al. (2015) [69]. This function should be minimised. A lower entrainer to distillate ratio means less amount of the entrainer is used.

5.1.3 CAMD formulation

The mathematical formulation of the CAMD is given in the following: eq. (5.1) is the objective function, eq. (5.2) is the octet rule, eq. (5.3) is the connectivity constraints, eq. (5.4) constraints the total number of groups making an entrainer while eq. (5.5) constraints the occurrence of functional groups. eq. (5.6) ensures the entrainer is liquid, eq. (5.7) ensures the entrainer is heavy, eq. (5.8) ensures that the boiling point is below the critical point (filter faulty structures off), and eq. (5.9) ensures that the energy is reduced. eq. (5.10) ensures affinity is towards methanol,eq. (5.11) ensure no azeotrope is formed between A-E, eq. (5.12)ensure no azeotrope is formed between B-E, eq. (5.13) ensures miscibility and ensures that no miscibility gap occurs eq. (5.14). eq. (5.15) and 5.16 calculate the minimum number of stages in the extractive and recovery columns respectively. eq. (5.17) determines the composition of the isovolatility line intersection at the side A-E, eq. (5.18) constraints the search direction for $x_m in$. eq. (5.19) calculates the E/D ratio while eq. (5.20) calculates the reflux ratio.

The objective function:

$$\min \qquad F = E/D \tag{5.1}$$

The Structural constraints

$$v \sum_k (2 - v_k)\, n_k = 2\, m \tag{5.2}$$

$$\sum_i n_i \geq n_j(v_j - 1) + 2 \tag{5.3}$$

$$3 \geq \sum_k n_k \geq 6 \tag{5.4}$$

$$1 \geq \sum_k n_k - n_{CH_3} - n_{CH_2} - n_{CH} - n_C \geq 2 \tag{5.5}$$

The pure property constraints

$$T_{m_0}\ \log \sum_k n_k\, T_{m_1,k} \leq 298K \tag{5.6}$$

$$368K \leq T_{b_0}\ \log \sum_k n_k\, T_{b_1,k} \leq 500K \tag{5.7}$$

$$T_{c_0}\ \log \sum_k n_k\, T_{c_1,k} \geq T_{b_0}\ \log \sum_k n_k\, T_{b_1,k} \tag{5.8}$$

$$h_{v_0} + \sum_k n_k\, h_{v_1,k} \leq 67kJ/mol \tag{5.9}$$

The mixture property constraint

$$S_{AB}^{\infty} \leq 2 \tag{5.10}$$

$$\log \alpha_{AE}\Big|_{x_E \approx 0} \times \log \alpha_{AE}\Big|_{x_E \approx 1} \geq 0 \tag{5.11}$$

$$\log \alpha_{BE}\Big|_{x_E \approx 0} \times \log \alpha_{BE}\Big|_{x_E \approx 1} \geq 0 \tag{5.12}$$

$$\sum_i x_i\ \log(\gamma_i) + \sum_i x_i\ \log(x_i) \leq 0 \tag{5.13}$$

$$\frac{\Delta G^{mix}}{RT}\Big|_{x_{dp+1,i}} - 2\,\frac{\Delta G^{mix}}{RT}\Big|_{x_{dp,i}} + \frac{\Delta G^{mix}}{RT}\Big|_{x_{dp-1,i}} \geq 0 \tag{5.14}$$

The process model:

$$N_{min,1} = \frac{\log S_1}{\log \alpha_{m,1}} \tag{5.15}$$

$$N_{min,2} = \frac{\log S_2}{\log \alpha_{m,2}} \tag{5.16}$$

$$\frac{K_A(x_A^{iso}, T)}{K_B^{\infty A,E}(x_A^{iso}, T)} = 1 \tag{5.17}$$

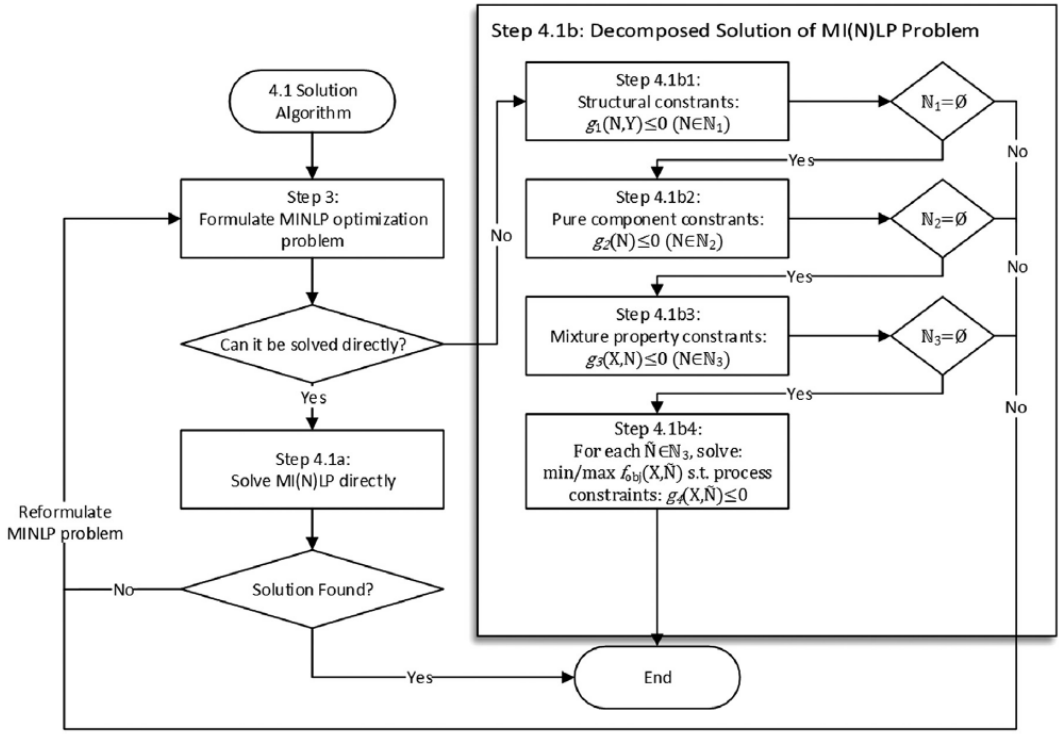
$$x_A^r\ \ \leq x_A^{iso} \tag{5.18}$$

$$E/D = 1 - \frac{1 - K_B^{\infty 1,2}}{x_A^r(K_A - K_B^{\infty 1,2})} \tag{5.19}$$

$$R_{min} = \frac{K_B^{\infty 1,2} - (E/D)_{min}}{1 - K_B^{\infty 1,2}} \tag{5.20}$$

5.1.4 Solution strategy

The solution strategy adopted is the decomposition method by Karunanithi et al (2005)[42]. All feasible structures are generated and their properties are evaluated in a sequential manner. In terms of the GAMS software, the solvers used are 'BARON" and "CONOPT".



The results are presented in an a chapter on their own.

Chapter 6
Results & Discussions

Two cases studies were investigated in this work. The first consist of testing already existing benchmark entrainer against the developed framework. The second consist of generating new entrainers through the CAMD framework.

6.1 Benchmark entrainers for acetone methanol

The benchmark entrainer considered are: water (H2O), dimethylsulfoxide (DMSO) and ethylene glycol (DOH). All three entrainer survive the evaluation of the pure property constraints, however water is eliminated due to the formation of azeotrope (as shown in the implementation chapter) and DMSO is eliminated due to the limit on the selectivity. The elimination process can be seen in fig. 6.1.

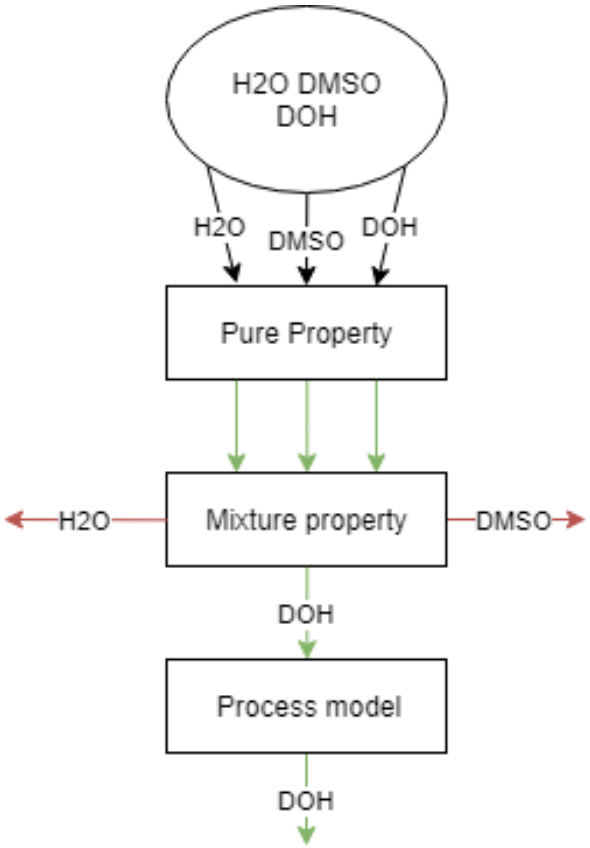


Figure 6.1: Evaluation of benchmark entrainer for acetone-methanol

The only surviving candidate is DOH. The result suggest that the criteria set on the selectivity might be too stringent. The elimination of water from the process cannot be avoided due to the limitation set by the UNIFAC thermodynamic model. It is however still

interesting to evaluate their performance in the process model and are thus re-included in the process model evaluation. The obtained results are summered in table 6.1.

Table 6.1: Process model evaluation for benchmark entrainers

Entrainer	S	N1	N2	(E/D)min	R
H2O	5.24	16	9	0.020	3.618
DMSO	1.46	26	3	0.619	1.385
DOH	2.24	21	2	1.021	1.376

The highest selectivity is exhibited by water followed by DOH. The number of stages in the extractive column is lowest for water followed by DOH. This could suggest that the number of stages in the extractive column decreases with increasing selectivity. This observation is in accordance with many observations and studies conducted in the literature however concerns about its validity have also been noted [60]. The separation in the recovery column is slightly harder than for the two other entrainer candidates (9 compared to 3 and 2). This could be due to the fact that the boiling point of water is the closest to methanol amongst the present candidates. The boiling point of DOH and DMSO are very close which could suggest the reason why the number of stages in the recovery column are very close. It seems based on the results that a low entrainer to distillate is neither associated with the lowest number of stages in any column nor have an effect on the reflux ratio. Water posses the highest reflux ratio and the lowest E/D. which could also suggest the low number of stages in the extractive column (an increase in the reflux ratio usually corresponds to a lower number of stages which only the case for water). Based on the results obtained in this case, it is not possible to make any clear tendencies on how the parameters of the process model interact. It is however important to note that the Fenske method and the ISS are based on totally different assumptions (Fenske assumes total reflux, and extractive distillation cannot operate under such assumptions). A larger amount of data is needed to be able to make any meaningful suggestions on the existence of any clear relation between the process parameters determined.

6.2 Entrainer design for acetone-methanol

A total of 476 possible structures have been generated by assembling the groups from "Category 2". Four "Category 1" structures are also considered resulting in a total of 480 feasible structures have been generated. An overview of these can be seen in the Appendix. Evaluating the pure property constraints indicated in table 5.2 eliminates 140 candidates while the mixture property control step including the restriction set on the selectivity, eliminates most candidates and only 18 candidates are left. The reduction of possible candidates is illustrated in fig. 6.2.

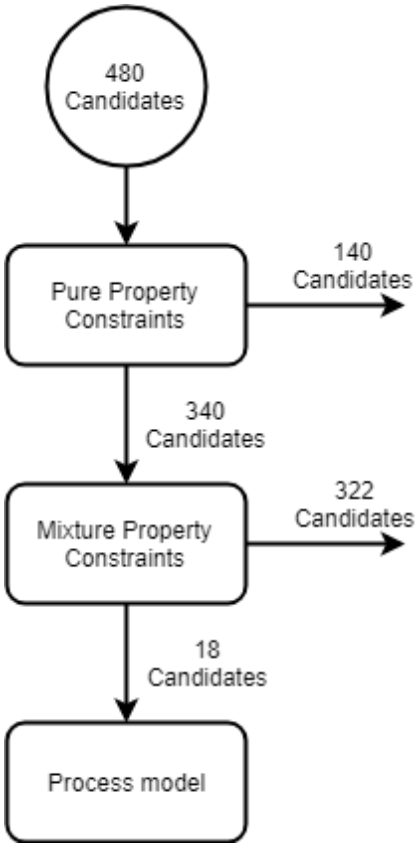


Figure 6.2: Continuous reduction of the solution space fr the acetone-methanol case

All 18 obtained candidates can be seen in table 6.2. None of the obtained candidates are carboxylic acids. However nearly obtained structures are alcohols (either contains OH or HCO). The obtained candidates are only identified by the groups present in the structure and not how the groups are aligned or connected. This i a notable disadvantage since the true identity of the candidate is not known. However it is possible to verify the candidate by manually assembling the molecule. A noticeable candidate in table 6.2 is "multsol1". This structure corresponds to ethylene glycol (OH-CH2-CH2-OH). Since ethylene glycol is defined as a group on its own, this candidate should not be considered in further studies although it is included in table 6.3. The candidate "multsol291" and "multsol292" can only make up unique structures (HCO only has one valency). Assembling the candidates "multsol291" result in the structure seen in fig. 6.3. The obtained structure corresponds to Succinaldehyde. The GCM predicted the boiling point to be 422 K , while to experimental value is only 331.15 K and should not ave survived the pure property check. This result clearly illustrates the limitations of using the predictive models.

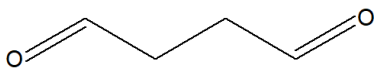


Figure 6.3: Multsol 291 structure corresponding to Succinaldehyde

Table 6.2: Entrainer candidate for the separation of acetone and methanol

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol1		2			2									
multsol105	2			1	1		1							
multsol107	2			1	2									
multsol146	2			1			2							
multsol291		2					2							
multsol292		2			1		1							
multsol369	1		1				2							
multsol372	1		1		1		1							
multsol373	1	1	1				2							
multsol381	1	1					1				1			
multsol431	1	1	1		1		1							
multsol456		1			1		1							
multsol457		1					2							
multsol461		3					2							
multsol468		3			1		1							
multsol470		1			2									
multsol471	1		1		2									

Another example of candidates resulting in a unique structure is "multsol461" which corresponds to glutaral. This compound is used or sterilisation of medicinal tools and in surgeries. Therefore, the prospects of it being used as an entrainer is not completely possible.

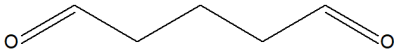


Figure 6.4: Multsol 461 structure corresponding to glutaral

It is however interesting to investigate whether glutaral is capable of acting as an entrainer. For this, a pseudo x-y diagram is constructed and can be seen in fig. 6.5.

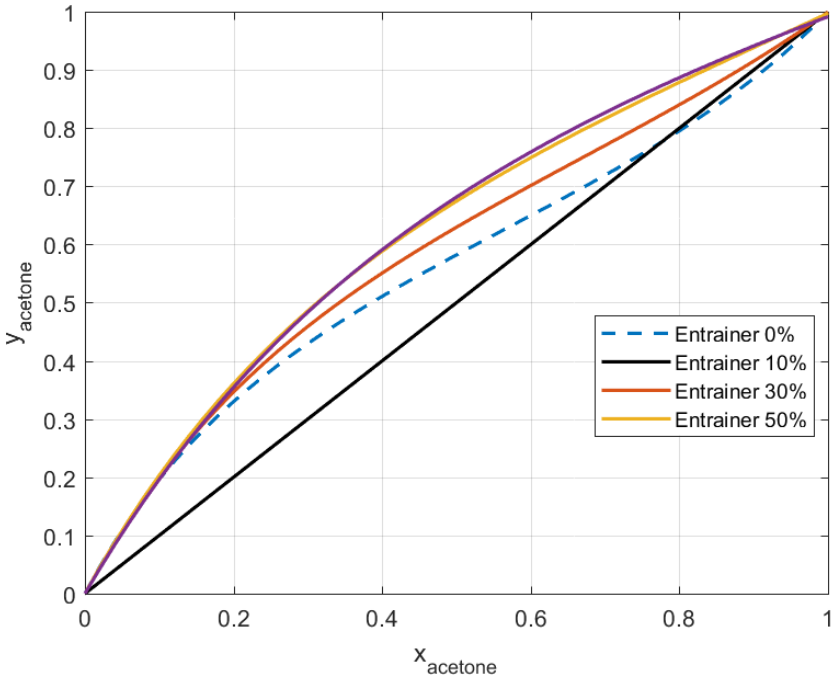


Figure 6.5: Pseudo-binary xy diagram for acetone methanol at various entrainer fraction

From fig. 6.5, it is evident that the entrainer does effect the relative volatilities. However it seems that an azeotrope is still present. A further investigation revealed that in fact the azeotrope is still present at 50% entrainer as can be seen in fig. 6.6. In fact it seems the azeotrope is becoming more prominent as the entrainer composition increases. The reason this is not captured by the model has not been determined since no experimental values for the boiling point or critical point were found. These could be the reason why the model does not capture this behaviour.

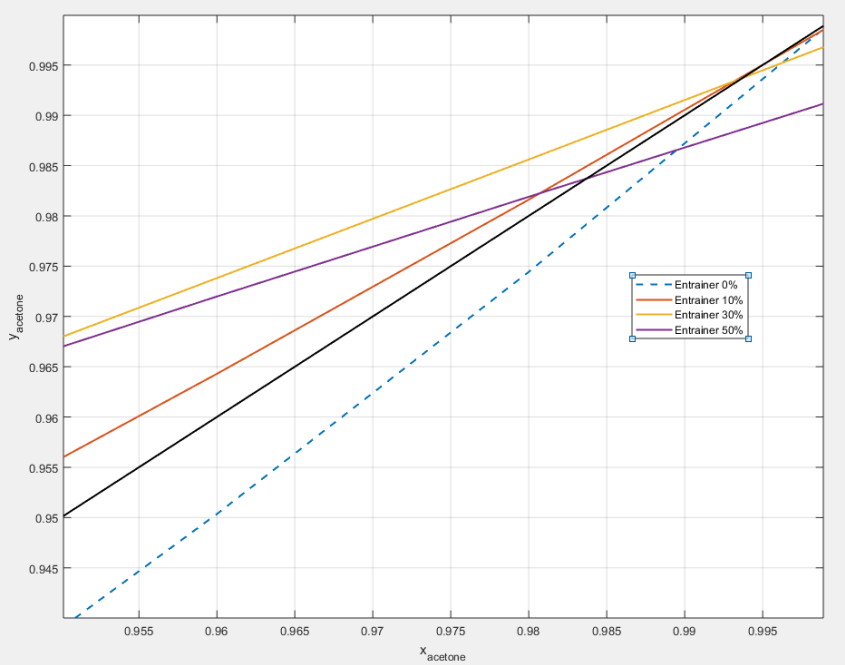


Figure 6.6: Zoom on Pseudo-binary xy diagram for acetone methanol at various entrainer fraction at high acetone liquid fractions

For the 19 obtained entrainer candidates, the process model was evaluated. In addition to the results from the process model, the selectivity, the capacity are also of interest due to their regular occurrence in the literature concerning extractive distillation. These results are reported in table 6.3. Based on the results obtained, not apparent clear conjuncture can be made concerning the E/D and R parameters. The highest value of E/D does neither correspond to the highest nor the lowest value of R. An interesting feature to investigate is if there is any clear correlation between any of these parameters. Figures featuring the parameters plot against each other can be found in the appendices (Appendix B). In most cases, no clear correlation or pattern could be concluded. However in the following, the most interesting trends are highlighted and discussed in details.

Table 6.3: Selected results from the CAMD procedure

Candidate	S	C	Nmin1	Nmin2	(E/D)min	R
DOH	2.24	1.24	21	2	1.021	1.376
multsol1	2.74	1.17	21	4	0.698	1.529
multsol105	2.36	1.92	25	3	0.718	1.859
multsol107	2.30	1.25	24	4	0.830	1.802
multsol146	2.18	2.41	27	4	0.530	1.948
multsol291	2.68	2.50	22	4	0.480	1.656
multsol292	2.86	1.89	21	4	0.588	1.535
multsol369	2.75	2.56	22	5	0.416	1.573
multsol372	2.93	1.91	22	5	0.416	1.573
multsol373	2.16	2.36	27	3	0.579	1.999
multsol381	2.00	2.01	30	9	0.848	2.666
multsol431	2.32	1.91	25	3	0.711	1.854
multsol456	4.01	1.79	17	3	0.569	1.178
multsol457	3.96	2.58	17	6	0.261	1.149
multsol461	2.12	2.23	29	2	0.609	2.056
multsol468	2.28	1.89	25	3	0.700	1.855
multsol470	3.65	1.03	17	6	0.504	1.459

Considering the selectivity since it is the most prominent reoccurring parameter when preforming entrainer design. The relation between the selectivity and the minimum number of stages in the extractive column is investigated and can be seen in fig. 6.7. For low values of the selectivity, the minimum number of stages in the extractive column has the its maximum value which decreases with increasing selectivity. Some of the lowest values for the number of stages i achieved the high values of the selectivity. This trends is not surprising and has been review in the literature chapter. The same conclusion as Momoh (1991) also hold true. In fact the lowest value for the number of stages does not correspond to the highest value of the selectivity [60]. However the trend of decreasing number of stages with increasing selectivities is undeniably present.

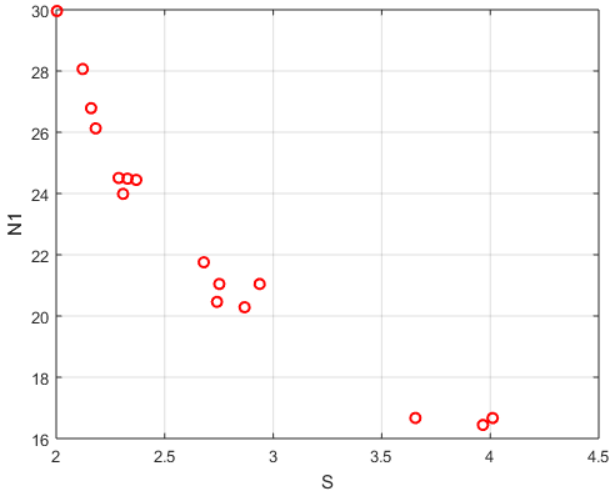


Figure 6.7: The selectivity Vs. the number of stages in the extractive column

Based on the figure obtained in fig. 6.8, it it seems that there is a tendency for E/D to decrease with increasing selectivity. However more data points are needed to make any clear conclusions.

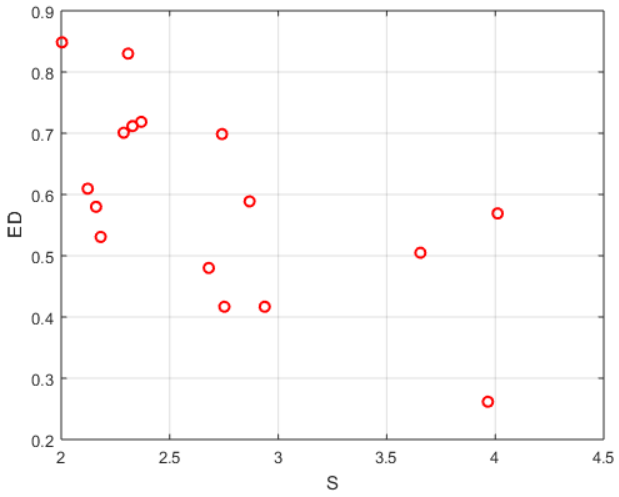


Figure 6.8: Selectivity Vs-. Entrainer to distillate ratio

The reflux ratio is clearly decreasing with increasing selectivity as seen in fig. 6.9. This could be due to the fact that a high selectivity reflects the affinity of methanol to associate preferentially with the entrainer and thus it becomes less volatile. This will result in the a top product that is more pure and thus less reflux is needed to achieve the intended purity.

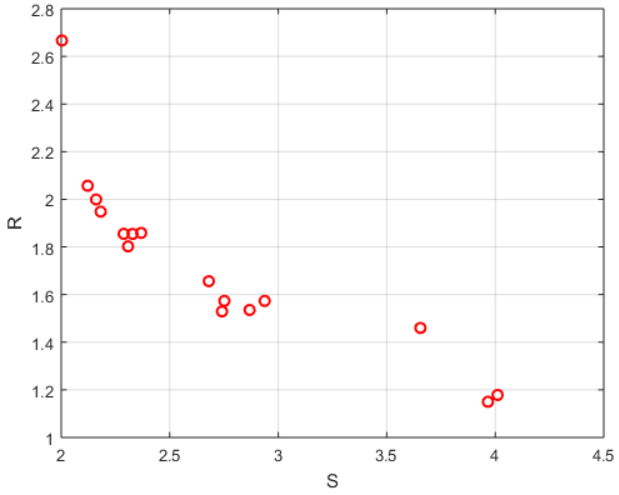


Figure 6.9: Selectivity Vs. reflux ratio

The trend of the reflux with regard to the number of stages in the extractive column can be seen in fig. 6.10. Usually, a minimal values of the reflux ratio corresponds to an infinite number of stages and vice versa. However, it seems from fig. 6.10 that increasing number of stages result in an increase in the reflux. This could be due to the fact that the Fenske equation used to calculate the minimum number of stages is not entirely applicable on extractive distillation. In fact one of the assumptions made was that the entrainer was

present throughout the column, which is not true since the entrainer is heavy and is not very volatile.

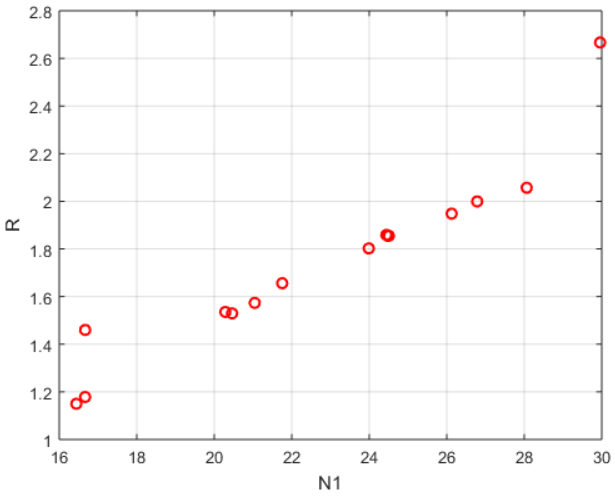


Figure 6.10: Number of stages in the extractive column Vs. The reflux ratio

In fig. 6.11 the relation between the entrainer to distillate flowrate and the reflux ratio are investigated. The figure indicates vaguely that there might be a relation between the tow parameters. In fact, the lowest value for E/D correspond to the lowest value of the reflux. The same is true for the highest values. This can be explained through the underlying assumption of the ISS method. The reflux ratio determined corresponds to the reflux in the top section. Based on the ISS assumption, only the top product (acetone) and the entrainer are present in this section. Therefore an increasing value of the E/D ratio necessitates additional reflux to achieve the desired purity.

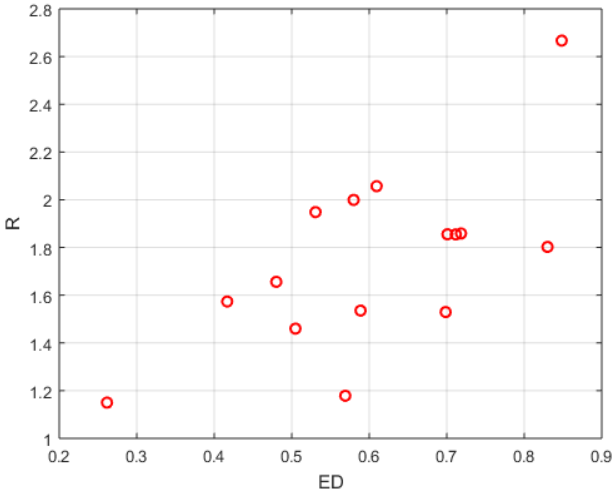


Figure 6.11: Entrainer to distillate ratio Vs. reflux ratio

Conclusions

A Computer-Aided Molecular Design (CAMD) framework has been developed to conduct molecular design for heavy entrainer in the separation of minimum boiling azeotropes. The Framework incorporates a series of models of different types: structural models, property models and thermodynamic models. Each model incorporated has been validated through a series of sanity tests and through working examples. All models implemented in this work were successfully implemented and validated. The current implementation does offer great advantages in terms of maintenance. The only "human" interaction needed is the definition of the azeotropic pairs, their pure component property and the basis set from which the entrainer should be constructed. This illustrates the strength of the predictive model. However there is always a risk associated with using them. Through the various implementations and case studies, the great weakness of property predictive models has been exposed. In many cases even small differences in the pure property prediction can have an immense effect. These small discrepancies have resulted in the formation of azeotropes that contradicts the actual real world case (formation of azeotrope between water and acetone) or the disappearance of azeotropes when in reality they are present. A novel shortcut method have been implemented in the CAMD framework. The short-cut is able to determine the minimal entrainer to distillate ratio as well as the corresponding reflux ratio. This work serves as a proof of concept that the ISS method can be incorporated in a CAMD framework. However, n the absence of any other attempts to incorporate this model into CAMD procedures, it has been difficult to to compare the performance of this model. However, the model did present some interesting results that have previously been noted in the literature. This observations include the decrease of the capital investment with increasing selectivities. Thus was observed through the decrease of the minimum number of stages in the extractive column with increasing selectivity. Furthermore, i was observed that with increasing selectivity the reflux ratio in the extractive column is reduced. Based on study conducted in this work, it can be concluded that the selectivity is by no mean an unreliable parameter in the search of suitable entrainers. However it must only be used as a form of threshold value and not a ranking parameter. The use of the ISS method by Petlyuk (2004) must be applied with care. In fact during the conduction of this work it became clear that the method is extremely sensitive to the vapour-liquid distribution coefficient at infinite dilution. This resulted in some difficulties when implementing the method in GAMS. This observation was also noted by Petlyuk (2015) [69]. His recommendation was to use experimental data to determine the value of the operating parameter E/D and R.

Future work

The future work consist of enhancing the ability to predict the pure compound properties. These are extremely important as they have great influence on the saturated vapour calculation. Increased accuracy will increase the reliability of the mode. This can be done by implementing the second order groups by Marrero and Gani [58].

The ISS method should also be further investigated. The objective of this work to implement the method in a CAMD framework, it should now be studied further and to investigate the possibility of using it on maximum boiling azeotropes.

The developed framework is only able to suggest the groups from which the potential entrainer candidate is made of. This can be cumbersome when validating the obtained encountering candidates and thus it might be advantageous to construct a model capable of performing such group assembling and suggesting complete structure.

Appendices

Appendix A - Feasible structures from the implementation.

Table 6.4: My caption

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol1		2			2									
multsol10	2	2	1										1	
multsol100	2		1		1						1			
multsol101	2	1	1		1						1			
multsol102	2	1	1		1									1
multsol103	2	1	1		1				1					
multsol104	2			1	1					1				
multsol105	2			1	1		1							
multsol106	2			1	1	1								
multsol107	2			1	2									
multsol108	2	1			1							1		
multsol109	2	1		1	1					1				
multsol11	2	2									1			
multsol110	2			1	1								1	
multsol111	2				1							1		
multsol112	2	1		1	2									
multsol113	2	1	1		1									
multsol114	2		2		1								1	
multsol115	2			1	1			1						
multsol116	2	1		1		1							1	
multsol117	2	1		1		1		1						
multsol118	2		1			1								1
multsol119	2		1			1			1					
multsol12	2	2								1		1		
multsol120	2		1			1					1			
multsol121	2	1	1			1					1			
multsol122	2	1	1			1			1					

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol123	2	1	1			1								1
multsol124	2			1		1							1	
multsol125	2			1		1		1						
multsol126	2			1		1	1							
multsol127	2	1		1		1	1							
multsol128	2			1		2								
multsol129	2	1				1						1		
multsol13	2	2	1							1				
multsol130	2					1						1		
multsol131	2	1	1			1								
multsol132	2		2			1				1				
multsol133	2			1		1				1				
multsol134	2	1	1				1							1
multsol135	2	1	1				1				1			
multsol136	2	1					1					1		
multsol137	2	1	1				1							
multsol138	2	1		1			1						1	
multsol139	2	1		1			1			1				
multsol14	2	2							1					
multsol140	2		1				1				1			
multsol141	2			1			1						1	
multsol142	2			1			1			1				
multsol143	2		1				1		1					
multsol144	2						1					1		
multsol145	2			1			1	1						
multsol146	2			1			2							
multsol147	2		1				1							1
multsol148	2		2				1						1	

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol149	2	1		1			1	1						
multsol15	2	2	1					1						
multsol150	2		2				2							
multsol151	2	1	1				1		1					
multsol152	2		1					1			1			
multsol153	2	1	1										1	
multsol154	2		1										1	1
multsol155	2	1		1									2	
multsol156	2			1									2	
multsol157	2	1	1										1	1
multsol158	2		1										1	
multsol159	2	1		1				1		1				
multsol16	2	2	1				1							
multsol160	2			1				1		1				
multsol161	2			1				1					1	
multsol162	2			1				2						
multsol163	2	1		1				1					1	
multsol164	2	1	1							1				1
multsol165	2	1								1		1		
multsol166	2	1	1							1				
multsol167	2		1							1				1
multsol168	2		1							1	1			
multsol169	2									1		1		
multsol17	2	2	1			1								
multsol170	2			1						1			1	
multsol171	2			1						2				
multsol172	2	1		1						1			1	
multsol173	2		2							1			1	

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol174	2	1	1							1	1			
multsol175	2		1							1				
multsol176	2	1	1					1						1
multsol177	2	1	1					1			1			
multsol178	2	1						1				1		
multsol179	2	1	1					1						
multsol18	2	2												2
multsol180	2		1					1	1					
multsol181	2		1					1						1
multsol182	2										1			1
multsol183	2		1								1		1	
multsol184	2	1	1								1		1	
multsol185	2	1									1			1
multsol186	2										2			
multsol187	2										1			
multsol188	2								1		1			
multsol189	2								1					1
multsol19	2	3							1					
multsol190	2		1						1				1	
multsol191	2		1						1	1				
multsol192	2	1												2
multsol193	2	1							1					1
multsol194	2	1							1		1			
multsol195	2	1	1						1				1	
multsol196	2	1	1						1	1				
multsol197	2	1										1	1	
multsol198	2													2
multsol199	2								2					

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol2	2	2				1						1		
multsol20	2	2	1		1									
multsol200	2								1					
multsol201	2							1				1		
multsol202	2	1												1
multsol203	2													1
multsol204	2		2					1					1	
multsol205	2	2									2			
multsol206	2	1									2			
multsol207	2		2					2						
multsol208	2	1	1					1	1					
multsol209	2	1									1			
multsol21	2		2		1			1						
multsol210	2		1					1						
multsol211	2		1				1							
multsol212	2	1		1		1				1				
multsol213	2		1			1								
multsol214	2	1		1						2				
multsol215	2	1		1	1			1						
multsol216	2		1		1									
multsol217	3	1		1				1						
multsol218	3		1						1					
multsol219	2		2			2								
multsol22	2		2		1		1							
multsol220	4			1					1					
multsol221	2	1		1		2								
multsol222	2	1							2					
multsol223	2	2							2					

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol224		1						1					1	
multsol225		1						1		1				
multsol226	1		1										2	
multsol227	1		1		1								1	
multsol228	1	1	1							1			1	
multsol229	1	1	1							2				
multsol23	2		2		1	1								
multsol230	1		1							1			1	
multsol231	1		1		1					1				
multsol232		1				1							1	
multsol233		1				1		1						
multsol234		1				2								
multsol235		4					1						1	
multsol236		4					1			1				
multsol237		4					1	1						
multsol238	1	1						1	1					
multsol239	1	1						1						1
multsol24	2		2		1					1				
multsol240	1							1						1
multsol241	1							1			1			
multsol242	1							1	1					
multsol243	1	1						1			1			
multsol244	1	1						1						
multsol245		4						1		1				
multsol246		4				1							1	
multsol247		4				1				1				
multsol248		1				1				1				
multsol249	1	2				1			1					

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol25	2		2			1							1	
multsol250	1	3				1								1
multsol251	1	2				1								1
multsol252	1	2				1					1			
multsol253		3				1				1				
multsol254		3				1		1						
multsol255		2				1							1	
multsol256		2				1				1				
multsol257		2				1		1						
multsol258		2				2								
multsol259		3				1							1	
multsol26	2		2			1		1						
multsol260		2				1	1							
multsol261		2			1	1								
multsol262	1	3				1								
multsol263	1	2				1								
multsol264		4				1	1							
multsol265		3				2								
multsol266		3				1	1							
multsol267	1	1				1					1			
multsol268	1	1				1								1
multsol269	1	1				1			1					
multsol27	2		2			1	1							
multsol270	1					1								1
multsol271	1					1					1			
multsol272	1	4				1								
multsol273	1	1							1	1				
multsol274	1								1	1				

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol275	1				1				1					
multsol276	1	1							1				1	
multsol277	1	1			1				1					
multsol278	1										1		1	
multsol279	1	1									1		1	
multsol28	2		2				1	1						
multsol280	1	1											1	1
multsol281	1	2					1							1
multsol282	1	2					1		1					
multsol283	1	2					1				1			
multsol284	1	3					1							1
multsol285		2					1						1	
multsol286		3					1			1				
multsol287		2					1			1				
multsol288		3					1	1						
multsol289		2					1	1						
multsol29	2		2				1			1				
multsol290		3					1						1	
multsol291		2					2							
multsol292		2			1		1							
multsol293	1	4					1							
multsol294	1	3					1							
multsol295	1	2					1							
multsol296	1	3					1		1					
multsol297	1	2						1			1			
multsol298	1	3						1						1
multsol299	1	3						1			1			
multsol3	2	2					1					1		

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol30	3		2										1	
multsol300	1	2						1						1
multsol301	1	2						1	1					
multsol302	1	3						1	1					
multsol303		3						1					1	
multsol304		2						2						
multsol305		3						1		1				
multsol306		2						1					1	
multsol307		2						1		1				
multsol308	1	2											1	
multsol309		4			1					1				
multsol31	3		2					1						
multsol310		3			1			1						
multsol311		2			1			1						
multsol312	1	3						1						
multsol313	1	2						1						
multsol314		3			1								1	
multsol315		2			1								1	
multsol316	1	3											1	1
multsol317	1	2											1	1
multsol318	1	3			1									1
multsol319	1	2			1									1
multsol32	3		2				1							
multsol320	1	3							1	1				
multsol321	1	3			1				1					
multsol322	1	2							1	1				
multsol323	1	3							1				1	
multsol324	1	2							1				1	

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol325	1	2			1				1					
multsol326	1	1	1			1							1	
multsol327	1	1	1			1				1				
multsol328	1	1	1			1	1							
multsol329		3											2	
multsol33	2		2					1		1				
multsol330		2											2	
multsol331	1	1	1			2								
multsol332	2	3												1
multsol333	1	3								1	1			
multsol334	1	3								1				
multsol335	1	2								1	1			
multsol336	1	2									1		1	
multsol337	1	2								1				1
multsol338		3								1			1	
multsol339		3								2				
multsol34	3		2			1								
multsol340		3			1					1				
multsol341		2								2				
multsol342	2	3									1			
multsol343	1	3									1		1	
multsol344	1	2			1						1			
multsol345		2								1			1	
multsol346		2			1					1				
multsol347	2											1	1	
multsol348	1	2								1				
multsol349	1		1			1				1				
multsol35	2		2							2				

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol350	1		1			1		1						
multsol351	1		1			1	1							
multsol352	1		1			1							1	
multsol353	1		1			2								
multsol354	1	3											1	
multsol355	1					1			1					
multsol356	1	3				1			1					
multsol357	3	2										1		
multsol358	1	3			1									
multsol359	1	2			1									
multsol36	2		2		2									
multsol360		4			1	1								
multsol361	1	1	1			1		1						
multsol362	1	3					1				1			
multsol363		3						2						
multsol364	1	1	1				1	1						
multsol365	1	1	1				1			1				
multsol366		1					1			1				
multsol367	1		1				1						1	
multsol368	1		1				1	1						
multsol369	1		1				2							
multsol37	3		1						1		1			
multsol370	1		1							2				
multsol371	1		1				1			1				
multsol372	1		1		1		1							
multsol373	1	1	1				2							
multsol374		1					1						1	
multsol375	2	1		1			2							

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol376	1	1	1				1						1	
multsol377	1	1					1		1					
multsol378	1						1							1
multsol379	1								1				1	
multsol38	3		1			1						1		
multsol380	1	1					1							1
multsol381	1	1					1				1			
multsol382	1						1		1					
multsol383	1						1				1			
multsol384	1	1					1							
multsol385	1	1	1					1		1				
multsol386	1		1					1					1	
multsol387	1		1					1		1				
multsol388	1		1					2						
multsol389	1		1		1			1						
multsol39	3			1		1					1			
multsol390	1	4						1						
multsol391	2	1		1				2						
multsol392	1	1	1					1					1	
multsol393		1					1	1						
multsol394	1									1				1
multsol395	1	1								1	1			
multsol396	1	1								1				1
multsol397	1	2	1			1							1	
multsol398	1	2	1			1				1				
multsol399	1	2	1			1	1							
multsol4	2	2						1				1		
multsol40	3	1		1		1								

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol400	1	2	1			1		1						
multsol401	1									1	1			
multsol402	1	2	1					1					1	
multsol403	1	2	1					1		1				
multsol404	1	2	1				1	1						
multsol405	1	4											1	
multsol406	1	4								1				
multsol407	1	1								1				
multsol408	1	2	1					2						
multsol409	1	2	1		1			1						
multsol41	3			1		1								1
multsol410	1	2	1							1			1	
multsol411		1								1			1	
multsol412	1	2	1				1			1				
multsol413	1	2	1										2	
multsol414	1	2	1				1						1	
multsol415	1												1	1
multsol416	1	2	1				2							
multsol417	1	2	1							2				
multsol418	1	1											1	
multsol419	1	2	1		1					1				
multsol42	3			1		1			1					
multsol420	1	1	1		1								1	
multsol421	1	2	1		1		1							
multsol422		4								1			1	
multsol423	1	1	1		1					1				
multsol424		4				2								
multsol425	1	2	1			2								

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol426		4											2	
multsol427	1	1	1										2	
multsol428	1	1	1		1			1						
multsol429	1	2	1		1								1	
multsol43	3			1		1								
multsol430		4						1					1	
multsol431	1	1	1		1		1							
multsol432	2	2			1							1		
multsol433		4				1		1						
multsol434	1	1	1					2						
multsol435		4			1			1						
multsol436	1	1	1		1	1								
multsol437	1		1		1	1								
multsol438	1	2	1		1	1								
multsol439		1											2	
multsol44	3								1			1		
multsol440	1				1									1
multsol441	1	2	1		2									
multsol442		4			1								1	
multsol443		4								2				
multsol444		1								2				
multsol445	4			1										1
multsol446	1	1			1									1
multsol447		1			1								1	
multsol448	1	3								1				1
multsol449	1	1			1									
multsol45	3	1		1	1									
multsol450	1	4			1									

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol451	1	1			1						1			
multsol452		1			1					1				
multsol453		1			1			1						
multsol454		1			1	1								
multsol455		1						2						
multsol456		1			1		1							
multsol457		1					2							
multsol458		1				1	1							
multsol459	1	1				1								
multsol46	3			1			1				1			
multsol460		4						2						
multsol461		3					2							
multsol462		4			1		1							
multsol463		4					2							
multsol464	1				1						1			
multsol465		4			2									
multsol466	1	3				1					1			
multsol467	1	3			1						1			
multsol468		3			1		1							
multsol469	1	1	1		2									
multsol47	3			1			1							1
multsol470		1			2									
multsol471	1		1		2									
multsol472		3			1	1								
multsol473	3		2		1									
multsol474		3			2									
multsol475	2	1							1					
multsol476	4											2		

Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol48	3	1		1			1							
multsol49	3		1				1					1		
multsol5	2	2							1					1
multsol50	3	1	1						1					
multsol51	3			1			1		1					
multsol52	3			1			1							
multsol53	3		1						1					1
multsol54	3			1				1						1
multsol55	3			1				1			1			
multsol56	3			1					1				1	
multsol57	3			1					1	1				
multsol58	3			1				1	1					
multsol59	3			1				1						
multsol6	2	2							1		1			
multsol60	3	1							1			1		
multsol61	3		1							1		1		
multsol62	3			1						1				1
multsol63	3	1		1						1				
multsol64	3		2							1				
multsol65	3			1						1	1			
multsol66	3			1						1				
multsol67	3	1										1		1
multsol68	3	1									1	1		
multsol69	3	1										1		
multsol7	2	2									1			1
multsol70	3										1	1		
multsol71	3		1									1	1	
multsol72	4		1									1		

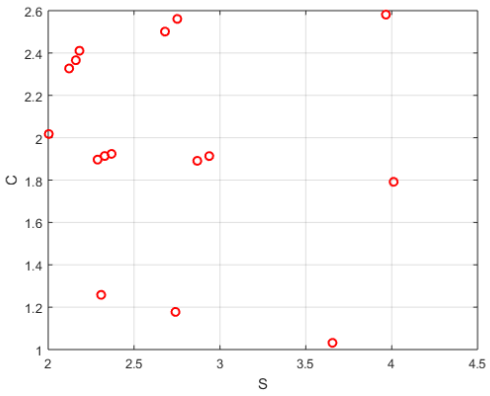
Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol73	3		1								2			
multsol74	3											1		1
multsol75	3											1		
multsol76	3			1	1									1
multsol77	3			1									1	1
multsol78	3	1		1									1	
multsol79	3			1							1		1	
multsol8	2	2												1
multsol80	3			1									1	
multsol81	3	1	1											1
multsol82	3		1								1			1
multsol83	3		1											1
multsol84	2		2										2	
multsol85	3		1											2
multsol86	3	1	1								1			
multsol87	3			1	1						1			
multsol88	4			1							1			
multsol89	3		1								1			
multsol9	2	2										1	1	
multsol90	3			1	1				1					
multsol91	3		1						2					
multsol92	3		1					1				1		
multsol93	3		1		1							1		
multsol94	3			1	1									
multsol95	2	1		1	1								1	
multsol96	2	1		1	1		1							
multsol97	2	1		1	1	1								
multsol98	2		1		1									1

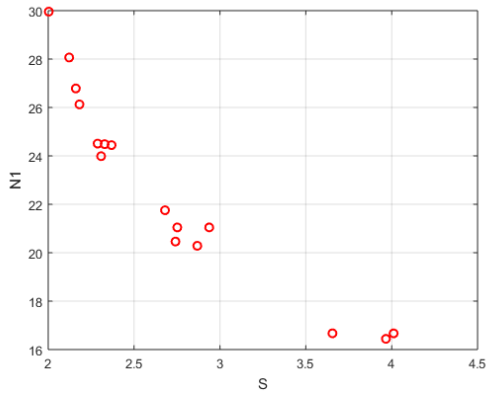
Table 6.4 continued from previous page

	CH3	CH2	CH	C	OH	CH3CO	HCO	CH3COO	CH2COO	HCOO	CH2O	CHO	COOH	COO
multsol99	2		1		1				1					

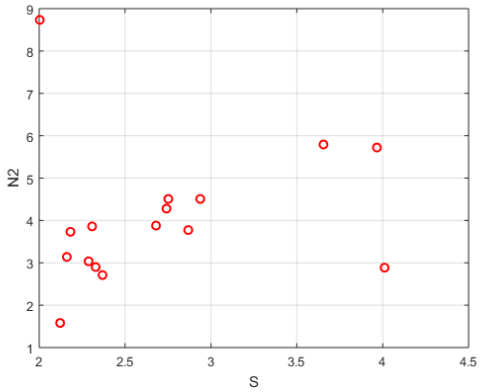
Appendix B - parameter plots



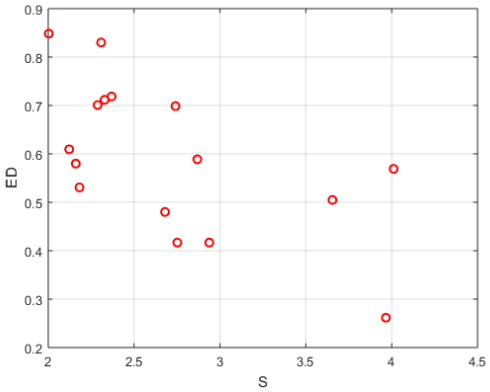
(a) S Vs C



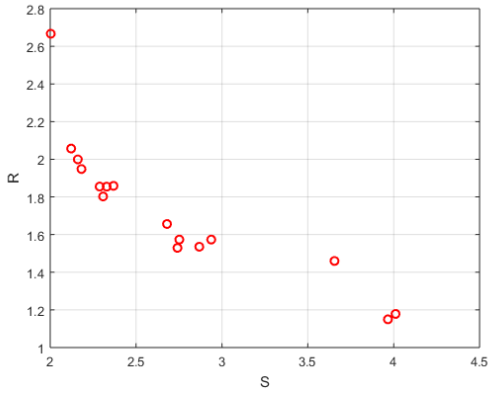
(b) S Vs. N1



(c) S Vs. N2

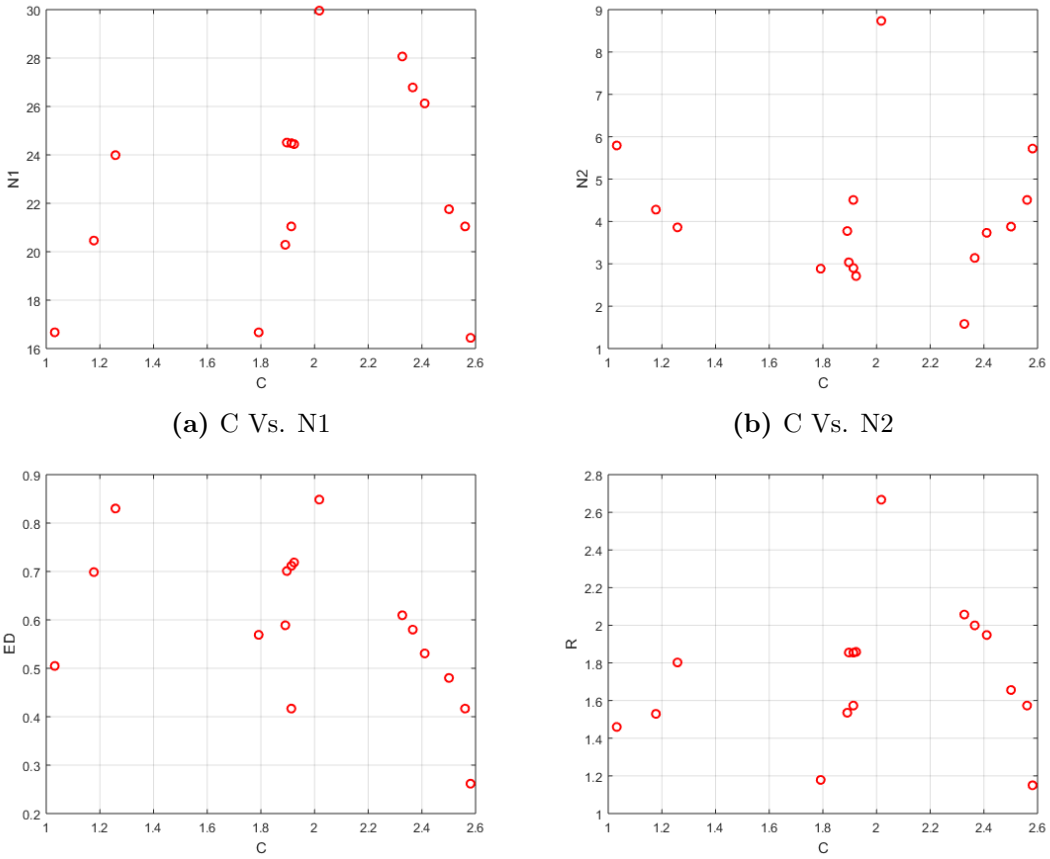


(d) S Vs. EDmin



(e) S Vs. R

Figure 6.12: Process model results against the Selectivity.



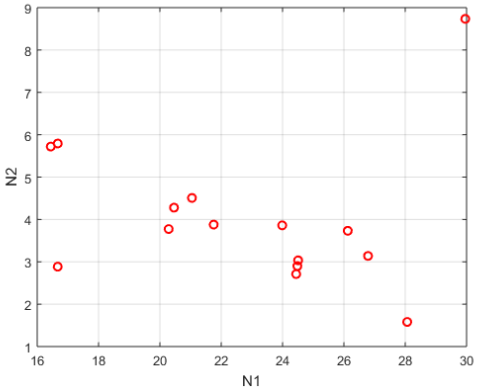
(a) C Vs. N1

(b) C Vs. N2

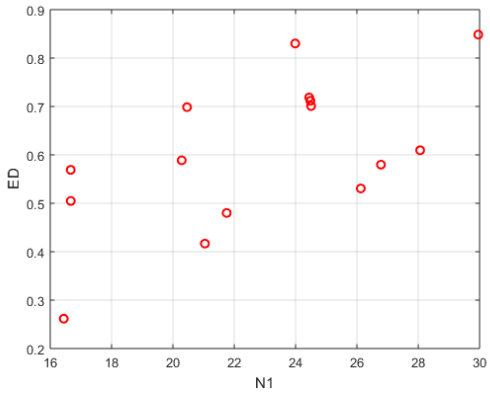
(c) C Vs. EDmin

(d) C Vs. R

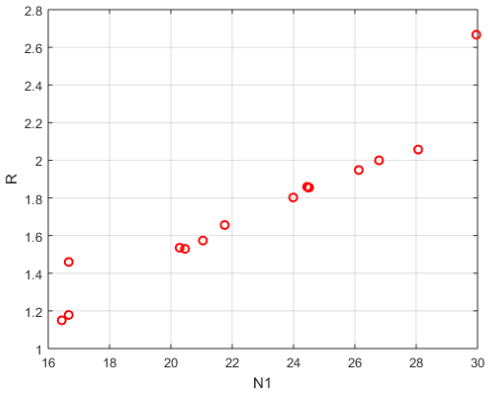
Figure 6.13: Process model results against the Capacity



(a) N1 Vs. N2

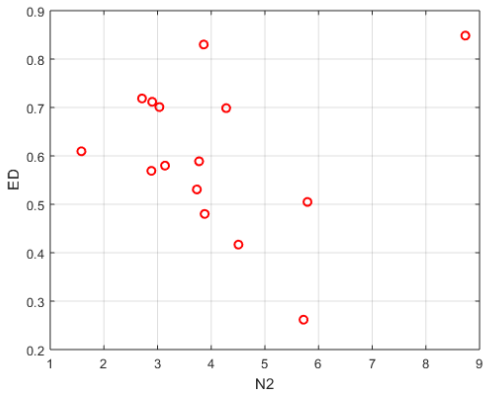


(b) N1 Vs. EDmin

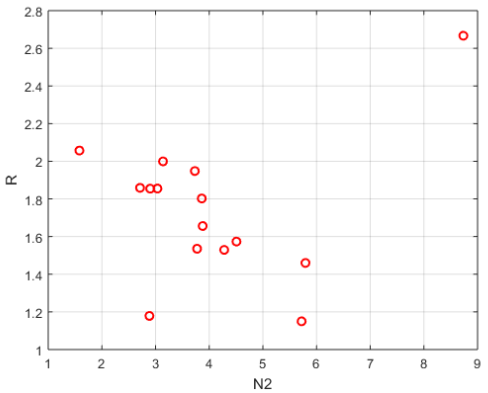


(c) N1 Vs. R

Figure 6.14: Process model results against the minimum number of stages in the extractive column

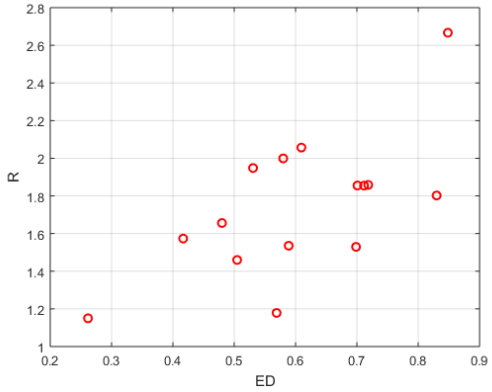


(a) N2 Vs. EDmin



(b) N2 Vs. R

Figure 6.15: Process model results against the minimum number of stages in the recovery column



(a) ED Vs. R

Figure 6.16: Minimum entrainer to distillate ratio to the reflux ratio

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